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GROUPES DE MONODROMIE EN GEOMETRIE ALGEBRIQUE

(SGA 7 I)

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INTRODUCTION

Soient S un schéma et $f : X \longrightarrow S$ un morphisme de schémas. Si f est propre et lisse, et que le nombre premier ℓ est inversible sur S, les groupes de cohomologie ℓ-adique $H^i(X_{\bar{s}}, \mathbb{Z}_\ell)$ des fibres géométriques de f forment un système local ℓ-adique sur S. Pour f seulement supposé propre, ces groupes sont les fibres d'un faisceau ℓ-adique $R^i f_* \mathbb{Z}_\ell$ sur S. La théorie des cycles évanescents met en relation la ramification de ce faisceau sur S et les singularités de f.

Nous ne considérerons que le cas où S est un trait hensélien (= spectre d'un anneau de valuation discrète hensélien). C'est en pratique le cas essentiel. Je renvoie à (I 2.1) pour une description heuristique de la théorie. Soient $S = \mathrm{Spec}(V)$, $k(\bar{\eta})$ une clôture algébrique du corps des fractions $k(\eta)$ de V et $k(\bar{s})$ la clôture algébrique correspondante du corps résiduel $k(s)$ ($k(\bar{s})$ est le corps résiduel du normalisé $\bar{V}$ de V dans $k(\bar{\eta})$). Dans le cas particulier où X est propre et plat sur S, de dimension relative n, et où f ne présente qu'un point de non lissité $x \in X_s$, on définit des $\mathrm{Gal}(\bar{\eta}/\eta)$-modules ϕ^i, de nature purement locale au voisinage de x, nuls pour $i \notin [0,n]$ (I 4.2), et on construit une suite exacte longue de $\mathrm{Gal}(\bar{\eta}/\eta)$-modules

$$\cdots \longrightarrow H^i(X_{\bar{s}}, \mathbb{Z}_\ell) \xrightarrow{sp} H^i(X_{\bar{\eta}}, \mathbb{Z}_\ell) \longrightarrow \phi^i \longrightarrow H^{i+1}(X_s, \mathbb{Z}_\ell) \longrightarrow \cdots$$

(le $\mathrm{Gal}(\bar{s},s)$-module $H^i(X_{\bar{s}}, Z_\ell)$ est regardé comme un $\mathrm{Gal}(\bar{\eta}/\eta)$-module avec action triviale de l'inertie ; sp est la flèche de spécialisation).

On donne aussi des critères, valables pour l'instant seulement en caractéristique 0, pour que les ϕ^i pour i petits soient nuls (par exemple, si X est de plus localement d'intersection complète, $\phi^i = 0$ pour $i \neq n$ (I. 4.5)).

Le théorème de monodromie affirme que l'action du sous-groupe d'inertie I de $Gal(\bar{\eta}/\eta)$ est quasi-unipotente : pour $T \in I$, il existe des entiers $N > 0$, $M > 0$ tels que l'endomorphisme $(T^M - I)^N$ de $H^i(X_{\bar{\eta}}, \mathbb{Z}_\ell)$ soit nul. On donne de ce théorème deux démonstrations. La première (I.1.2), de nature arithmétique, s'applique dès que les groupes de cohomologie considérés ont des propriétés de finitude raisonnables. Le point clef est que, lorsque $k(s)$ est de type fini sur son sous-corps premier, l'action de I est quasi-unipotente pour toute représentation ℓ-adique de $Gal(\bar{\eta}/\eta)$. La seconde démonstration, plus géométrique, requiert la pureté et la résolution des singularités : elle n'est pour l'instant valable qu'en caractéristique 0 ou pour un H^1. Elle apporte de précieuses informations sur l'exposant de nilpotence N ($N \leqslant i+1$ pour un H^i). Ainsi qu'on le verra ultérieurement, elle se prête bien, sur $\mathbb{C}$, à la comparaison avec la théorie transcendante.

Ces résultats, appliqués au H^1 des variétés abéliennes, i.e. à leur module de Tate, permettent d'étudier la réduction mod p de celles-ci. On donne ainsi deux démonstrations du théorème de réduction stable des variétés abéliennes selon lequel, après ramification, la fibre spéciale connexe du modèle de Néron est extension d'une variété abélienne par un tore. La première (I. 6) reprend la méthode de la démonstration arithmétique du théorème de monodromie. La seconde (IX 3.6) s'appuie sur une analyse beaucoup plus fine du modèle de Néron, et de ses propriétés de polarisation.

Les exposés I à V de Grothendieck n'ont pas été rédigés. Ils ont été résumés dans un exposé I . Les résultats énoncés y sont démontrés de façon succinte, mais essentiellement complète.

L'exposé II applique la méthode des pinceaux de Lefschetz et (I 5.3) à l'étude du groupe fondamental. Pour S une surface sur un corps algébriquement clos k, d'exposant caractéristique p, on montre que le groupe profini $\pi_1^{(p)}(S,s)$, complété en dehors de p du groupe fondamental, est de pro-(p)-présentation finie.

Comme expliqué plus haut, les exposés III à V n'existent pas.

L'exposé VI contient, avec quelques compléments, la théorie des déformations de Schlessinger. On y prend soin de tenir compte des automorphismes infinitésimaux des objets qu'on classifie, et de ne pas passer trop brutalement au foncteur des classes d'isomorphie d'objets. On y donne aussi une nouvelle construction des $\underline{\mathrm{Ext}}^i(L_{X/S}, -)$ ($L_{X/S}$ complexe cotangent relatif de X/S). Cet exposé sert dans le reste du séminaire surtout via l'étude qui y est faite des déformations des singularités quadratiques ordinaires.

Les exposés VII et VIII sont consacrés à la théorie des biextensions. Cette théorie joue un rôle essentiel dans l'exposé IX, pour exprimer ce qu'il advient d'une polarisation quand on passe d'une variété abélienne à un modèle de Néron de celle-ci. Cet exposé IX contient la démonstration du théorème de réduction stable des variétés abéliennes, et diverses applications.

La suite de ce séminaire : SGA 7 II, par P. Deligne et N. Katz, paraîtra ultérieurement.

Bures sur Yvette, mai 1972,

P. DELIGNE

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SGA 7

Exposé I

Résumé des premiers exposés de A. Grothendieck

rédigé par P. Deligne

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0. Préliminaires

0.0. Terminologie. Rappelons les définitions suivantes :

(0.0.1) trait : spectre d'un anneau de valuation discrète. On note souvent η et s les points génériques et fermés d'un trait.

(0.0.2) strictement local : pour un anneau : local hensélien à corps résiduel séparablement clos. Pour un schéma : spectre d'un tel anneau.

(0.0.3) point géométrique de S (dans cet exposé) : morphisme $\bar{x} : \mathrm{Spec}(\bar{k}) \longrightarrow S$ d'image $x \in S$, $k(\bar{x}) =_{\mathrm{dfn}} \bar{k}$ étant une clôture séparable de $k(x)$. Pour un trait hensélien $S = \mathrm{Spec}(V)$, on note souvent $\bar{\eta}$ un point géométrique générique (i.e. d'image η) de S. Le corps résiduel de l'anneau de valuation $\bar{V}$ clôture intégrale de V dans $k(\bar{\eta})$ est une extension algébrique séparablement close de $k(s)$ et définit un point géométrique $\bar{s}$ de S localisé en s.

(0.0.4) <u>localisé strict de</u> X <u>en le point géométrique</u> $\bar{x}$: SGA 4 VIII 4.4.

(0.0.5) <u>essentiellement de type fini sur</u> S : pour un S-schéma local (resp. local hensélien, resp. strictement local) : spectre d'un anneau local (resp. de l'hensélisé d'un anneau local, resp. localisé strict) d'un schéma de type fini sur S .

(0.0.6) $F \longmapsto F(n)$: twist à la Tate.

(0.0.7) Un endomorphisme T d'un espace vectoriel de dimension finie est <u>quasi-unipotent</u> si ses valeurs propres sont des racines de l'unité, i.e. s'il existe $N \geqslant 1$ $M \geqslant 1$ tels que $(T^N - 1)^M = 0$.

0.1. <u>Rappels sur la cohomologie ℓ-adique</u>

Soit X un schéma de type fini sur un corps algébriquement clos k d'exposant caractéristique p et ℓ un nombre premier premier à p .

Les groupes de cohomologie ℓ-adique de X ont été définis dans SGA 4 et SGA 5.

a) $H^i(X,\mathbb{Z}/\ell^n)$ est le $i^{\text{ème}}$ groupe de cohomologie du site étale X_{et} de X à valeurs dans $\mathbb{Z}/\ell^n$;

b) par définition, $H^i(X,\mathbb{Z}_\ell) = \varprojlim_n H^i(X,\mathbb{Z}/\ell^n)$;

c) par définition, $H^i(X,\mathbb{Q}_\ell) = H^i(X,\mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$.

Les groupes de cohomologie ℓ-adique à supports propres de X sont définis de même à partir des $H^i_c(X,\mathbb{Z}/\ell^n)$ de SGA 4 XVII.

Dans chacun des cas suivants, on sait prouver que ces groupes $H^i(X)$ ont des propriétés de finitude raisonnables :

(0.1.1) en cohomologie à support propre ;

(0.1.2) pour X propre sur k ;

(0.1.3) pour X lisse sur k (grâce à a) et à la dualité de Poincaré SGA 4 XVII

(0.1.4) lorsqu'on dispose de la résolution des singularités à la Hironoka pour les schémas de type fini sur k , de dimension $\leqslant \dim(X)$;

(0.1.5) pour $i = 0, 1$.

Dans tous ces cas, les $H^i(X,\mathbf{Z}/\ell^n)$ sont finis, les $H^i(X,\mathbf{Z}_\ell)$ de type fini, les $H^i(X,\mathbf{Q}_\ell)$ de dimension finie et (sauf peut-être pour (0.1.5)) on dispose de suites exactes courtes scindées

$$0 \longrightarrow H^i(X,\mathbf{Z}_\ell) \otimes \mathbf{Z}/\ell^n \longrightarrow H^i(X,\mathbf{Z}/\ell^n) \longrightarrow \mathrm{Tor}_1(H^{i+1}(X,\mathbf{Z}_\ell),\mathbf{Z}/\ell^n) \longrightarrow 0 .$$

De plus, la démonstration du théorème de finitude fournit chaque fois une démonstration d'un théorème de constructibilité générique : si k est le corps des fractions d'un schéma noéthérien intègre S et que X est la fibre générique de $f_1 : X_1 \longrightarrow S$ (de type fini), il existe un ouvert non vide $U \subset S$ tel que, sur U, les $R^i f_* \mathbf{Z}/\ell^n$ soient localement constants de formation compatible à tout changement de base.

0.2. Ne supposons plus k algébriquement clos, et soit $\bar{k}$ une clôture algébrique (ou séparable, cela reviendrait au même) de k. Nous considérerons les groupes de cohomologie "géométriques" $H^i(X_{\bar{k}})$ ($X_{\bar{k}}$ est le schéma déduit de X par extension des scalaires de k à $\bar{k}$). Par transport de structure, le groupe de Galois $\mathrm{Gal}(\bar{k}/k)$ agit sur ces groupes. En $\mathbf{Q}_\ell$-cohomologie, lorsqu'on est dans l'un des cas (0.1.1) à (0.1.5), on obtient ainsi des représentations continues de $\mathrm{Gal}(\bar{k}/k)$ dans un groupe linéaire $GL(n,\mathbf{Q}_\ell)$.

0.3. <u>Rappels sur les traits</u> (0.0.1)

Pour les démonstrations, on renvoie à [4]. Soient S un trait hensélien, η, s, V, $\bar{V}$, $\bar{\eta}$, $\bar{s}$ comme en (0.0.1) (0.0.3). On note p l'exposant caractéristique de $k(s)$ et $\mathrm{Gal}(\bar{\eta}/\eta)$ (resp. $\mathrm{Gal}(\bar{s}/s)$) le groupe de Galois $\mathrm{Gal}(k(\bar{\eta})/k(\eta))$ (resp. ...). Par transport de structure, $\mathrm{Gal}(\bar{\eta}/\eta)$ agit sur $k(\bar{s})$, d'où une application de $\mathrm{Gal}(\bar{\eta}/\eta)$ dans (en fait sur) $\mathrm{Gal}(\bar{s}/s)$, de noyau le <u>groupe d'inertie</u> I. Le sous-corps de $k(\bar{\eta})$ défini par I est une extension non ramifiée maximale de $k(\eta)$.

Le groupe profini I admet un plus grand sous-groupe P qui soit un pro-p-groupe. Le quotient I/P est le <u>groupe d'inertie modéré</u>. On a canoniquement

$$I/P \simeq \varprojlim_n \mu_n(k(\bar{s})) =_{\mathrm{dfn}} \hat{\mathbf{Z}}(1)(k(\bar{s})) .$$

Si ϕ_n est l'application canonique de I/P dans $\mu_n(\bar{k})$, et que u est un élément de $\bar{V}$ de valuation $1/n$, on a pour $\sigma \in I$

$$\sigma(u) \equiv \phi_n(u).u \qquad \text{(mod. éléments de valuation} > 1/n).$$

En résumé, on a :

$$(0.3.1) \quad \begin{aligned} &\mathrm{Gal}(\bar{\eta}/\eta) \supset I \supset P , \\ &\mathrm{Gal}(\bar{\eta}/\eta)/I \simeq \mathrm{Gal}(\bar{s}/s) , \\ &I/P \simeq \hat{\mathbb{Z}}(1)(k(\bar{s})) , \\ &P : \text{pro-p-groupe } (\{e\} \text{ si } p = 1) , \end{aligned}$$

et l'action (par automorphismes intérieurs dans $\mathrm{Gal}(\bar{\eta}/\eta)$) de $\mathrm{Gal}(\bar{\eta}/\eta)/I$ sur I/P s'identifie à l'action naturelle de $\mathrm{Gal}(\bar{s}/s)$ sur $\hat{\mathbb{Z}}(1)(k(\bar{s}))$.

0.4. Soit plus généralement S strictement local régulier, D un diviseur régulier dans S et p' l'exposant caractéristique du point générique de D . Soient η un point géométrique générique de S et $\bar{s}$ le point géométrique localisé au point fermé s , qui s'en déduit. D'après le lemme d'Abhyankar, le groupe fondamental $\pi_1(S-D,\bar{\eta})$ admet un dévissage analogue à (0.3.1) :

$$(0.4.1) \quad \begin{aligned} &\pi_1(S-D,\bar{\eta}) \supset I \supset P \\ &\pi_1(S-D,\bar{\eta})/I \simeq \mathrm{Gal}(\bar{s}/s) \\ &I/P \simeq \hat{\mathbb{Z}}(1)(k(\bar{s})) \\ &P : \text{sans quotient d'ordre premier à } p' . \end{aligned}$$

L'action par automorphismes intérieurs de $\pi_1(S-D,\bar{\eta})/I$ sur I/P s'identifie à l'action naturelle de $\mathrm{Gal}(\bar{s}/s)$ sur $\hat{\mathbb{Z}}(1)(k(\bar{s}))$.

0.5. La désingularisation de Néron

La méthode de désingularisation de Néron fournit le résultat suivant :

Lemme 0.5.1. Soit $S \longrightarrow S_o$ un morphisme de traits henséliens, avec $S = \mathrm{Spec}(V)$ $S_o = \mathrm{Spec}(V_o)$ et π une uniformisante de V_o . On suppose que π est une uniformisante de V , et que les extensions $k(s)/k(s_o)$ et $k(\eta)/k(\eta_o)$ sont séparables. Alors, V est limite inductive de V_o-algèbres henséliennes lisses et essentiellement de type fini B_i sur V_o .

Pour construire les V_o-algèbres voulues, on prend $V_o \subset B \subset V$ avec B de type fini, et on désingularise $\mathrm{Spec}(B)$ à la Néron ([1] § 4).

Soit $S = \mathrm{Spec}(V)$ un trait hensélien.

(0.5.2) Si S est d'égale caractéristique, d'uniformisante π, on peut appliquer (0.5.1) pour $S_o = \mathbf{F}_p[\pi]_{(\pi)}$. L'extension s/s_o est séparable car $\mathbf{F}_p$ est parfait, l'extension η/η_o l'est car π, étant une uniformisante, n'est pas une puissance $p^{\text{ième}}$ (si $p \neq 1$) donc fait partie d'une p-base. On trouve que S est limite de spectres de $\mathbf{F}_p[\pi]_{(\pi)}$-algèbre henséliennes lisses essentiellement de type fini.

(0.5.3) Si S est complet, d'inégale caractéristique, à corps résiduel parfait k, V est une extension d'Eisenstein de l'anneau des vecteurs de Witt (= anneau de Cohen) $W(k)$

$$V \simeq W(k)\ [\pi]\ /(\pi^n + \sum_0^{n-1} a_i\ \pi^i) \quad .$$

Si on perturbe un peu les a_i, on trouve une extension isomorphe. On peut donc supposer les a_i dans $W(k')$, avec k' de type fini sur $\mathbf{F}_p$. Soit k_o la clôture parfaite de k' dans k et

$$V_o = W(k_o)\ [\pi]/(\pi^n + \sum_0^{n-1} a_i\ \pi^i) \quad .$$

Le lemme 0.5.1 s'applique à V/V_o, et le corps résiduel de V_o est une extension radicielle d'un corps de type fini sur $\mathbf{F}_p$.

1. Démonstration arithmétique du théorème de monodromie

Le lecteur trouvera dans [5], Appendice, une démonstration du résultat suivant de A. Grothendieck.

Proposition 1.1. Soient S un trait hensélien, ℓ un nombre premier premier à l'exposant caractéristique p de $k(s)$ et ρ une représentation continue de $\mathrm{Gal}(\bar{\eta}/\eta)$ dans $GL(n,\mathbb{Q}_\ell)$. On suppose remplie la condition suivante :

$(*_\ell)$ Aucune extension finie de $k(s)$ ne contient toutes les racines de l'unité d'ordre une puissance de ℓ.

Alors, *il existe un sous-groupe ouvert* I_1 *du groupe d'inertie* I *tel que* $\rho(\sigma)$ *soit unipotent pour* $\sigma \in I_1$.

La condition $(\star_\ell)$ est automatiquement vérifiée pour $k(s)$ de type fini sur le corps premier (ou radiciel sur un tel corps).

Le théorème suivant résulte aussitôt de 1.1.

Théorème de monodromie 1.2. *Avec les notations précédentes, soit* X *un schéma de type fini sur* η *et* H *un espace de cohomologie de* $X_{\bar{\eta}}$, *à coefficients dans* $\mathbb{Q}_\ell$, *de l'un des types* (1.1.1) *à* (1.1.5). *Si la condition* $(\star_\ell)$ *est vérifiée, l'action d'un quelconque élément de* I *sur* H *est quasi-unipotente*.

Les résultats de passage à la limite 0.5 permettent de se débarasser de l'hypothèse $(\star_\ell)$.

Variante 1.3. *La condition* $(\star_\ell)$ *dans* 1.2 *est superflue*.

Soit H'_o un espace de $\mathbb{Z}_\ell$-cohomologie de $X_{\bar{\eta}}$ qui donne naissance à H , et $H_o = H'_o/\text{torsion}$. On a par hypothèse $H = H_o \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, et l'action de $\mathrm{Gal}(\bar{\eta}/\eta)$ se déduit de

$$\rho : \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow GL(H_o) \quad .$$

Soit $I_{\ell'}$ le plus grand sous-groupe premier à ℓ de I . Son image dans $GL(H_o)$ es un groupe de Lie ℓ-adique premier à ℓ , donc est finie.

Procédons à une extension de trait $S' \longrightarrow S$. Ceci ne change pas H_o (SGA 4 XVI 6.6) et $\rho(I')$ est d'indice fini dans $\rho(I)$, car I'/P' est d'indice fini dans I/P et ce qui précède. Il suffit donc de prouver le théorème sur S' : on peut supposer que

a) $\mathrm{Gal}(\bar{\eta}/\eta)$ agit trivialement sur $H_o/\ell H_o$;

b) si S est d'inégale caractéristique, S est complet à corps résiduel parfait (EGA 0_{III} 10.3.1).

Posons $S = \mathrm{Spec}(V)$, définissons V_o comme en (0.5.2) et (0.5.3) et appliquons (0.5.1). On a

$$V = \varinjlim B_i$$

avec $S_i = \mathrm{Spec}(B_i)$ essentiellement lisse de type fini sur V_o . Soit $D_i \subset S_i$ d'équation $\pi = 0$. Pour i assez grand, H_o provient d'un $\mathbb{Z}_\ell$-faisceau constant tordu $\mathcal{H}_i$ sur $S_i - D_i$ (d'après la "constructibilité générique" (0.1)) et $\mathcal{H}_i/\ell\mathcal{H}_i$ est constant.

Les dévissages (0.3.1) et (0.4.1) donnent lieu à des diagrammes commutatifs :

(1.3.1)
$$\begin{tikzcd} \mathrm{Gal}(\bar\eta/\eta) \arrow[d] & \supset & I \arrow[d] & \supset & P \arrow[d] \\ \pi_1(S_i-D_i,\bar\eta) & \supset & I_i & \supset & P_i \end{tikzcd} \quad \text{et} \quad \begin{tikzcd} I/P \arrow[r, "\sim"] \arrow[d] & \hat{\mathbb{Z}}(1) \arrow[d, equal] \\ I_i/P_i \arrow[r, "\sim"] & \hat{\mathbb{Z}}(1) \end{tikzcd}$$

et la représentation H_o de $\mathrm{Gal}(\bar\eta/\eta)$ se déduit d'une action de $\pi_1(S_i-D_i,\bar\eta)$ sur H_o , triviale sur $H_o/\ell H_o$. Le groupe $\mathrm{Ker}(GL(H) \longrightarrow GL(H/\ell H))$ est un pro-ℓ-groupe ; P_i agit donc trivialement, et on applique la variante suivante de 1.1.

<u>Variante</u> 1.4. <u>Reprenons les notations de</u> 0.4 <u>et soit</u> ρ <u>une représentation continue de</u> $\mathrm{Gal}(\bar\eta/\eta)$ <u>dans</u> $\mathrm{Gal}(n,\mathbb{Q}_\ell)$. <u>On suppose que</u> $k(s)$ <u>vérifie</u> $(\star_\ell)$ <u>et que</u> $\rho(P)$ <u>est fini</u>. <u>Alors</u>, <u>l'action de</u> I <u>est quasi-unipotente</u>.

2. <u>Cycles évanescents</u>

2.1. Soit S un trait strictement local, et reprenons les notations de 0.3. Par hypothèse, on a ici $s = \bar s$. Soit $f : X \longrightarrow S$ un schéma de type fini sur S . Le formalisme des cycles évanescents est un outil pour étudier la relation entre les cohomologies de X_s et de $X_{\bar\eta}$, et l'action du groupe d'inertie sur celle de $X_{\bar\eta}$, à partir d'informations <u>locales</u> (sur X) sur le morphisme f .

Un résultat clef de ce type est le théorème de spécialisation SGA 4 XVI 2.2, selon lequel pour f propre et lisse X_s et $X_{\bar\eta}$ ont même cohomologie.

Dans la théorie transcendante, si $f : X \longrightarrow D$ est un morphisme propre d'un espace analytique X dans le disque D , les cycles évanescents apparaissent comme suit :

a) Quitte à rapetisser D , on peut supposer que X est un fibré topologique au-dessus du disque épointé $D^\star$ (théorème d'isotopie de Thom). Les fibres

X_t ($t \in D^*$) sont en particulier toutes isomorphes.

b) Il n'est pas déraisonnable de se représenter la fibre spéciale X_s comme déduite de "la" fibre générale X_t par contraction de polyèdres dans X_t : on aurait $\pi : X_t \longrightarrow X_s$. La méthode des cycles évanescents consiste alors à étudier la différence entre les cohomologies de X_s et X_t à l'aide de la suite spectrale de Leray de π .

La théorie géométrique expliquée ci-dessous est très proche de cette théorie transcendante ; il faut remplacer le point général $t \in D$ par $\bar{\eta}$, le point géométrique générique de S . Inversement, on peut calquer la théorie transcendance sur la théorie géométrique, en remplaçant t ou $\bar{\eta}$ par le revêtement universel $\tilde{D}^*$ de D^* , et X_t ou $X_{\bar{\eta}}$ par $X \times_D \tilde{D}^*$. Ce faisant, on se débarasse de nombreux ε et η , mais on perd parfois de l'information.

2.2. Fixons un anneau de torsion Λ dans lequel p soit inversible (par exemple $\Lambda = \mathbf{Z}/\ell^n\mathbf{Z}$, ℓ premier à p). Soient les morphismes :

$$\begin{array}{ccccc} X_s & \xrightarrow{\ i\ } & X & \xleftarrow{\ \bar{j}\ } & X_{\bar{\eta}} \\ \downarrow{\scriptstyle f_s} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f_{\bar{\eta}}} \\ s & \longrightarrow & S & \longleftarrow \eta \longleftarrow & \bar{\eta} \end{array}$$

Les faisceaux de cycles évanescents ψ_i sur X_s sont les

(2.2.1) $$\psi^i = i^* R^i \bar{j}_* \Lambda \quad .$$

Si f est <u>propre</u>, on a (SGA 4 XII 5.5)

(2.2.2) $$i^* \ : \ H^p(X, R^q \bar{j}_* \Lambda) \xrightarrow{\ \sim\ } H^p(X_s, i^* R^q j_* \Lambda) \quad ,$$

et la suite spectrale de Leray pour $\bar{j}$ se récrit

(2.2.3) $$H^p(X_s, \psi^q) \Longrightarrow H^{p+q}(X_{\bar{\eta}}, \Lambda) \quad .$$

Lorsque rien ne se passe à l'infini, on peut parfois obtenir cette suite spectrale sans hypothèse de propreté. Qu'il faille quelque hypothèse à l'infini se voit déjà sur le cas trivial $X_s = \emptyset$.

On tire aussitôt des définitions :

Proposition 2.3. *Soient* $\bar{x}$ *un point géométrique de* X_s *(par exemple un point fermé)* $X_{(\bar{x})}$ *l'henselisé strict de* X *en* $\bar{x}$ *et* $X_{(\bar{x})\bar{\eta}}$ *la fibre géométrique générique de la projection de* $X_{(\bar{x})}$ *sur* S *. On a*

$$\psi^i = H^i(X_{(\bar{x})\bar{\eta}}, \Lambda)$$

Corollaire 2.4. *Là où* f *est lisse,* $\psi^i = 0$ *pour* $i > 0$ *et* $\psi^\circ = \Lambda$.

C'est SGA 4 XV 2.1.

Le groupe $\mathrm{Gal}(\bar{\eta}/\eta)$ (égal par hypothèse au groupe d'inertie) agit par transport de structure sur les ψ^i , et

Scholie 2.5. *La suite spectrale* 2.2.3 *est* $\mathrm{Gal}(\bar{\eta}/\eta)$*-équivariante*.

Variante 2.6. Soient $\bar{S} = \mathrm{Spec}(\bar{V})$ et $\bar{X} = X \times_S \bar{S}$. $X_{\bar{\eta}}$ est le complément de $X_s = X_{\bar{s}}$ dans $\bar{X}$. Posons

$$\phi^i = \underline{H}^{i+1}_{X_s}(\bar{X}, \Lambda)$$

(faisceaux de cohomologie à support). On a une suite exacte de faisceaux

$$0 \longrightarrow \phi^{-1} \longrightarrow \Lambda \longrightarrow \psi^\circ \longrightarrow \phi^\circ \longrightarrow 0$$

et, pour $i > 0$, des isomorphismes $\psi^i \xrightarrow{\sim} \phi^i$. Pour f lisse, les ϕ^i sont nuls.

Posons $\phi^i_{gl} = H^i_{X_s}(\bar{X}, \Lambda)$ (analogue global des ϕ^i). On dispose d'une suite spectrale

(2.6.1) $$H^p(X_s, \phi^q) \Longrightarrow \phi^{p+q}_{gl}$$

et d'une suite exacte longue de cohomologie qui, pour f propre, se récrit

(2.6.2) $$\cdots \longrightarrow H^i(X_s, \Lambda) \longrightarrow H^i(X_{\bar{\eta}}, \Lambda) \longrightarrow \phi^i_{gl} \longrightarrow \cdots$$

Variante 2.7. Soit K_t l'extension modérément ramifiée maximale de K dans $\bar{K}$. On pose $X_t = X \times_S \mathrm{Spec}(K_t)$. Soit $\bar{j}_t$ la projection de X_t dans X . On définit des variantes modérées ("tame") des ψ^i en posant

$$\psi_t^i = R^i \bar{j}_{t*} \Lambda$$

Utilisant que P est un pro-p-groupe, avec p inversible dans Λ, on trouve que

(2.7.1) $$H^*(X_t,\Lambda) = H^*(X_{\bar{\eta}},\Lambda)^P \quad ,$$

(2.7.2) $$\psi_t^i = \psi^{iP} \quad .$$

Pour f propre, la suite spectrale de Leray de $\bar{j}_t$ se déduit de **2**.2.3 par passage aux P-invariants :

(2.7.3) $$H^p(X_s,\psi_t^q) \Longrightarrow H^{p+q}(X_t,\Lambda) = H^{p+q}(X_{\bar{\eta}},\Lambda)^P \ .$$

3. <u>Démonstration géométrique du théorème de monodromie</u>

Soit Λ comme en 2.2.

<u>Conjecture</u> 3.1 <u>(pureté)</u>. <u>Soient</u> A <u>un anneau excellent strictement local</u> (0.2) <u>régulier et</u> D <u>un diviseur régulier de</u> Spec(A) <u>défini par un paramètre</u> x . <u>Soit</u> U = Spec(A) - D = Spec(A[1/X]). <u>On a</u>

(3.1.1) $$H^q(U,\Lambda) = \begin{cases} \Lambda & \text{pour } q = 0 \\ \Lambda(-1) & \text{pour } q = 1 \text{ (notation (0.0.6))} \\ 0 & \text{pour } q > 1 \end{cases}$$

Voici des cas où cette conjecture est connue (SGA 4 XIX) :

pour $q = 0$ ou 1 , pour A de caractéristique 0 , pour A d'égale caractéristique p lorsqu'on dispose de la résolution des singularités en dimension $\leq \dim(A)$ et en caractéristique p , ou pour (Spec(A),D) un couple lisse (théorème de pureté relatif SGA 4 XVI 3.7).

Lorsqu'on dispose de la pureté, et que $D = \sum_{i\in B} D_i$ est un diviseur à croisements normaux dans Spec A , défini par une partie d'un système régulier de paramètre, la cohomologie de U = Spec(A) - D est donnée par

$$(3.1.2)\qquad \begin{aligned} H^{o}(U,\Lambda) &= \Lambda \\ H^{1}(U,\Lambda) &= \Lambda(-1)^{B} \\ H^{q}(U,\Lambda) &= \overset{q}{\Lambda}\, H^{1}(U,\Lambda) \qquad \text{(donné par le cup-produit)} \end{aligned}$$

Ces formules sont identiques à celles qui, sur $\mathbb{C}$, donnent la cohomologie de $D^{*B} \times D^{k}$.

3.2. Soit S un trait strictement local. On garde les notations (0.0.1) (0.0.3). Soit $f : X \longrightarrow S$ un morphisme de type fini. On suppose que X est régulier, que X_{η} est lisse et que X_{s} est un diviseur à croisements normaux dans X . On supposera que ce diviseur est (globalement, et non seulement localement pour la topologie étale) somme de diviseurs réguliers : $(X_{s})_{red} = \sum_{i\in B} D_{i}$ (le cas général peut se récupérer par localisation). Notons m_{i} leurs multiplicités : $X_{s} = \sum_{i\in B} m_{i}\, D_{i}$.

Soient x un point de X_{s} , $\bar{x}$ un point géométrique de X localisé en x (par exemple x fermé, $x = \bar{x}$) , $C = \{i \mid x \in D_{i}\}$ et K le complexe concentré en degrés 0 et -1

$$K : \mathbb{Z}^{C} \xrightarrow{\ d\ } \mathbb{Z} : d((n_{j})) = \sum_{j} m_{j}\, n_{j} \quad .$$

<u>Théorème</u> 3.3. (<u>utilise la pureté</u>). <u>Les groupes de cycles évanescents modérés</u> $\psi^{i}_{t\bar{x}}$ <u>sont les suivants</u> :

(i) <u>On a un isomorphisme canonique de</u> Λ-<u>algèbres graduées</u>

$$\psi^{*}_{t\,\bar{x}} \simeq \overset{*}{\bigwedge}(H_{1}(K) \otimes \Lambda(-1)) \otimes_{\Lambda} \psi^{o}_{t\bar{x}} \quad .$$

(ii) $\psi^{o}_{t\,\bar{x}}$ <u>est une</u> Λ-<u>algèbre</u> Λ^{E} , <u>pour</u> E <u>un ensemble sur lequel le groupe d'inertie modéré</u> $I_{t} = I/P = \hat{\mathbb{Z}}(1)(k(s))$ <u>agit transitivement</u>, <u>de nombre d'éléments le plus grand diviseur premier à</u> p <u>de</u> $\# \, H_{o}(K)$.

Voici la méthode de démonstration.

a) Soient $X_{(\bar{x})}$ le localisé strict de X en $\bar{x}$, $U = X_{(\bar{x})} - X_{(\bar{x})s}$ (le complément d'un diviseur à croisements normaux), t_{j} une équation locale de D_{j} , $X_{n} = X_{(x)}[(t_{j}^{1/n})]$ (pour $(n,p) = 1$; c'est un schéma régulier) et U_{n} l'image réciproque de U dans X_{n} . D'après 3.1.2, on a $H^{q}(U_{n},\Lambda) = \overset{q}{\bigwedge}(\Lambda(-1)^{C})$; pour $m = x\, d$, l'application "image réciproque" de $H^{q}(U_{n},\Lambda)$ dans $H^{q}(U_{m},\Lambda)$ est la multiplication par d^q . Posant $\hat{\tilde{U}} = \varprojlim_{(n,p)=1} U_n$, on a donc

$$(3.3.1) \qquad \begin{cases} H^0(\hat{\tilde{U}},\Lambda) = \Lambda \\ H^q(\hat{\tilde{U}},\Lambda) = \varinjlim_n H^q(U_n,\Lambda) = 0 \quad \text{pour} \quad q \neq 0 \end{cases} .$$

b) $\hat{\tilde{U}}$ est un prorevêtement de U de groupe $\hat{\mathbf{Z}}(1)(k(s))^C$. Par définition, on a $\psi_t{}^{*}_{\bar{x}} = H^*(U_{n_t},\Lambda)$; $\hat{\tilde{U}}$ est un prorevêtement de groupe $H_1(K) \otimes \hat{\mathbf{Z}}(1)(k(s))$ de chacune des composantes connexes de U_{n_t} , et celles-ci sont permutées transitivement par le groupe d'inertie modéré. De (3.3.1) et de la suite spectrale d'Hochschild-Serre on tire alors les formules 3.3.

En termes imagés, on peut dire que $U_{\bar{\eta}}$ est la fibre d'un morphisme δ d'un $K(\hat{\mathbf{Z}}(1)(k(s))^C,1)$ cohomologique U dans le $K(\hat{\mathbf{Z}}(1)(k(s)),1)$ cohomologique η , ce morphisme δ induisant $d = \Sigma\, m_j\, n_j : \hat{\mathbf{Z}}(1)(k(s))^C \longrightarrow \hat{\mathbf{Z}}(1)(k(s))$.

<u>Corollaire</u> 3.4. (<u>utilise la pureté</u>). <u>Supposons</u> f <u>propre</u>. <u>Soit</u> N <u>le plus petit commun multiple des multiplicités</u> m_j ($j \in B$). <u>Soit</u> $q \geq 0$, <u>et</u> q' <u>la borne inférieure de</u> q <u>et du nombre maximum de diviseurs</u> D_j <u>passant par un même point</u>. <u>Pour</u> T <u>dans le groupe d'inertie modéré</u> $I_t = I/P$, <u>on a</u>

$$(T^N - I)^{q'} = 0 \quad \underline{\text{sur}} \quad H^q(X_{\bar{\eta}},\Lambda)^P \quad .$$

Résulte aussitôt de 3.3 et de la variante 2.7 de 2.5.

Les résultats de pureté requis sont disponibles pour les H^1 , en caractéristique 0 , ou pour S essentiellement de type fini sur un corps.

<u>Théorème</u> 3.5. <u>Soit</u> C <u>une courbe projective et lisse sur le corps des fractions</u> $k(\eta)$ <u>d'un anneau de valuation discrète</u> V . <u>Le groupe d'inertie</u> I <u>agit sur</u> $H^1(C_{\bar{\eta}},\mathbf{Q}_\ell)$ <u>par automorphismes quasi-unipotents d'échelon</u> 2 : <u>il existe</u> N <u>tel que, pour</u> $T \in I$, <u>on ait</u>

$$(T^N - I)^2 = 0 \quad \text{sur} \quad H^1(C_{\bar{\eta}},\mathbf{Q}_\ell) \quad .$$

On se ramène à supposer V strictement local. L'assertion est qu'il existe un sous-groupe ouvert I_1 de I tel que $(T - I)^2 = 0$ sur H^1 pour $T \in I_1$.

Il nous est loisible d'étendre les scalaires de $k(\eta)$ à une extension finie (remplacer V par son normalisé dans cette extension). Puisque l'image du pro-p-groupe P dans un groupe de Lie ℓ-adique $GL(n,Q_\ell)$ est finie, on peut en particulier supposer que P agit trivialement.

Soit C_o un modèle minimal de C sur V (pour l'existence de C_o, basée sur le théorème d'Abhyankar, cf. la discussion dans [3] § 2, p. 87 ; on pourrait ici se ramener au cas V excellent en passant au complété de V : invoquer SGA 4 XVI 1.6). Soit C_1 un modèle régulier de C, déduit de C_o par éclatements, de fibre spéciale un diviseur à croisements normaux. Les arguments 3.3, 3.4 s'appliquent à C_1, ψ_t^o, ψ_t^1 et $H^1(C_{\bar{\eta}},\mathbb{Z}/\ell^n)^P = H^1(C_{\bar{\eta}},\mathbb{Z}/\ell^n)$ avec N indépendant de n, et 3.5 en résulte.

Remarque 3.6. Soit A une variété abélienne sur $k(\eta)$. Puisque A est quotient d'une jacobienne, on trouve encore que l'action de I sur $H^1(A_\eta,\mathbb{Q}_\ell)$ ou $T_\ell(A)$ est quasi-unipotente d'échelon 2.

Remarque 3.7. L'action de I sur $H^1(X_{\bar{\eta}},\mathbb{Q})$ est en fait quasi-unipotente d'échelon 2 pour tout schéma X de type fini sur η.

Le théorème 3.5 (ou 3.6) est à la base de la démonstration du théorème de réduction stable pour les variétés abéliennes qui sera donnée dans l'exposé IX § 3 .

3.8. Il est clair que lorsqu'on dispose de la pureté et de la résolution des singularités (par exemple en caractéristique 0) la méthode précédente démontre le théorème de monodromie, et fournit des bornes pour l'exposant de nilpotence. Ainsi, en égale caractéristique 0 , on trouve que pour X propre et lisse sur η et T dans un sous-groupe ouvert de I , on a

$$(T-1)^{q+1} = 0 \quad \text{sur} \quad H^q(X_{\bar{\eta}}, \mathbb{Z}_\ell) \qquad (T \in I_1 \subset I).$$

4. Critères de nullité pour les faisceaux de cycles évanescents

4.1. Soient S un trait strictement local et $f : X \longrightarrow S$ un schéma de type fini sur S . Posons $n = \dim(X_\eta)$.

Si X' est l'adhérence schématique de X_η dans X , on a :

a) Sur $X_s - X'_s$, $\Phi^{-1} = \Lambda$ et les autres Φ^i sont nuls ;

b) sur X'_s , les ϕ^i et ψ^i relatifs à X/S ou à X'/S coïncident, et $\phi^{-1} = 0$

Puisque X' est plat sur S , ceci permet souvent de se ramener au cas X est plat sur S .

Théorème 4.2.

(i) On a $\psi^i = 0$ pour $i > n$.

(ii) Plus précisément, pour x un point de X_s dont l'adhérence soit de dimension k ($k = \deg tr(k(x)/k(s))$) et pour $\bar{x}$ un point géométrique localisé x , on a

$$\psi^i_{\bar{x}} = 0 \quad \text{pour} \quad i > n-k \text{ , i.e.}$$

$$d(\psi^i) \leq n-i \qquad \text{(notation de SGA 4 XIV 2.1).}$$

On a $\psi^i_{\bar{x}} = H^i(X_{(\bar{x})\bar{\eta}}, \Lambda)$, et $X_{(\bar{x})\bar{\eta}}$ est limite projective de schémas affines de dimension $\leq n$ sur $k(\bar{\eta})$. L'assertion (i) résulte donc du théorème de Lefschetz affine (SGA 4 XIV 3.2) par passage à la limite.

Pour prouver (ii), on peut supposer que X est plat sur S (4.1) et qu existe un S-morphisme quasi-fini et plat $g : X \longrightarrow \mathbb{A}^k_S$ tel que $g(x)$ soit le p générique de $\mathbb{A}^k_s$. Soit S' le trait localisé strict de $\mathbb{A}^k_S$ en $\bar{x}$. On a

a) $S'_{\bar{\eta}}$ est le spectre d'un corps, et $Gal(\overline{\eta'}/s'_{\bar{\eta}})$ est un p-groupe Q .

b) D'après (i) pour g , on a $H^i(X_{(x)\overline{\eta'}}, \Lambda) = 0$ pour $i > n-k$.

c) On a $H^i(X_{(x)\bar{\eta}}, \Lambda) = H^i(X_{(x)\overline{\eta'}}, \Lambda)^Q$, d'où le théorème.

Corollaire 4.3. Si f est propre et plat et que la dimension du lieu de non liss de f dans X_s est $\leq d$, le morphisme de spécialisation $H^i(X_s) \longrightarrow H^i(X_{\bar{\eta}})$ es un isomorphisme pour $i > n+d+1$ et un épimorphisme pour $i = n+d+1$.

4.4. Pour prouver, dans certains cas, que les ϕ^i pour i petit sont nuls, on s'appuyera sur la théorie de la dualité locale (directement ou via des théorèmes de Lefschetz locaux de SGA 2 XIV). Cet outil n'est disponible qu'en caractéristique 0 (ou en égale caractéristique p lorsqu'on dispose de la résolution des singularités à la Hironaka dans les dimensions considérées).

Théorème 4.5. Supposons S de caractéristique 0 . Soient $f : X \longrightarrow S$ un morphisme de type fini et x un point fermé de X_s . On suppose que

(a) x est un point de non lissité isolé dans sa fibre

(b) X est de profondeur géométrique relative $\geqslant n$ en x : au voisinage de x , X s'identifie à un sous-schéma défini par k équations dans un schéma lisse de dimension relative N en x , avec $n \leqslant N-k$.

Alors $\phi_x^i = 0$ pour $i < n$.

Soit X_x le localisé strict de X en x et $V = X_x - \{x\}$. Pour tout trait S' fini sur S , soient de même $X'_x = X_x \times_S S'$ et $V' = V \times_S S'$. Pour S le normalisé de S dans $\bar{\eta}$, soient $\bar{X}_x = X_x \times_S \bar{S}$, $\bar{V} = V \times_S \bar{S}$. On a

(1) Si $\phi^i = 0$ sur $(X_x)_s$ (i.e. en toute générisation de x dans X_s) pour $i \leqslant m$, alors

$$H^{i-1}_{\{x\}}(\bar{X}_x) = \tilde{H}^i(\bar{V}) \xrightarrow{\sim} \tilde{H}^i(\bar{V}_{\bar{\eta}}) = \phi^i \qquad \text{pour } i \leqslant m .$$

Les ϕ^i sont en effet des faisceaux de cohomologie à support dans V_s de $\bar{V}$; sous les hypothèses de 4.5, (a) est d'application pour $m = \infty$ (SGA 4 XV 2.1).

(2) L'hypothèse (b) de 4.5 est stable par changement de base. D'après SGA 2 XIV 5.6, elle implique que

$$\tilde{H}^i(V') = H^{i-1}_{\{x\}}(X'_x) = 0 \qquad \text{pour } i < n ;$$

on conclut en notant que $\tilde{H}^i(\bar{V})$ est limite inductive des $\tilde{H}^i(V')$.

D'après 4.2. et 4.5, on a

Corollaire 4.6. Pour S un trait hensélien de caractéristique 0 , X/S une intersection complète relative de dimension relative n et $x \in X_s$ un point de non lissité isolé dans sa fibre, on a $\phi_x^i = 0$ pour $i \neq n$.

Corollaire 4.7. Sous les hypothèses de 4.6, ϕ_x^n est un Λ-module plat.

Pour Λ' une Λ-algèbre finie, un general non sense (SGA 4 XVII 5.2.11) fournit en effet, pour les ϕ_Λ^i , relatifs à Λ' ,

$$\phi^{n-i}_{\Lambda'} = \mathrm{Tor}^{\Lambda}_{i}(\Lambda', \phi^{n}_{\Lambda})$$

et on applique 4.6 à Λ' .

Variante 4.8. Dans 4.5, supprimons la condition (a) et soit Z le lieu de non de f . On peut encore conclure

$$\phi^{i}_{x} = 0 \quad \text{pour } i < n - \dim Z_s \quad .$$

On procède par récurrence descendante sur $\dim \overline{\{x\}}$. L'hypothèse de récurrence permet d'appliquer 4.5 (1) avec $m = n - \dim Z_s$, et on achève la démonstration comme précédemment.

4.9. Il est possible d'obtenir un critère de nullité utilisant seulement des informations sur les fibres géométriques de f . Dans le cas des intersections complètes, le résultat est toutefois d'une unité moins bon que 4.5.

Proposition 4.10. Sous les hypothèses générales de 4.5, supposons que

(a) x est un point de non lissité isolé dans sa fibre.

(b) Pour tout point y de $X_{\bar{\eta}}$, d'adhérence dans $X_{\bar{\eta}}$ de dimension $d(y)$, qui soit une générisation de x , on a $\mathrm{prof\ et}_y(X_{\bar{\eta}}) \geq n - d(g)$, i.e. $H^i_{\{y\}}(X_{\bar{\eta}}) = 0$ pour $i \leq n - d(g)$

(c) $\mathrm{prof\ et}_x(X_s) \geq n$,

Alors, $\phi^i_x = 0$ pour $i < n-1$.

Gardons les notations de 4.5 ; on peut appliquer 4.5 (1) avec $m = \infty$. D'après le théorème de Lefschetz local (SGA 2 XIV), l'hypothèse (b) fournit

$$H^i(V') \xrightarrow{\sim} H^i(V_s) \qquad (\ i < n-1\ )$$

$$H^i(V') \hookrightarrow H^i(V_s) \qquad (\ i = n-1\ )$$

et on conclut par (c) que $H^i(\bar{V}) = \varinjlim H^i(V') = 0$ pour $i \leq n-1$, d'où 4.10.

4.11. Comme en 4.8, on peut donner de 4.10 une variante ne supposant pas (a).

5. Action de la monodromie sur les π_1

Dans cet § (= exposé de Grothendieck du 19/11/1968), nous ferons usage du théorème de réduction semi-stable pour les courbes. Artin et Winters [2] ont donné une démonstration directe de ce théorème, qu'on peut aussi déduire [3] du théorème analogue pour les variétés abéliennes (IX § 3, utilisant 3.6). Nous utiliserons aussi la théorie de Schlessinger (exposé VI de Rim).

5.1. Soient k un corps algébriquement clos, C une courbe projective sur k et $x_1 \dots x_n$ un ensemble fini de points fermés de C. On dit que $X = (C ; x_1, \dots, x_n)$ est <u>stable</u> si

(a) C est réduite et ses seules singularités sont des points doubles à tangentes distinctes ;

(b) les x_i sont distincts, et des points lisses de C ;

(c) si C_1 est une composante irréductible de C, et que E est l'ensemble des points de la normalisée C'_1 de C_1 au-dessus d'un point singulier de C_1 ou d'un des x_i, alors, si C'_1 est de genre 0 (resp. 1), E a au moins 3 (resp 1) éléments.

La condition (c) équivaut à chacune des suivantes :

(c') X n'a pas d'automorphismes infinitésimaux non triviaux.

(c") Si ω est le faisceau inversible dualisant, le faisceau inversible $\omega' = \omega(\Sigma x_i)$ est ample.

Un S-schéma C, muni de sections $x_1, \dots, x_n$ est une <u>courbe stable</u> sur S si ses fibres géométriques sont des courbes stables. Dans [3] §§ 1,2, seul le cas $n = 0$ est considéré. L'extension des résultats de loc. cit. au cas général est triviale :

(1) Pour X/k stable, $H^1(C, \omega'^{\otimes n}) = 0$ pour $n \geq 2$ et $\omega'^{\otimes n}$ est très ample pour $n \geq 3$ (cf. [3] 1.2).

(2) Le schéma formel M des modules de X est lisse, et C reste singulier au-dessus d'un diviseur à croisements normaux de M.

(3) Pour X et Y stables sur S, $\underline{\mathrm{Isom}}_S(X,Y)$ est fini et non ramifié sur S (isomorphismes compatibles aux points marqués).

(4) Pour X/k stable, $\mathrm{Aut}_k(X) \longrightarrow \mathrm{Aut}_k \mathrm{Pic}^\circ(C)$ est injectif.

On a enfin

Proposition 5.1.1. *Soit $X_\eta = (C_\eta\ ;\ x_1, \ldots, x_n)$ stable sur le point générique η d'un trait S, avec C_η lisse. Il existe une extension finie séparable η' de η et une courbe stable X' sur le normalisé S' de S dans η telle que* $X' \otimes_{S'} \eta' = X_\eta \otimes_\eta \eta'$.

Cela résulte du théorème de réduction semi-stable habituel qui permet de prendre η' tel que la fibre spéciale géométrique du modèle minimal C'' de $C_{\eta'}$ sur S' vérifie (a). On éclate ensuite, de façon itérée s'il le faut, des points lisses de C'' pour que (b) soit vérifiée. On contracte alors les chaînes de composantes rationelles lisses de self intersection deux ne contenant aucun $\bar{x}_i$ de la fibre spéciale et on obtient C' (cf. [3], preuve de 2.3 p. 88).

On peut vérifier comme dans loc. cit. que C' existe dès que la jacobienne de C_η a réduction stable. L'hypothèse de lissité sur C_η est par ailleurs superflue.

5.2. Soit X_η un schéma géométriquement connexe de type fini sur le corps des fractions d'un trait strictement local S. Pour ξ un point géométrique de $X_{\bar\eta}$ on a une suite exacte

(5.2.1) $$0 \longrightarrow \pi_1(X_{\bar\eta},\xi) \longrightarrow \pi_1(X,\xi) \longrightarrow \mathrm{Gal}(\bar\eta/\eta) \longrightarrow 0 ,$$

d'où un homomorphisme de $\mathrm{Gal}(\bar\eta/\eta)$ dans le groupe des automorphismes extérieurs de $\pi_1(X_{\bar\eta})$. Passant au plus grand quotient premier à p $\pi_1^{(p)}(X_{\bar\eta})$ de $\pi_1(X_{\bar\eta})$, on trouve

(5.2.2) $$\mathrm{Gal}(\bar\eta/\eta) \longrightarrow \mathrm{Aut\ ext}\,(\pi_1^{(p)}(X_{\bar\eta})) .$$

Supposons que X_η ait un point rationnel x, et soit $\bar{x}$ le point géométrique $\bar\eta \longrightarrow \eta \xrightarrow{x} X_\eta$. Pour $\xi = \bar{x}$, la suite exacte (5.2.2) splitte canoniquement, d'où

(5.2.3) $\quad [x] : \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow \mathrm{Aut}\,(\pi_1(X_{\bar{\eta}},\bar{x})) \longrightarrow \mathrm{Aut}\,(\pi_1^{(p)}(X_{\bar{\eta}},\bar{x}))$.

Théorème 5.3. *L'image de* P *par* 5.2.2 (*ou* 5.2.3, *cela revient au même*) *est finie*.

5.3.1. *Réduction au cas où* X_η *est géométriquement normal et intègre*.

Soient X'_η le normalisé de X_η , $(X'_{i\eta})_i$ la famille des composantes connexes de X'_η , $X''_\eta = X'_\eta \times_{X_\eta} X'_\eta$ et $(X''_{j\eta})_j$ les composantes connexes de X''_η . Quitte à passer à une extension finie de $k(\eta)$, on peut supposer que les $X'_{i\eta}$ sont géométriquement normaux et intègres, avec un point rationnel x_i , et que les $X''_{j\eta}$ sont géométriquement connexes, avec un point rationnel y_j . Pour $k = 1$ ou 2 et $p_k(y_j) \in X'_{i\eta}$, choisissons une classe de chemins ℓ_{kij} de $\mathrm{pr}_k(\bar{y}_j)$ à $\bar{x}_i$: c'est un élément d'un torseur sous $\pi_1(X'_{i\bar{\eta}},\bar{x}_i)$. Soit $\bar{\ell}_{kij}$ le chemin "premier à p " correspondant (le même modulo $\mathrm{Ker}(\pi \longrightarrow \pi^{(p)})$. On peut prendre $\bar{\ell}_{kij}$ invariant par P : l'ensemble des chemins premiers à p est $\varprojlim$ d'ensembles finis d'ordre premier à p sur lesquels le pro-p-groupe P agit.

Dès lors, si un sous-groupe d'indice fini P' de P agit trivialement sur les $\pi_1^{(p)}(X'_{i\bar{\eta}},\bar{x}_i)$, il résulte des formules de Van Kampen (i.e. de ce que $X'_{i\bar{\eta}} \longrightarrow X_{i\bar{\eta}}$ est de descente effective pour les revêtements étales) que P' agit trivialement sur $\pi_1^{(p)}(X_{\bar{\eta}},\bar{x})$.

5.3.2. Pour $X_{\bar{\eta}}$ normal et U un ouvert, $\pi_1(U)$ s'envoie sur $\pi_1(X_{\bar{\eta}})$; on peut donc supposer $X_{\bar{\eta}}$ lisse et affine. Coupant par les sections hyperplanes *génériques* (ceci remplace S par le localisé strict de S au point générique de la fibre spéciale d'un $\mathbb{A}^n_S$) et appliquant Bertini, on se ramène à supposer que X_η est *lisse, géométriquement connexe et de dimension un*. Quitte à passer à une extension finie de $k(\eta)$, on peut supposer que X_η est le complément dans une courbe projective et lisse $\bar{X}_\eta$ d'un ensemble fini de points rationnels x_i ,(dont il nous est loisible d'augmenter le nombre. Appliquant le théorème de réduction semi-stable (5.1.1), on se ramène à supposer qu'il existe une courbe stable $X = (C\ ;\ x_1\ ,\dots,\ x_n)$ sur S telle que $X_\eta = C_\eta - \{x_1\ ,\dots,\ x_n\}$. On applique alors le résultat suivant

Théorème 5.4. Soient S un schéma strictement local, $f : \bar{X} \longrightarrow S$ un morphisme propre et plat, de fibre spéciale une courbe dont les seuls points de non lissité sont des points doubles ordinaires et Y un sous-schéma de $\bar{X}$, réunion d'un nombre fini de sections disjointes contenues dans l'ouvert de lissité. Soit x un point rationnel de $X = \bar{X} - Y$. Soit U l'ouvert de S au-dessus duquel X est lisse, et ξ un point géométrique de U. Alors l'image de $\pi_1(U,\xi)$ dans $\mathrm{Aut}(\pi_1^{(p)}(X_\xi, x\xi))$ est abélienne et première à p.

On prendra garde qu'on ne peut faire agir $\pi_1(U,\xi)$ sur $\pi_1^{(p)}(X_\xi,x\xi)$ que parce que les $\pi_1^{(p)}$ vérifient le théorème de spécialisation.

On se ramène au cas S complet, puis au cas universel où S est un schéma formel de modules pour la fibre spéciale, munie de ses sections. S est alors un anneau de séries formelles sur un anneau de Cohen : $S = \mathrm{Spec}(W[[t_1 \dots t_n]])$ et U est le complément d'un diviseur à croisements normaux d'équation $t_1 \dots t_m = 0$. Dans ce cas universel, d'après le lemme d'Abhyankar, $\pi_1(U,\xi)$ est abélien et premier à p, d'où le théorème.

Remarque 5.5. On peut démontrer un résultat analogue à 5.4 en dimension relative quelconque en considérant une section plane générique de dimension un et en appliquant Bertini.

6. Appendice : démonstration arithmétique du théorème de réduction stable (par P. Deligne)

Soient S un trait et A_η une variété abélienne sur $k(\eta)$, de dimension $d \neq 0$. Pour $S' \longrightarrow S$ un morphisme local de traits, on note $A_{\eta'}$ la variété abélienne sur $k(\eta')$ déduite de A_η par extension des scalaires. Soient $\mathcal{A}_{S'}$ le modèle de Néron de $A_{\eta'}$ sur S', $\mathcal{A}_{S's'}$ sa fibre spéciale et $\mathcal{A}^\circ_{S's'}$ la composante neutre de $\mathcal{A}_{S's'}$. Le théorème de réduction stable est le suivant.

Théorème 6.1. Pour S' convenable, $A_{\eta'}$ a réduction stable, i.e. $\mathcal{A}^{\circ}_{S's'}$ est extension d'une variété abélienne par un tore.

Il résulte assez facilement de 6.1. qu'on peut prendre S' tel que $k(\eta')$ soit une extension séparable finie de $k(\eta)$. Supposons tout d'abord que le corps résiduel de S est de type fini sur le corps premier (le même argument marcherait pour $k(s)$ radiciel sur un tel corps).

Soient ℓ un nombre premier premier à l'exposant caractéristique résiduel p, $T_\ell(A)$ le module de Tate, $V_\ell(A) = T_\ell(A) \otimes \mathbb{Q}_\ell$ et $\rho : \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow GL(T_\ell(A))$ la représentation naturelle. Une extension des scalaires préliminaire nous ramène au cas où le groupe d'inertie I agit de façon unipotente (1.2) donc à travers son plus grand pro-ℓ-groupe quotient $\mathbb{Z}_\ell(1)$ (0.3). Soient T l'image dans $GL(T_\ell(A))$ d'un générateur de $\mathbb{Z}_\ell(1)$ et r le plus grand entier $\geqslant 0$ tel que $(T-1)^r \neq 0$. On a :

Lemme 6.2. L'application

$$\sigma \in I,\ v \in V_\ell(A) \longmapsto (\rho(\sigma)-1)^r\,(v)$$

induit des morphisme et isomorphisme

(6.2.1)
$$\begin{array}{ccc} V_\ell(A)_I \otimes \mathbb{Q}_\ell(r) & \xrightarrow{\ \ M\ \ } & V_\ell(A)^I \\ \downarrow & & \uparrow \\ V_\ell(A)/\mathrm{Ker}(T-1)^r \otimes \mathbb{Q}_\ell(r) & \xrightarrow{\ \ \sim\ \ } & \mathrm{Im}(T-1)^r \end{array} .$$

$V_\ell(A)_I$ et $V_\ell(A)^I$ sont des $\mathrm{Gal}(\bar{s}/s)$-modules et M est galoisien.

6.3. On dit qu'une représentation ℓ-adique $\tau : \mathrm{Gal}(\bar{s}/s) \longrightarrow GL(V)$ est de poids n ($n \in \mathbb{Z}$) si la condition suivante est vérifiée :

(*) Pour un modèle convenable X de $k(s)$ (X = une variété irréductible de type fini sur le corps premier, avec $k(s) = k(X)$), τ se factorise par $\pi_1(X,\bar{s})$ et pour x un point fermé de X, les valeurs propres de $\tau(F_x)$ sont des nombres algébriques dont les conjugués complexes sont tous de valeur absolue $|k(x)|^{n/2}$ (F_x désigne le Frobenius géométrique ϕ_x^{-1}).

Si V (resp. W) est de poids n (resp. m) et que $n \neq m$, on a $\mathrm{Hom}_{\mathrm{Gal}}(V,W) = 0$.

6.4. Il résulte de la propriété universelle du modèle de Néron que $V_{\ell}(A)^{I} = V_{\ell}(\mathcal{A}^{\circ}_{S,s})$ Si a est la dimension du plus grand quotient abélien de $\mathcal{A}^{\circ}_{S,s}$ et m la dimension de la partie multiplicative de $\mathcal{A}^{\circ}_{S,s}$, d'après Weil, $V_{\ell}(A)^{I}$ est donc extension d'une représentation de dimension $2a$ et poids -1 par une représentation de dimension m et poids -2.

6.5. Soit A^{*} la variété abélienne duale de A. Rappelons que $V_{\ell}(A^{*})$ et $V_{\ell}(A)$ (donc $V_{\ell}(A^{*})^{I}$ et $V_{\ell}(A)_{I}$ sont en dualité à valeurs dans $\mathbb{Q}_{\ell}(1)$. Si a^{*} et m^{*} sont définis pour A^{*} comme a et m pour A, il résulte de 6.4 que $V_{\ell}(A)_{I}$ est extension d'une représentation de dimension m^{*} et de poids 0 par une représentati de dimension $2a^{*}$ et de poids -1.

6.6. Le module galoisien $\mathbb{Q}_{\ell}(r)$ est de poids $-2r$. Puisque $\mathrm{Im}(T-1)^{r} \neq 0$, on déduit de 6.4, 6.5 et de l'isomorphisme 6.2 que $r \leq 1$. Si $r = 0$, on a

$$\left\{\begin{array}{l} a + m \leqslant d \\ 2a + m = 2d, \end{array}\right.$$

donc $m = 0$, $a = d$ et A_{η} a bonne réduction. Supposons que $r = 1$. On tire alors de 6.2 que $\mathrm{Im}(T-1)$ est de poids -2 et que $V_{\ell}(A)/V_{\ell}(A)^{I}$ est de poids 0 ; ces espaces ont même dimension $m_1 \leqslant m$.

On a alors

$$\dim V_{\ell}(A)^{I} = 2d - m_1 = 2a + m,$$

$$2 \dim \mathcal{A}_{S,s} \geqslant 2(a+m) \geqslant 2a + m + m_1 = 2d = 2 \dim \mathcal{A}_{S,s},$$

de sorte que A_{η} a réduction stable $(\dim \mathcal{A}_{S,s} = a + m)$. On a obtenu en même temps que $m_1 = m$, i.e. que $\mathrm{Im}(T-1) \subset V_{\ell}(A)^{I} \simeq V_{\ell}(\mathcal{A}^{\circ}_{S,s})$ est le V_{ℓ} de la partie multiplicative de $\mathcal{A}^{\circ}_{S,s}$.

6.7. Pour traiter le cas général, nous utiliserons (0.5) (cf. 1.3). On se ramène à supposer que

(a) les conditions de (0.5.2) ou (0.5.3) sont vérifiées ;

(b) $\mathrm{Gal}(\bar{\eta}/\eta)$ agit trivialement sur A_ℓ ;

(c) l'action de I sur $T_\ell(A)$ est unipotente.

Sous ces hypothèses, nous prouverons que A_η a réduction stable. L'hypothèse (a) permet d'appliquer (0.5.1) pour avoir $S = \varprojlim S_i$ comme en 1.3, dont nous reprenons les notations. Soit s_i le point fermé de S_i .

Pour i assez grand, la composante neutre $\mathcal{A}^\circ$ du modèle de Néron $\mathcal{A}_S$ provient d'un schéma en groupe lisse à fibres connexes $\mathcal{A}_i$ sur S_i , et $(\mathcal{A}_i)_\ell$ est constant sur $S_i - D$. Le groupe $\pi_1(S_i - D_i, \bar{\eta})$ agit sur $T_\ell(A)$, et agit trivialement sur $T_\ell(A)/\ell\, T_\ell(A) \simeq A_\ell$. Comme en 1.3, on voit que P_i agit trivialement. On a alors par (1.3.1) :

$$T_\ell(A)_I = T_\ell(A)_{I_i} , \qquad T_\ell(\mathcal{A}_{i,s_i}) = T_\ell(\mathcal{A}_s^\circ) = T_\ell(A)^I = T_\ell(A^{I_i})$$

et (6.2.1) est un diagramme de $\mathrm{Gal}(\bar{s}_i/s_i)$-modules auquel on peut appliquer les arguments 6.4 à 6.6.

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SGA 7

Exposé II

Propriétés de finitude du groupe fondamental

par Michèle RAYNAUD

Soient k un corps séparablement clos de caractéristique $p \geq 0$, X un schéma connexe de type fini sur k et $\Pi_1^{(p)}(X)$ le plus grand quotient "premier à p" de $\Pi_1(X)$. On montre, dans cet exposé, que, si X est une surface, $\Pi_1^{(p)}(X)$ est topologiquement de présentation finie et qu'il en est de même pour X quelconque si l'on admet la résolution des singularités.

Dans le cas d'une surface, la méthode de démonstration, due à J.P. Murre, consiste à se ramener au cas où l'on a une fibration au-dessus de P^1 ayant les propriétés (i), (ii) et (iii) de la proposition ci-dessous.

Proposition 2.1. Soient k un corps algébriquement clos, S une surface projective, irréductible, lisse sur k. Alors on peut trouver un éclatement

$$g : S' \longrightarrow S \quad ,$$

où S' est une surface régulière, munie d'une fibration

$$f : S' \longrightarrow P^1 \quad ,$$

tels que f ait une section σ, et que les conditions suivantes soient satisfaites :

(i) f est lisse aux points de $\sigma(P^1)$.

(ii) les fibres de f sont des courbes géométriquement intègres n'ayant que des singularités ordinaires.

(iii) il existe un ouvert non vide U de P^1 tel que les fibres de f aux points de U soient lisses.

La proposition résulte de l'existence d'un plongement projectif $S \longrightarrow P^r$ tel qu'on puisse trouver un pinceau de Lefschetz par rapport à ce plongement (XVII 2.5). Soit en effet $D = \{H_t\}_{t \in P^1}$ un tel pinceau d'hyperplans ; rappelons que l'on a les propriétés suivantes (loc. cit. 2.2) :

a) l'axe Δ de D coupe transversalement S ; le sous-schéma fermé $\Delta \cap$ est donc réduit et formé d'un ensemble fini de points fermés de S, $x_1,\ldots,x_n$.

b) il existe un ouvert non vide U de P^1 tel que, pour tout point $t \in$ H_t coupe transversalement S.

c) pour tout point $t \in P^1-U$, H_t coupe transversalement S sauf en un point qui est un point singulier quadratique ordinaire.

Par chaque point x de S n'appartenant pas à Δ passe un unique hyper plan H_t, et l'application qui, à x, associe $t \in P^1$, est une application rati nelle a de S dans P^1 de domaine de définition $S-(\Delta \cap S)$.

On considère l'éclatement $g : S' \longrightarrow S$ du sous-schéma fermé $S \cap \Delta$ d S. Le schéma S' est régulier et la fibre S'_{x_i} au-dessus d'un point x_i s'iden tifie au fibré projectif défini par l'espace tangent à S en x_i, donc est isomo phe à P^1 (EGA IV 19.4.2, 19.4.4). De plus l'application rationnelle $f = ag$ est partout définie (XVIII 3.1). Enfin, on peut associer à chaque point x_i la sectio de f qui, à un point $t \in P^1$, fait correspondre le vecteur tangent en x_i à $H_t \cap S$.

Vérifions les conditions (i), (ii) et (iii). On note que, pour chaque $t \in P^1$, les courbes S'_t et $S_t = H_t \cap S$ sont isomorphes ; le morphisme $S'_t \longrightarrow S_t$ est en effet un isomorphisme sauf peut-être en les points x_i, point où S_t est régulière ; comme de plus S'_t est intersection complète dans S', S'_t est intègre, donc isomorphe à S_t.

Les conditions (i) et (ii) ne sont autres que les conditions a) et b). En f est lisse aux points de σ car f est plat et car les fibres S'_t sont géomét quement régulières aux points $\sigma(t)$.

Corollaire 2.2. Les notations étant celles de 2.1, soit Y un diviseur à croisements normaux dans S, $Y' = Y \times_S S'$. Alors on peut choisir f, g, U tels que $Y' \times_{P^1} U$ soit un diviseur à croisements normaux relativement à U.

On sait que, s'il existe un pinceau de Lefschetz pour un plongement $S \longrightarrow P^r$, les pinceaux de Lefschetz forment un ouvert de la variété des droites de P^r (XVII 3.2.1). Il en est de même de l'ensemble des pinceaux de Lefschetz dont l'axe ne rencontre pas Y et tels qu'il existe un ouvert non vide U_1 de P^1 tel que l'intersection de Y et H_t soit lisse si $t \in U_1$. Il suffit alors de construire f et g à partir d'un pinceau satisfaisant à ces deux conditions et de remplacer par $U \cap U_1$.

2.3.0. Soit G un groupe profini. Rappelons que G est dit topologiquement de génération finie s'il existe un entier n tel que G soit isomorphe à un quotient du pro-groupe libre à n générateurs $F(n)$; soit $K = \mathrm{Ker}(F(n) \longrightarrow G)$; on dit que G est topologiquement de présentation finie si, de plus, on peut trouver un nombre fini d'éléments $k_1,\ldots,k_r$ de K tels que le plus petit sous-groupe invariant fermé de K, contenant $k_1,\ldots,k_r$, soit égal à K.

Théorème 2.3.1. Soient k un corps séparablement clos de caractéristique $p \geqslant 0$, X un schéma connexe de type fini sur k. On suppose que les schémas de type fini et de dimension $\leqslant \dim(X)$ sur une clôture algébrique de k sont fortement désingularisables (SGA 5 I 3.1.5) (condition réalisée si $\dim X = 2$ d'après [1] ou si $k = 0$ d'après [2]). Alors le groupe fondamental $\Pi_1^{(p)}(X)$ est topologiquement de présentation finie.

On peut supposer k algébriquement clos d'après SGA 1 IX 4.10.

1) Réduction au cas où X est une surface régulière.

Si $f : X' \longrightarrow X$ est un morphisme de type fini, on pose $X'' = X' \times_X S'$ et on note $(X'_a)_{a \in A}$ et $(X''_b)_{b \in B}$ l'ensemble des composantes connexes de X' et X'' respectivement. Lorsque f est un morphisme de descente effective pour la catégorie des revêtements étales, le groupe fondamental $\Pi_1(X)$ peut se calculer explicitement en terme des groupes fondamentaux $\Pi_1(X'_a)$ et $\Pi_1(X''_b)$; il en résulte (SGA I IX 5.2, 5.3) que, si les $\Pi_1(X'_a)$ sont topologiquement de génération finie, il en est de même de $\Pi_1(X)$; si les $\Pi_1(X'_a)$ sont topologiquement de présentation finie et les $\Pi_1(X''_b)$ topologiquement de génération finie, $\Pi_1(X)$ est topologiquement de présentation finie.

Comme X est désingularisable, on peut trouver un schéma régulier X' et un morphisme propre birationnel $f : X' \longrightarrow X$; comme f est un morphisme de descente effective pour la catégorie des revêtements étales, il suffit de prouver le théorème lorsqu'on fait l'hypothèse supplémentaire que X est régulier. On en déduit en effet que $\Pi_1(X)$ est topologiquement de génération finie ; sachant que $\Pi_1(X')$ est topologiquement de présentation finie et les $\Pi_1(X''_b)$ topologiquement de génération finie, on en déduit que $\Pi_1(X)$ est topologiquement de présentation finie.

Soit maintenant $(U_i)_{i \in I}$ un recouvrement fini de X par des ouverts affines et soit $f : \coprod U_i \longrightarrow X$ le morphisme canonique. Comme f est un morphisme de descente effective pour la catégorie des revêtements étales, on voit qu'il suffit de prouver le théorème pour les U_i ; on est donc ramené au cas où X est affine et lisse.

On suppose désormais X affine, lisse sur k, et $\dim X > 2$. On peut alors appliquer le théorème de Lefschetz pour les schémas quasi-projectifs ([3] V 7.4) ; si Y est une section hyperplane "assez générale" de X, on a un isomorphisme

$$\Pi_1(Y) \simeq \Pi_1(X) \quad .$$

Il suffit donc de prouver le théorème pour Y, ce qui nous ramène au cas où $\dim X = 2$.

2) <u>Cas où X est une surface lisse sur</u> k

D'après le théorème de désingularisation forte des surfaces, on peut trouver une surface S lisse, intègre, projective et une immersion ouverte $i : X \longrightarrow S$ telle que le diviseur $Y = S - X$ soit à croisements normaux. On applique le corollaire 2.2 dont on reprend les notations.

Montrons qu'il suffit de prouver que le groupe fondamental de $X' = X\times_S S'$ est topologiquement de présentation finie. Comme X et X' sont réguliers et comme on a

$$\operatorname{codim}(X-\bigcup_{1\leq i\leq n}\{x_i\},X) \geq 2 \quad ,$$

tout revêtement étale de X' induit sur $X-\bigcup_{1\leq i\leq n}\{x_i\}$ un revêtement étale qui se prolonge de façon unique en un revêtement étale de X (SGA 2 I.11) ; ceci montre que l'on a un isomorphisme

$$\Pi_1(X')\simeq \Pi_1(X) \quad .$$

On considère le diagramme cartésien

$$\begin{tikzcd} X' \arrow[d, "h"'] & X'_U \arrow[l] \arrow[d, "h_U"] \\ P^1 & U \arrow[l] \end{tikzcd} \quad ,$$

où $h = f|X'$. On note L l'ensemble des nombres premiers distincts de p et $\bar{\eta}$ un point géométrique au-dessus du point générique η de P^1. Vérifions que les hypothèses de SGA 1 XIII 4.8 sont satisfaites. Le morphisme h a une section σ car il en est ainsi de f. Prouvons la condition a) de SGA 1 XIII 4.7 ; les fibres de h étant géométriquement intègres, le morphisme h est 0-acyclique et localement 0-acyclique (SGA 4 XV 4.1, 1.16) ; il reste à montrer que, pour tout faisceau d'ensembles localement constant F sur X', (F,h) est cohomologiquement propre en dimension ≤ 0. Si ξ est un point fermé de P^1 et si Z désigne la fermeture intégrale de $\operatorname{Spec} \mathcal{O}_{P^1,\xi}$ dans $\bar{\eta}$ le schéma $X'_Z = X'\times_{P^1} Z$ vérifie la condition (S_2) aux points fermés de X' et est régulier en tout autre point, donc est normal ; par suite un revêtement étale de X'_Z dont la restriction à $X'_{\bar{\eta}}$ est connexe est nécessairement connexe. Ceci démontre notre assertion, d'où a). Comme X'_U est le complémentaire dans S'_U d'un diviseur à croisements normaux relativement à U, h_U est localement 1-asphérique pour L (SGA 4 XV 2.1) et, pour tout faisceau de L-groupes constant fini F sur X'_U, (F,h_U) est cohomologiquement propre en dimension ≤ 1 (SGA 1 XIII 2.4), d'où la condition b). Comme P^1 est normal et $T = P^1 - U$ de codimension 1 , on a

$$\text{prof ét}_T(S) \geqslant 2 \quad ,$$

d'où la condition c). La condition

$$\text{prof hop}_x(X') \geqslant 3$$

est valable en tout point fermé de X' d'après le théorème de pureté, d'où la condition e). Enfin, pour tout point $t \in P^1$, h est lisse en $\sigma(t)$, donc est localement 1-asphérique pour L en $\sigma(t)$, d'où la condition d).

D'après SGA 1 XIII 4.7, si a est un point géométrique de $X'_{\bar{\eta}}$, on a une suite exacte

$$1 \longrightarrow \Pi_1^{(p)}(X'_{\bar{\eta}},a) \longrightarrow \Pi_1'(X'_U,a) \longrightarrow \Pi_1(U,a) \longrightarrow 1 \quad ,$$

et le choix de la section σ définit un scindage de cette suite exacte. Notons K l'image de $\Pi_1(U,a)$ dans $\text{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}},a))$; le groupe des coinvariants sous K , $\Pi_1^{(p)}(X'_{\bar{\eta}},a)_K$, est le quotient de $\Pi_1^{(p)}(X'_{\bar{\eta}},a)$ par le sous-groupe invariant fermé engendré par les éléments du type $x^{-1}qx$, où $x \in \Pi_1^{(p)}(X'_{\bar{\eta}},a)$, $q \in K$. D'après loc. cit. (4.7.4), on a un isomorphisme

$$\Pi_1^{(p)}(X',a) \simeq \Pi_1^{(p)}(X'_{\bar{\eta}},a) \quad .$$

Soit $P^1 - U = \coprod_{1 \leqslant i \leqslant n} \{t_i\}$ et, pour chaque i , soit $\bar{T}_i$ le localisé strict de P^1 en un point géométrique $\bar{t}_i$ au-dessus de t_i et s_i le point générique de $\bar{T}_i$. Un revêtement étale de U , dont la restriction à chaque s_i est triviale, se prolonge en un revêtement étale de P^1 , donc est trivial. Pour chaque i , soit $\bar{k}(s_i)$ la clôture algébrique de $k(s_i)$; l'assertion ci-dessus revient à dire que $\Pi_1(U,a)$ est égal au sous-groupe invariant fermé engendré par les images des groupes de Galois, $\text{Gal}(\bar{k}(s_i),k(s_i))$, dans $\Pi_1(U,a)$. D'après I 5.3, l'image K_i de $\text{Gal}(\bar{k}(s_i),k(s_i))$ dans $\text{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}},a))$ est finie. Comme K est le plus petit sous-groupe invariant fermé contenant tous les K_i , et comme $\Pi_1^{(p)}(X'_{\bar{\eta}},a)$ est topologiquement de présentation finie (SGA 1 XIII 3.2), ceci prouve que $\Pi_1^{(p)}(X'_{\bar{\eta}},a)_K$, donc aussi $\Pi_1^{(p)}(X)$ qui lui est isomorphe, est topologiquement de présentation finie.

Remarque 2.3.2. La résolution des singularités n'est pas nécessaire pour démontrer le théorème 2.3.1 dans le cas où X est l'ouvert complémentaire d'un diviseur à croisements normaux d'un schéma projectif, lisse sur k .

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S G A 7

Exposé VI

FORMAL DEFORMATION THEORY

by D. S. RIM

1. Formal existence theorem

We recall the notion of cofibered category (S.G.A.1.IV). Let $\underline{C}$ be a fixed category. A cofibered category $\underline{A}$ over $\underline{C}$ is, by definition, a category $\underline{A}$ together with a functor $P : \underline{A} \to \underline{C}$ such that

Ax.1. For any map f in $\underline{C}$ and any object a in $\underline{A}$ with $P(a) =$ the source of f, there exists an f-morphism $\alpha : a \to b$ which is cocartesian (i.e. for any f-morphism $\alpha' : a \to b'$ there exists a unique T-morphism $\tau : b \to b'$ in $\underline{A}$ such that $\alpha' = \tau \cdot \alpha$ where $T =$ the target of f).

Ax.2. A composit of cocartesian morphism in $\underline{A}$ is again cocartesian.

The following properties of a cofibered category $P : \underline{A} \to \underline{C}$ follow immediately from the definitions:

(cancellation) Let $a \underset{\beta}{\overset{\alpha}{\rightrightarrows}} a'$ be any two cocartesian maps such that $P(\alpha) = P(\beta)$. If there exists a cocartesian map $b \xrightarrow{\gamma} a$ such that $\alpha\gamma = \beta\gamma$, then $\alpha = \beta$.

(factorization) Given cocartesian maps $a \xrightarrow{\alpha} a'$, $a \xrightarrow{\beta} a''$, there exists a cocartesian map $a' \xrightarrow{\gamma} a''$ such that $\gamma\alpha = \beta$ if and only if there exists $f : P(a') \to P(a'')$ in $\underline{C}$ such that $f \cdot P(\alpha) = P(\beta)$. Furthermore γ is unique.

Let $\underline{A}$ be a cofibered category over $\underline{C}$. For each object S in $\underline{C}$ we denote by $\underline{A}(S)$ the subcategory of $\underline{A}$ whose objects are the objects a in $\underline{A}$ such that $P(a) = S$, and $\mathrm{Hom}_{\underline{A}(S)}(a,a') = \{\alpha \in \mathrm{Hom}_{\underline{A}}(a,a') \mid P(\alpha) = \mathrm{id}_S\}$. A cofibered groupoid over $\underline{C}$ is a cofibered category $\underline{A}$ over $\underline{C}$ such that $\underline{A}(S)$

is a groupoid for every object S in $\underline{C}$. We recall that a groupoid is a category in which every morphism is an isomorphism. Thus, a cofibered groupoid over $\underline{C}$ is a cofibered category $P : \underline{A} \to \underline{C}$ such that every map in $\underline{A}$ is cocartesian. Furthermore, for every map $\alpha \in F\ell(\underline{A})$, α is an isomorphism in $\underline{A}$ if and only if $P(\alpha)$ is an isomorphism.

Throughout the rest of this section, we fix a complete noetherian local ring Λ with the residue field k , and we set $\underline{C}_\Lambda$ to be the category of artinian local Λ-algebras with the same residue field k , where the maps are local homomorphisms over Λ . From 1.2 on, a cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is always assumed to have the property that $\underline{A}(k) = \{e\}$, where $\{e\}$ denotes the category in which $\mathrm{Hom}_{\{e\}}(a, b)$ consists of one and only one element for any two objects a, b in $\{e\}$. A map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will be called simply a $\underline{C}_\Lambda$-map if $p(a') = p(a)$ and $p(\alpha) = 1_{p(a)}$. A map $a' \xrightarrow{\alpha} a$ on $\underline{A}$ will be called surjective, by abuse of language, if $p(\alpha) : p(a') \longrightarrow p(a)$ is surjective in $\underline{C}_\Lambda$.

Ex.1.1.(a) Let $F : \underline{C} \longrightarrow$(categories) be a functor. Define the category $\underline{F}$ as follows: an object in $\underline{F}$ is a pair (S,a) where $S \in ob(\underline{C})$ and $a \in obF(S)$, and a map $(S,a) \to (S',a')$ is a pair (f,u') where $f : S \to S'$ is a map in $\underline{C}$ and $u' : F(f)a \to a'$ is a map in $F(S')$. Composition of two maps $(S,a) \xrightarrow{(f,u')} (S',a') \xrightarrow{(g,u'')} (S'',a'')$ is given by $(gf, u''\cdot(F(g)u')) : (S,a) \to (S'',a'')$. With respect to the functor $P : \underline{F} \to \underline{C}$ given by $(S,a) \mapsto S$, $\underline{F}$ becomes a cofibered category over $\underline{C}$. If $F(S)$ is a groupoid for every $S \in ob(\underline{C})$, then $\underline{F}$ is a cofibered groupoid over $\underline{C}$. In particular, a functor $F : \underline{C} \to$ (groups) yields a cofibered group, and a functor $F : \underline{C} \to$ (Ens) yields a cofibered discrete groupoid. We also note that a natural transformation $\varphi : F \to G$, where $F,G : \underline{C} \to$ (categories) are functors, induces a cocartesian functor $\varphi : \underline{F} \to \underline{G}$ over $\underline{C}$.

Ex.1.1.(b) Let X , Y be schemes over Λ together with a fixed isomorphism $X \underset{\Lambda}{\otimes} k \xleftrightarrow{\tau} Y \underset{\Lambda}{\otimes} k$. Then the functor $\mathrm{Isom}_{X,Y} : \underline{C}_\Lambda \to$ (groupoids) given by $\mathrm{Isom}_{X,Y}(S) =$ the set of isomorphisms $X \underset{\Lambda}{\otimes} S \xleftrightarrow{\sigma} Y \underset{\Lambda}{\otimes} S$ such that $\sigma \underset{\Lambda}{\otimes} k = \tau$ **yields a cofibered groupoid** $P : \underline{\mathrm{Isom}}_{X,Y} \to \underline{C}_\Lambda$.

Ex.1.1.(c) Let X_0 be a fixed scheme over k , and let $\underline{M}_{X_0}$ be the category consisting of deformations of X_0 over $\underline{C}_\Lambda$ (see §4). Thus an object in $\underline{M}_{X_0}$ is a pair (R,X) where R is in $\underline{C}_\Lambda$ and X is a deformation of X_0 to R . Then the functor $P : \underline{M}_{X_0} \to \underline{C}_\Lambda$ given by $(R,X) \rightsquigarrow R$ defines a cofibered groupoid over $\underline{C}_\Lambda$.

Ex.1.1.(d) For any groupoid X we denote by $[X]$ the discrete groupoi consisting of the isomorphism classes of objects in X . If $P : \underline{F} \to \underline{C}_\Lambda$ is a cofibered groupoid, we obtain a functor $[\underline{F}] : \underline{C}_\Lambda \to$ (Ens) given by $[\underline{F}](S) = [\underline{F}(S)]$, and in turn a cofibered discrete groupoid, which is, by abuse of notation, denoted by $[\underline{F}]$.

Definition 1.2. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called semi-homogeneous if the following properties hold:

(1) every diagram

$$\begin{tikzcd} a'' \arrow[r, dashed] \arrow[d, dotted] & a' \arrow[d] \\ a_1 \arrow[r] & a \end{tikzcd}$$

of solid arrows in $\underline{A}$ where $a' \to a$ is surjective can be complete into a commutative diagram including dotted arrows.

(2) Let $R \underset{k}{\times} k[\varepsilon] \overset{\pi_1}{\underset{\pi_2}{\rightrightarrows}} \begin{matrix} R \\ k[\varepsilon] \end{matrix}$ be the projection maps, where $k[\varepsilon]$ is the ring of dual numbers over k , and let $\begin{matrix} a' \xrightarrow{\alpha} a \\ b' \xrightarrow[\beta]{} a \end{matrix}$ be π_1-map in $\underline{A}$. Then there exists a $\underline{C}_\Lambda$ -map $\gamma : a' \to b'$ with $\alpha = \beta\cdot$ if and only if $(\pi_2)_*(a') \approx (\pi_2)_*(b')$.

Remark 1.3.(a) A reformulation of the notion in terms of 2-categories will be given later in 2.6(b). We note that if $\underline{A}$ is semi-homogeneous. Then so is $[\underline{A}]$ and in particular 1.2(2) entails the canonical map

$$[\underline{A}(R \times_k k[\varepsilon])] \longrightarrow [\underline{A}(R)] \times [\underline{A}(k[\varepsilon])]$$

to be bijective.

<u>Remark</u> 1.3.(b) Consider a commutative diagram

$$\begin{tikzcd}[column sep=small] & a' \arrow[dl, "v_1"'] \arrow[dr, "v_2"] & \\ a_1 \arrow[dr, "u_1"'] & & a_2 \arrow[dl, "u_2"] \\ & a & \end{tikzcd}$$

in a cofibered category $\underline{A}$ over $\underline{C}_\Lambda$. Set $R = P(a)$, $R' = P(a')$ and $R_i = P(a_i)$ for $i = 1, 2$. If we set a'' to be the "direct image" under the map $(P(v_1), P(v_2)) : R' \to R_1 \times_R R_2$, we obtain a commutative diagram

$$\begin{tikzcd}[column sep=small] & a' \arrow[ddl, "v_1"'] \arrow[d] \arrow[ddr, "v_2"] & \\ & a'' \arrow[dl] \arrow[dr] & \\ a_1 \arrow[dr, "u_1"'] & & a_2 \arrow[dl, "u_2"] \\ & a & \end{tikzcd}$$

Thus 1.2(1) can be replaced by

(1)': every diagram

$$\begin{tikzcd}[column sep=small] a_1 \arrow[dr] & & a_2 \arrow[dl] \\ & a & \end{tikzcd} \quad \text{in } \underline{A} \text{ above } \quad \begin{tikzcd}[column sep=small] R_1 \arrow[dr] & & R_2 \arrow[dl] \\ & R & \end{tikzcd} \quad \text{in } \underline{C}_\Lambda \text{ , where}$$

$R_2 \to R$ is surjective can be completed to a commutative diagram

$$\begin{tikzcd}[column sep=small] & a' \arrow[dl] \arrow[dr] & \\ a_1 \arrow[dr] & & a_2 \arrow[dl] \\ & a & \end{tikzcd} \quad \text{above} \quad \begin{tikzcd}[column sep=small] & R_1 \times_R R_2 \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ R_1 \arrow[dr] & & R_2 \arrow[dl] \\ & R & \end{tikzcd}$$

where π_i $(i = 1,2)$ are projections.

<u>Remark</u> 1.3.(c) If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, then $[\underline{A}(k[\varepsilon])]$ is canonically provided with a vector space structure over k , where $k[\varepsilon]$ is the ring of dual numbers over k . Indeed, let $\underline{\underline{V}}$ be the category of

finite-dimensional vector spaces over k . Then any functor $H : \underline{\underline{V}} \to (\mathrm{Ens})$ preserving the product is canonically factored through

$$\begin{tikzcd} \underline{\underline{V}} \arrow[rr, "H"] \arrow[dr, "\bar{H}"'] & & (\mathrm{Ens}) \\ & \underline{\underline{V}} \arrow[ur] & \end{tikzcd}$$

where $\underline{\underline{V}} \to (\mathrm{Ens})$ is the forgetful functor. On the other hand, the functor $G : \underline{\underline{V}} \longrightarrow \underline{C}_\Lambda$ given by $W \longmapsto k[W] = k \oplus W$, with $W^2 = 0$, transforms products in $\underline{\underline{V}}$ into fibre products over k in $\underline{C}_\Lambda$. If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, then $[\underline{A}(k[V] \times_k k[W])] \longrightarrow [\underline{A}(k[V])] \times [\underline{A}(k[W])]$ is bijective by 1.3(a), and the functor $V \longmapsto [\underline{A}(k[V])]$ commutes with **products**, **hence the result.**

Definition 1.4. (1) *Let* $R \in \mathrm{ob}(\underline{C}_\Lambda)$. *A small extension of* R *in* $\underline{C}_\Lambda$ *is a surjective map* $R' \xrightarrow{f} R$ *in* $\underline{C}_\Lambda$ *such that* $(\mathrm{Ker}\ f)^2 = 0$ *and* $\mathrm{Ker}\ f \simeq k$. *In other words,* $R' \xrightarrow{f} R$ *in* $\underline{C}_\Lambda$ *is a small extension of* R *if*

$$0 \longrightarrow k \xrightarrow{t} R' \xrightarrow{f} R \longrightarrow 0$$

is exact for some $t \in R'$ *with* $t^2 = 0$.

(2) *Let* $\underline{A}$ *be a cofibered groupoid over* $\underline{C}_\Lambda$, *and let* a *be an object in* $\underline{A}$. *A small prolongation of* a *is an arrow* $a' \xrightarrow{\alpha} a$ *in* $\underline{A}$ *such that* $P(\alpha)$ *is a small extension in* $\underline{C}_\Lambda$. *An arrow* $a' \xrightarrow{\alpha} a$ *in* $\underline{A}$ *will be called* "*versal for small prolongations of* a " *if, for every small prolongation* $a_1' \longrightarrow a$ *in* $\underline{A}$, *there exists a commutative diagram*

$$\begin{tikzcd} a' \arrow[rr] \arrow[dr, "\alpha"'] & & a_1' \arrow[dl] \\ & a & \end{tikzcd}$$

Proposition 1.5. *Let* $\underline{A}$ *be a semi-homogeneous cofibered groupoid over* $\underline{C}_\Lambda$.
(1) *Let* $R \times_k k[\varepsilon] \overset{\pi_1 \to R}{\underset{\pi_2 \to k[\varepsilon]}{\rightrightarrows}}$ *be projection maps in* $\underline{C}_\Lambda$, *and let* $a' \xrightarrow{\alpha} a$

be a π_1-map. Then there exists a map $a \xrightarrow{\beta} a'$ such that $\alpha\beta = 1_a$ if and only if there exists a map $a \longrightarrow (\pi_2)_* a'$.

(2) Let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$, and consider a commutative diagram

$$\begin{tikzcd} & a'' \arrow[dl, "\alpha'"'] \arrow[dr, "\alpha_1'"] & \\ a' \arrow[dr, "\alpha"'] & & a' \arrow[dl, "\alpha_1"] \\ & a & \end{tikzcd} \qquad \text{above} \qquad \begin{tikzcd} & R' \times_R R' \arrow[dl] \arrow[dr] & \\ R' \arrow[dr] & & R' \arrow[dl] \\ & R & \end{tikzcd}$$

Define the canonical map $\tilde{f} : R' \times_R R' \longrightarrow k[\mathrm{Ker}\, f]$ by $\tilde{f}(r_1', r_2') = r_1'(0)+(r_1'-r_2')$. If there exists a map $a' \longrightarrow (\tilde{f})_*(a'')$, then there exists a map $\gamma : a' \longrightarrow a$ such that $\alpha = \alpha_1 \cdot \gamma$.

Proof (1) The part "only if" is clear. Now assume that there exists a map $\varphi : a \longrightarrow (\pi_2)_* a'$. If we set $h : R \longrightarrow R \times_k k[\epsilon]$ by $h = 1 \times P(\varphi)$, then the composite map $R \xrightarrow{h} R \times_h k[\epsilon] \xrightarrow{\pi_1} R$ is the identity, and therefore we get maps $a \underset{\alpha_1}{\overset{\gamma}{\rightleftarrows}} f_* a$ such that $\alpha_1 \cdot \gamma = 1_a$ where $P(\gamma) = h$ and $P(\alpha_1) = \pi_1$. On the other hand, $\pi_2 \cdot h = P(\varphi)$ and hence there exists a π_2-map $h_* a \longrightarrow (\pi_2)_* a'$. This means that $(\pi_2)_*(h_* a) = (\pi_2)_* a'$, and therefore by 1.2(2) we have a $\underline{C}_\Lambda$-map $u : h_* a \longrightarrow a'$ such that $\alpha \cdot u = \alpha_1$. Define $\beta : a \longrightarrow a'$ by $\beta = u \cdot \gamma$. Then $\alpha \cdot \beta = \alpha \cdot u \cdot \gamma = \alpha_1 \cdot \gamma = 1_a$.

(2) The map $1 \times \tilde{f} : R' \times_R h' \longrightarrow R' \times_k k[\mathrm{Ker}\, f]$ is an isomorphism compatible with the projections on the first factor, and $\pi_2 \cdot (1 \times \tilde{f}) = \tilde{f}$ where $\pi_2 : R' \times_k k[\mathrm{Ker}\, f] \longrightarrow k[\mathrm{Ker}\, f]$ is the projection map. We note that $k[\mathrm{Ker}\, f] \simeq k[\epsilon]$ since $R' \xrightarrow{f} R$ is a small extension. Therefore, if there exists a map $a' \longrightarrow (\tilde{f})_* a'' = (\pi_2)_* [(1\ \tilde{f})_* a'']$, then by (1) above there exists a map $\gamma' : a' \longrightarrow a''$ such that $\alpha' \cdot \gamma' = 1_{a'}$. Define $\gamma : a' \longrightarrow a$ by $\gamma = \alpha_1' \cdot \gamma'$. We then have $\alpha_1 \cdot \gamma = \alpha$.

Corollary 1.6. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\wedge$ and consider a diagram

$$\begin{tikzcd} a' \arrow[dr, "\alpha"'] & & a_1 \arrow[dl, "\alpha_1"] \\ & a & \end{tikzcd}$$

in $\underline{A}$ where α_1 is a small prolongation. Assume that a is versal for small prolongations of e. Then there exists a map $a' \xrightarrow{\rho} a_1$ such that $\alpha = \alpha_1\rho$ in $\underline{A}$ if and only if there exists a map $f : P(a') \to P(a_1)$ such that $P(\alpha) = P(\alpha_1)\cdot f$ in $\underline{C}_\wedge$.

Proof: Assume that $P(\alpha) = P(\alpha_1)\cdot f$ where $f : P(a') \to P(a_1)$. Replacing a' by its direct image under f, we may and shall assume that $P(\alpha) = P(\alpha_1)$. Set $R' = P(a_1)$, $R = P(a)$. Since $\underline{A}$ is semi-homogeneous, we get a commutative diagram

$$\begin{tikzcd} & a'' \arrow[dl, "\beta"'] \arrow[dr, "\beta_1"] & \\ a' \arrow[dr, "\alpha"'] & & a_1 \arrow[dl, "\alpha_1"] \\ & a & \end{tikzcd} \quad \text{above} \quad \begin{tikzcd} & R'\times_R R' \arrow[dl] \arrow[dr] & \\ R' \arrow[dr] & & R' \arrow[dl] \\ & R & \end{tikzcd}$$

Since a is versal for small prolongations of e, so is a' and therefore our statement follows from 1.5(2).

Definition 1.7. For each object R in $\underline{C}_\wedge$ we put $\overline{\Omega}_R = M/(m_\wedge R+M^2)$ where $m_\wedge$, M are the maximal ideals of $\wedge$, R respectively. It is a finite-dimensional vector space over k, and $R \longmapsto \overline{\Omega}_R$ defines a functor $\overline{\Omega} : \underline{C}_\wedge \to \underline{\underline{V}}$ where $\underline{\underline{V}}$ is the category of finite-dimensional vector spaces over k. If $\underline{A}$ is a cofibered category over $\underline{C}_\wedge$, then the composite functor $\underline{A} \xrightarrow{P} \underline{C}_\wedge \to \underline{\underline{V}}$ is, by abuse of notation denoted by $\overline{\Omega}$ again.

Proposition 1.8. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\wedge$. Then, given an object a in $\underline{A}$ versal for small prolongations of e, there exists $a' \to a$ such that

(i) a' *is versal for small prolongations of* a

(ii) $p(a') \longrightarrow p(a)$ *is surjective; and the kernel is annihilated by the maximal ideal of* $p(a)$.

(iii) $\overline{\Omega}_{a'} \longrightarrow \overline{\Omega}_a$ *is an isomorphism*.

Proof: Set $R = P(a)$ and consider an exact sequence $0 \to J \to S \to R \to 0$ where S is a formal power-series ring over Λ in a finite number of variables, and $J \subset M^2$ where M is the maximal ideal of S . Let Σ be the set of ideals J_1 in S such that (i) $J \supset J_1 \supset MJ$ and (ii) there exists π_1-morphism $a_1 \to a$ in $\underline{A}$ where $\pi_1 : S/J_1 \to R$ is the canonical map. The property 1.2(1) of $\underline{A}$ together with the fact that J/MJ is a finite dimensional vector space over k entails that there exists a smallest ideal in the family Σ . Let it be J' and choose a π'-morphism $a' \overset{g}{\to} a$ where $\pi' : S/J' \to R$ is the canonical map. Note that $\overline{\Omega}_{a'} \to \overline{\Omega}_a$ is an isomorphism since $J' \subset J \subset M^2$. We claim that $a' \overset{g}{\to} a$ is versal for small prolongations of a . Indeed, let $a_1 \to a$ be a small prolongation. If we put $R_1 = P(a_1)$, we obtain an exact commutative diagram

$$\begin{array}{ccccccccc}
0 & \to & J/MJ & \to & S/MJ & \to & R & \to & 0 \\
 & & \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle h_1} & & \| & & \\
0 & \to & k & \to & R_1 & \to & R & \to & 0
\end{array}$$

If $h = 0$, then $\pi_1 : R_1 \to R$ admits a cross-section so that h_1 induces $f : S/J' \to R_1$. If $h \neq 0$, then $h_1 : S/MJ \to R_1$ is surjective and hence h_1 induces $f : S/J' \to R_1$ by the definiton of the ideal J' . Therefore in either cases we obtain a commutative diagram

$$\begin{tikzcd}
S/J' \arrow[rr, "f"] \arrow[dr, "\pi'"'] & & R_1 \arrow[dl, "\pi_1"] \\
 & R &
\end{tikzcd}$$

Since a is versal for small prolongations of e , it follows from 1.6 that we obtain a commutative diagram

$$\begin{tikzcd} a' \arrow[rr, "\rho"] \arrow[dr] & & a_1 \arrow[dl] \\ & a & \end{tikzcd}$$

Let $P : \underline{A} \to \underline{C}_\Lambda$ be a cofibered category (in groupoids). It is then clear that we obtain a cofibered category $\mathrm{pro}(P) : \mathrm{pro}(\underline{A}) \to \mathrm{pro}(\underline{C}_\Lambda)$ (in groupoids) **(pour la définition de la catégorie de pro-objets $\mathrm{pro}(\underline{A})$, voir A. Grothendieck, Sém. Bourbaki 195).**

$$\begin{tikzcd} \underline{A} \arrow[r] \arrow[d, "P"'] & \mathrm{pro}(\underline{A}) \arrow[d, "\mathrm{pro}(P)"] \\ \underline{C}_\Lambda \arrow[r] & \mathrm{pro}(\underline{C}_\Lambda) \end{tikzcd}$$

Now let $\hat{\underline{C}}_\Lambda$ be the full subcategory of pro-$\underline{C}_\Lambda$ in which
$\mathrm{ob}(\hat{\underline{C}}_\Lambda) = \{(R_i)_{i\in\underline{N}}\ ,\ R_i \in \mathrm{ob}(\underline{C}_\Lambda) \mid \varphi_i : R_i \to R_{i-1}$ is surjective for all i and induces isomorphisms $\underline{m}_i/\underline{m}_\Lambda+\underline{m}_i^2 \to \underline{m}_{i-1}/\underline{m}_\Lambda+\underline{m}_{i-1}^2$ for all sufficiently large $i\ \}$ where $\underline{N}$ stands for the ordered category of natural numbers, and $\underline{m}_i$ is the maximal ideal of R_i . We set $\hat{\underline{A}}$ to be the full subcategory of pro-$\underline{A}$ in which

$$\mathrm{ob}(\hat{\underline{A}}) = \{a \in \mathrm{ob}(\mathrm{pro}(\underline{A})) \mid \mathrm{pro}(P)\ (a) \in \mathrm{ob}(\hat{\underline{C}}_\Lambda)\}$$

We then obtain a cofibered category $\hat{P} : \hat{\underline{A}} \to \hat{\underline{C}}_\Lambda$ (in groupoids) with a commutative diagram

$$\begin{tikzcd} \underline{A} \arrow[r, hook] \arrow[d, "P"'] & \hat{\underline{A}} \arrow[d, "\hat{P}"] \\ \underline{C}_\Lambda \arrow[r, hook] & \hat{\underline{C}}_\Lambda \end{tikzcd}$$

We note that the category $\hat{\underline{C}}_\Lambda$ is canonically equivalent to the category of complete local Λ-algebras of formally finite type over Λ with the fixed residue field k . Henceforth we shall make such identification.

Definition 1.9. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. An object ξ in $\hat{\underline{A}}$ is called a quasi-initial object of $\hat{\underline{A}}$ if for every object a in $\hat{\underline{A}}$ there exists a map $\xi \to a$ in $\hat{\underline{A}}$. A quasi-initial object ξ of $\hat{\underline{A}}$ is called minimal if the morphism $\xi : h_R \to [\hat{A}]$ induces a bijection $\xi(k[\varepsilon]): \mathrm{Hom}_{\Lambda\text{-alg}}(R,k[\varepsilon]) \to [\underline{A}(k[\varepsilon])]$, where $R = P(\xi)$. An object ξ in $\hat{A}$ is called a versal formal object of $\underline{A}$ if every surjective map $\alpha : a' \to a$ in $\underline{A}$ induces a surjective map $\mathrm{Hom}_{\hat{A}}(\xi,a') \to \mathrm{Hom}_{\hat{A}}(\xi,a)$.

A versal formal object ξ of $\underline{A}$ is called minimal if the morphism $\xi : h_R \to [\hat{A}]$ induces a bijection $\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R,k[\varepsilon]) \to [\underline{A}(k[\varepsilon])]$ where $R = P(\xi)$.

Proposition 1.10. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$.

(i) every (minimally) versal formal object of $\underline{A}$ is a quasi-initial (minimal) object of $\hat{A}$.

(ii) If ξ is an object in $\hat{A}$ such that $\tilde{\xi}(k[\varepsilon]): \mathrm{Hom}_{\Lambda\text{-alg}}(R,k[\varepsilon]) \to [\underline{A}(k[\varepsilon])]$ is injective, where $P(\xi) = R$, then every endomorphism of ξ in $\hat{A}$ is an automorphism. In particular a quasi-initial minimal object of $\hat{A}$, if it exists, is unique up to (non-canonical) isomorphism.

Proof: (i) let ξ be a versal formal object of $\underline{A}$. We must show that, for any object a in $\hat{A}$, we have a map $\xi \to a$. By the definition of $\hat{A}$, there exists a sequence of surjective maps in $\underline{A}$: $\cdots \to a_n \xrightarrow{\alpha_n} a_{n-1} \to \cdots \to a_1 \xrightarrow{\alpha_1} a_0$ such that $a = \varprojlim_n a_n$. Assume that there exists maps $\varphi_i : \xi \to a_i$ such that $\alpha_i\varphi_i = \varphi_{i-1}$ for all $i < n$. Since ξ is a versal formal object of $\underline{A}$ and $\alpha_n : a_n \to a_{n-1}$ is surjective in $\underline{A}$, we obtain a map $\xi \xrightarrow{\varphi_n} a_n$ such that $\alpha_n\varphi_n = \varphi_{n-1}$. If we set $\varphi = \varprojlim_n \varphi_n$, we have $\varphi : \xi \to a$.

(ii) Let $\xi \xrightarrow{\alpha} \xi$ be a map in $\hat{A}$. Then for every $\lambda \in \mathrm{Hom}_{\Lambda\text{-alg}}(R,k[\varepsilon])$ we have $\lambda_*\xi = \lambda_*\alpha\xi \simeq (\lambda\cdot P(\alpha))_*\xi$ and hence by the hypothesis on ξ we have that $\lambda = \lambda\cdot P(\alpha)$ for all $\lambda \in \mathrm{Hom}_{\Lambda\text{-alg}}(R,k[\varepsilon])$, and therefore $P(\alpha) : R \to R$ is surjective. However it is trivial to see that every surjective endomorphism of a noetherian ring is always an automorphism and hence $P(\alpha)$ is an automorphism of R and therefore α is an automorphism of ξ .

Theorem 1.11. (Schlessinger) Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then $\underline{A}$

admits a minimally-versal formal object if and only if

(i) $\underline{A}$ is semi-homogeneous

(ii) $[\underline{A}(k[\varepsilon])]$ is a finite-dimensional vector space over k (cf. 1.3(c)).

Proof: (Necessity) Let ξ in $\widehat{\underline{A}}$ be a minimally-versal formal object of $\underline{A}$, and consider a diagram

$$\begin{tikzcd} a_1 \arrow[dr] & & a_2 \arrow[dl, "\alpha"] \\ & a & \end{tikzcd}$$

in $\underline{A}$ where α is surjective. Since ξ is a quasi-initial object of $\widehat{\underline{A}}$, there exists a map $\xi \to a_1$. Since α is surjective, the composit map $\xi \to a_1 \to a$ can be lifted to a_2 i.e., we obtain a commutative diagram

$$\begin{tikzcd} & \xi \arrow[dl] \arrow[dr] & \\ a_1 \arrow[dr] & & a_2 \arrow[dl, "\alpha"] \\ & a & \end{tikzcd}$$

which proves the condition 1.2(1). We now prove the condition 1.2(2). Let

$$\begin{tikzcd} S \times k[\varepsilon] \arrow[r, shift left, "\pi_1"] \arrow[r, shift right, "\pi_2"'] & S,\ k[\varepsilon] \end{tikzcd}$$

be the projections,

and let

$$\begin{tikzcd} a' \arrow[dr, "\alpha"'] & & b' \arrow[dl, "\beta"] \\ & a & \end{tikzcd}$$

be π_1-maps in $\underline{A}$. By the definition of ξ, we obtain a commutative diagram

$$\begin{tikzcd} & \xi \arrow[dl, "u"'] \arrow[dr, "v"] & \\ a' \arrow[dr, "\alpha"'] & & b' \arrow[dl, "\beta"] \\ & a & \end{tikzcd}$$

and in particular we have $\pi_1 \cdot P(u) = \pi_1 \cdot P(v)$. Assume now that $(\pi_2)_* a' \approx (\pi_2)_* b'$. We then have $\pi_2 \cdot P(u) = \pi_2 \cdot P(v)$ because of the minimality of

ξ , and hence $P(u) = \pi_1 P(u) \times \pi_2 P(u) = \pi_1 P(v) \times \pi_2 P(v) = P(v)$.
Since $\underline{A}$ is a cofibered groupoid, we obtain a unique isomorphism $a' \overset{\sigma}{\to} b'$ in $\underline{A}(S \times k[\varepsilon])$ such that $\sigma u = v$. Then $\beta\sigma u = \alpha u$ entails that $\beta\sigma = \alpha$ since $\underline{A}$ is a cofibered category.

(Sufficiency) Let $c_1, c_2, \ldots, c_n$ be objects in $\underline{A}(k[\varepsilon])$ which form a k-basis of the vector space $[\underline{A}(k[\varepsilon])]$, and choose a_1 in $\underline{A}(k[\varepsilon] \underset{k}{\times} \ldots \underset{k}{\times} k[\varepsilon])$ such that $(\pi_j)_* a_1 \approx c_j$. Then $\tilde{a}(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \to [\underline{A}(k[\varepsilon])]$ is bijective, where $R_1 = P(a_1)$. In particular $a_1 \to e$ is versal for small prolongations of e . Choose $a_2 \to a_1$ which is versal for small prolongations of a_1 such that $\overline{\Omega}_{a_2} \to \overline{\Omega}_{a_1}$ is bijective. We then choose $a_3 \to a_2$, versal for small prolongations of a_2 such that $\overline{\Omega}_{a_3} \to \overline{\Omega}_{a_2}$, is bijective etc., and set $\xi = \varprojlim_n a_n$. We claim that ξ is a minimally-versal formal object of $\underline{A}$. Indeed, since $\overline{\Omega}_{a_n} \to \overline{\Omega}_{a_{n-1}}$ is bijective for all n , $\overline{\Omega}_\xi \to \overline{\Omega}_{a_1}$ is bijective so that $\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon] \to [\underline{A}(k[\varepsilon])]$ is bijective. Now consider a diagram

$$\begin{tikzcd} & a' \arrow[d, "\alpha"] \\ \xi \arrow[r, "h"] & a \end{tikzcd}$$

where α is surjective in $\underline{A}$. We must show that h can be lifted to a' . For this, one may assume, by induction, that a' is a small prolongation of a . Since $P(a)$ is an artinian ring, $h : \xi \to a$ is factored as a composit map $\xi \to a_q \longrightarrow a$ for some q . Since $\underline{A}$ is semi-homogeneous, we get a diagram

$$\begin{tikzcd} & a'' \arrow[r] \arrow[d, "\alpha'"] & a' \arrow[d, "\alpha"] \\ \xi \arrow[r] & a_q \arrow[r] & a \end{tikzcd}$$

where α' is also a small prolongation. We then obtain a commutative diagram

$$\begin{tikzcd} & & a'' \arrow[d, "\alpha'"] \\ \xi \arrow[r] & a_{q+1} \arrow[r] \arrow[ur] & a_q \end{tikzcd}$$

and therefore we are done.

Remark 1.12. Let

$$\cdots \longrightarrow a_n \longrightarrow a_{n-1} \longrightarrow \cdots \longrightarrow a_2 \longrightarrow a_1 \longrightarrow a_o = e$$

be an infinite sequences of arrows in $\underline{A}$ such that

$$\begin{cases} a_i \text{ is versal for small prolongations of } a_{i-1} \\ P(a_i) \longrightarrow P(a_{i-1}) \text{ is surjective} \\ \overline{\Omega}_{a_i} \dashrightarrow \overline{\Omega}_{a_{i-1}} \text{ is bijective for } i \gg 0 \end{cases}$$

Then the above proof shows that $\varprojlim a_i$ is a versal formal object of $\underline{A}$.

Remark 1.13. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. If $\underline{A}$ has the property 1.11(i) and (ii), then so does $[\underline{A}]$, and $\xi \in \mathrm{ob}(\hat{\underline{A}})$ is a minimally-versal formal object of $\underline{A}$ if and only if it is so for $[\underline{A}]$ by 1.10(ii).

Remark 1.14. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, admitting a minimally-versal formal object ξ . Then an invariant of $R = P(\xi)$, which is a complete local Λ-algebra of formally finite type over Λ , is automatically an invariant of $\underline{A}$ over $\underline{C}_\Lambda$. Therefore, for example, we shall say that $\underline{A}$ is (formally) smooth over $\underline{C}_\Lambda$ if R is (formally) smooth over Λ i.e., a formal power-series algebra over Λ , or we say that $\underline{A}$ is irreducible over $\underline{C}_\Lambda$ if $\mathrm{Spec}(R)$ is irreducible etc. In the next section we introduce the notion of "smoothness" of a functor $\underline{A} \to \underline{B}$ over $\underline{C}_\Lambda$, where $\underline{A}$, $\underline{B}$ are cofibered categories in groupoids over $\underline{C}_\Lambda$. It turns out (cf. 2.4) that "$\underline{A}$ is (formally) smooth over $\underline{C}_\Lambda$" is equivalent to " $P : \underline{A} \to \underline{C}_\Lambda$ is a smooth functor" , and therefore no confusion arises. We also define the dimension of $\underline{A}$ over $\underline{C}_\Lambda$ by $\dim_{\underline{C}_\Lambda} \underline{A} = \dim k \otimes_\Lambda R$. For this we note the following fact: Let $\underline{C}_k$ be the category of artinian local k-algebras with the same residue field k . Then $\underline{C}_k \to \underline{C}_\Lambda$ is a fully-faithful imbedding, whose (left) adjoint functor $\underline{C}_\Lambda \to \underline{C}_k$ is given by $R \to k \otimes_\Lambda R$. Let $\underline{A}|\underline{C}_k$ be the pull-back of $P : \underline{A} \to \underline{C}_\Lambda$ under $\underline{C}_k \to \underline{C}_\Lambda$. Then it is trivial to see that if ξ is a minimally-versal formal object for $\underline{A}$ over $\underline{C}_\Lambda$, then $k \otimes_\Lambda \xi$ is a minimally-versal formal object for $\underline{A}|\underline{C}_k$ over $\underline{C}_k$, where $\xi \to k \otimes_\Lambda \xi$ is a cocartesian arrow above $R \to k \otimes_\Lambda R$. It follows that $\dim_{\underline{C}_\Lambda} \underline{A} = \dim_{\underline{C}_k} (\underline{A}|\underline{C}_k)$.

1.15. In the remaining of this paragraph, we biefly explain how to modify the preceeding theory in order to allow for residue field extensions.

Let Λ and K be complete noetherian local rings, with K a Λ-algebra. We assume that

(a) the natural map from Λ to K is a local homomorphism;

(b) the residue field K' of K is an extension of finite type of the residue field k of Λ .

The most interesting case is when $K = K'$ and K is a finite inseparable extension of k .

We denote by $C^\wedge_{\Lambda,K}$ the category whose objects are the complete noetherian local rings R , provided with a structure of Λ-algebra and with a Λ-epimorphism $s : R \longrightarrow K$.

We denote by $C_{\Lambda,K}$ the subcategory of $C^\wedge_{\Lambda,K}$ consisting of the (R, s) such that $\mathrm{Ker}(s)$ is a R-module of finite length. The category $C^\wedge_{\Lambda,K}$ can be identified with a full subcategory of $\mathrm{Pro}(C_{\Lambda,K})$.

If K is artinian, so is any object in $C_{\Lambda,K}$.

For any finite dimensional vector space V over K' we denote by $D_V(K)$ the algebra of dual numbers over K defined by V ; its elements are of the form $k + v\,\varepsilon$ (with $k \in K$, $v \in V$ and $\varepsilon^2 = 0$: $(k + v\,\varepsilon)(k' + v'\,\varepsilon) = (k\,k') + (k\,v' + k'\,v)\varepsilon$). The map $s : D_V(K) \to K$: $s(k + v\,\varepsilon) = k$, turns $D_V(K)$ into an object of $C_{\Lambda,K}$. The role played before by $k[\varepsilon]$ will be played here by $D_{K'}(K)$.

As explained in 1.8, any cofibered groupoid $\underline{A}$ over $C_{\Lambda,K}$ extends in a natural way to a cofibered category $\underline{A}^\wedge$ over $C^\wedge_{\Lambda,K}$. A map in $\underline{A}$ or $\underline{A}^\wedge$ will be called <u>surjective</u> if the corresponding homomorphism of algebras in $C_{\Lambda,K}$ or in $C^\wedge_{\Lambda,K}$ is.

<u>We assume that</u> $\underline{A}(K) = \{e\}$.

<u>Definition</u> 1.16. <u>A cofibered groupoïd</u> $\underline{A}$ <u>over</u> $C_{\wedge,K}$ <u>is called semi-homogeneous if the following properties hold</u>:

(I) <u>every diagram</u>

$$\begin{array}{ccc} a'' & \dashrightarrow & a' \\ \downarrow & & \downarrow \\ a_1 & \longrightarrow & a \end{array}$$

<u>of solid arrows in</u> $\underline{A}$, <u>where</u> $a' \longrightarrow a$ <u>is surjective, can be completed into a commutative diagram including dotted arrows;</u>

(II) <u>For any</u> R , <u>the natural map</u>

$$[\underline{A}(R \times_K D_{K'}(K))] \to [\underline{A}(R) \times \underline{A}(D_{K'}(K))] \simeq [\underline{A}(R)] \times [\underline{A}(D_{K'}(K))]$$

<u>is bijective</u>.

If $\underline{A}$ is semi-homogeneous, so is $[\underline{A}]$.

1.17. Let us first consider the restriction of $\underline{A}$ to the full subcategory of $C_{\wedge,K}$ whose objects are the $D_V(K)$. Let Ω be the vector space $\Omega^1_{K/\wedge} \otimes_K K'$, and $d : K \to \Omega$ the derivative. A map $f : D_V(K) \to D_W(K)$ can be written:

$$f(k + v\,\varepsilon) = k + (u(v) + D\,d\,k)$$

with $u \in \mathrm{Hom}_{K'}(V, W)$ and $D \in \mathrm{Hom}_{K'}(\Omega, W)$. If we identify f with (u, D) , the composition law is

$$(u, D) \circ (u', D') = (uu', D + u\,D') \quad .$$

The condition (II) entails

(II') the functor from K'-vector spaces to sets

$$V \longmapsto [\underline{A}(D_V(K))]$$

commutes with products.

<u>Lemma</u> 1.18. <u>Assume condition</u> (II') <u>is satisfied by</u> $\underline{A}$. <u>Then</u>

(1) $[\underline{A}(D_{K'}(K))]$ <u>carries a unique vector space structure, such that</u> $[\underline{A}(D_V(K))] \simeq V \otimes_{K'} [\underline{A}(D_{K'}(K))]$ <u>as a functor in</u> V .

(2) <u>There is a linear map</u> $\gamma : \mathrm{Hom}(\Omega, K') \to [\underline{A}(D_{K'}(K))]$ <u>such that, for any</u> $D \in \mathrm{Hom}(\Omega, K')$,

$$[\underline{A}((k, D))] : [\underline{A}(D_{K'}(K))] \to [\underline{A}(D_{K'}(K))] \text{ is } x \longmapsto x + \gamma(D) .$$

(3) <u>For any</u> $(u, D) : D_V(K) \longrightarrow D_W(K)$, <u>the map</u>

$$(u, D) : [\underline{A}(D_V(K))] \to [\underline{A}(D_W(K))] \text{ i.e.}$$

$$(u, D) : V \otimes [\underline{A}(D_{K'}(K))] \to W \otimes [\underline{A}(D_{K'}(K))] \ \underline{\text{is}}$$

$$x \longmapsto u(x) + \gamma(D) .$$

Let us introduce the notations

$$F(V) = [\underline{A}(D_V(K))]$$

$$A = F(K')$$

$$F(u,D) : F(V) \to F(W) = \text{the map induced by } (u,D) : D_V(K) \to D_W(K)$$

$$F(D) : F(V) \to F(V) : F(D) = F(1, D) .$$

By (II') and the fact that K' is a K'-vector space object in the category of K'-vector spaces, $F(K')$ is a K'-vector space in (Ens), hence a K'-vector space, and (1) is clear. Let us now identify $F(V)$ and $V \otimes A$.

We deduce from the identity

$$(u, D) = (1, D) \circ (u, 0) \text{ that}$$

$$F(u, D) = F(D) \circ (u \otimes 1_A) .$$

The functoriality of F amounts to

$$F(0) = \mathrm{Id}$$

$$F(D) \circ F(D') = F(D + D')$$

$$(u \otimes 1) \circ F(D) = F(u\, D) \circ (u \otimes 1_A)$$

For any $D : \Omega \to V$ and any $n \in N$, if i and p are the maps

$$i : V \hookrightarrow V \oplus K'^n \qquad p : V \oplus K'^n \longrightarrow K^n ,$$

then $p\, i\, D = 0$ and the following diagramm is commutative

$$\begin{array}{ccccc} V \otimes A & \longrightarrow & (V \oplus K'^n) \otimes A & \longrightarrow & K'^n \otimes A \\ \downarrow{\scriptstyle F(D)} & & \downarrow{\scriptstyle F(i,D)} & & \| \\ V \otimes A & \longrightarrow & (V \oplus K'^n) \otimes A & \longrightarrow & K'^n \otimes A \end{array}$$

Let $x : K^n \to V$ be a map, $x_i = x(e_i)$, and let a_i be a family of elements of A. We have $(1_V + x) \circ i \circ D = D$. Hence, if

$$F(i\, D)\ (\Sigma\, e_i\, a_i) = \Sigma\, e_i\, a_i + a \qquad (a \in V \otimes H)\ , \text{ then}$$

$$F(D)\ (\Sigma\, x_i\, a_i) = \Sigma\, x_i\, a_i + a$$

and a does not depend on the x_i. Hence,

$$F(D)\ (y) = y + G(D) \qquad , \text{ with}$$

$$G(D + D') = G(D) + G(D')$$

$$u\, G(D) = F(u\, D) \qquad (\text{for any } u : V \to W)\ .$$

One easily checks that any such G is defined, for some $\gamma \in \Omega \otimes A$, by the formula

$$G(D) = D \otimes 1_A\ (\gamma)\ ,$$

as promised in (2) (3).

Definition 1.19. (1) An object ξ in $\hat{A}$ is called versal formal object if every surjective map $\alpha : a' \longrightarrow a$ in A induces a surjective map $\mathrm{Hom}_{\hat{A}}(\xi, a') \longrightarrow \mathrm{Hom}_{\hat{A}}(\xi, a)$.

(2) A formal versal object ξ in $\hat{A}$, with image (R, s) in $C_{\hat{\Lambda},K}$, is called minimal if for any two maps $\varphi_1, \varphi_2 : R \rightrightarrows D_{K'}(K)$, if $\varphi_1(\xi)$ is isomorphic to $\varphi_2(\xi)$, then there exists a derivation D of K into K' such that $(1, D)(\varphi_1) = \varphi_2$, i.e. $\varphi_2 - \varphi_2 = D\, s$.

Theorem 1.20. If A is semi-homogeneous and if

$$\dim_{K'} ([A(D_{K'}(K))]) < \infty\ ,$$

then A admits a minimally versal formal object, and any two of them are isomorphic.

With the notations of 1.18, let H^* be a subspace of $[\underline{A}(D_{K'}(K))]$, supplementary to the image of γ, and let H be the dual of H^*. We define $R_o = D_H(K)$. Let ξ_o be an object in $\underline{A}(R_o)$, whose image in $[\underline{A}(R_o)] \simeq H \otimes [\underline{A}(D_{K'}(K))] \simeq \mathrm{Hom}(H^*,[\underline{A}(D_{K'}(K))])$ is the inclusion of H^* in $[\underline{A}(D_{K'}(K))]$. Then

(1) The maps $\mathrm{Hom}_{C_{\wedge,K}}(D_H(K), D_V(K)) \longrightarrow [\underline{A}(D_V(K))]$ are surjective.

(2) The minimality condition 1.19(2) is valid for (R_o, ξ_o).

Choose now some $S \in \hat{C}_{\wedge,K}$, formally smooth over $\wedge$, and an epimorphism $t : S \longrightarrow R_o$ such that

$$\Omega_{R_o/\wedge} \otimes K' \xrightarrow{\ \sim\ } \Omega_{S/\wedge} \otimes K' .$$

If M is the maximal ideal of S, we inductively define, for $n \geq 1$, $R_n = S/I_n$ and $\xi_n \in \mathrm{Ob}(\underline{A}(R_n))$ by the properties that

(a) I_{n+1} is the intersection of the ideals $I : I_n M \subset I \subset I_n$, such that ξ_n extends to an object ξ in $\underline{A}(S/I)$.

(b) ξ_{n+1} is an extension of ξ_n.

If $R = \varprojlim R_n$ and $\xi = \varprojlim \xi_n$, then one checks that (R, ξ) is a minimally-versal formal object, and the only one up to isomorphism.

2. Prorepresentable cofibered groupoids.

Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$, admitting a minimally-versal formal object ξ. In general we can not recover $\underline{A}$ (up to $\underline{C}_\wedge$-equivalence) out of ξ alone. One has to know when two maps $R \overset{f}{\underset{g}{\rightrightarrows}} S$ yield a $\underline{C}_\wedge$-isomorphism $\xi(f) \longrightarrow \xi(g)$, where $R = P(\xi)$. As an application of 1.11 and 1.12 we now consider the question of "recovering $\underline{A}$" up to $\underline{C}_\wedge$-equivalence. Indeed, we shall have a quite satisfactory answer applicable to a wide class of cofibered groupoids.

A number of cofibered groupoids which usually occur in practice enjoy a slightly stronger property than those of semi-homogeneous ones, as it will be defined explicitly in 2.5. To investigate them it is convenient to rephrase what we have done so far in terms of "smooth functors" . All the notations and terminologies in the previous section remain in force throughout in this section.

Definition 2.1. Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. We say that φ is smooth if every surjective map $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram

$$(*) \qquad \begin{tikzcd} \varphi(a') \arrow[r] \arrow[dr, "\varphi(\alpha)"'] & b' \arrow[d, "\beta"] \\ & \varphi(a) \end{tikzcd}$$

for some $\alpha : a' \longrightarrow a$ in $\underline{A}$. We say that φ is minimally smooth if it is smooth and $\varphi(k[\varepsilon]) : [\underline{A}(k[\varepsilon])] \longrightarrow [\underline{B}(k[\varepsilon])]$ is bijective.

Remark 2.2(a) Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. We note that " φ is smooth" $\Leftrightarrow$ "every surjective map $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed into a commutative diagram $(*)$ such that $\varphi(a') \longrightarrow b'$ is a $\underline{C}_\Lambda$-isomorphism" $\Leftrightarrow$ "every small prolongation $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed into a commutative diagram $(*)$ ".

Remark 2.2(b) For any cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$, the natural functor $\underline{A} \longrightarrow [\underline{A}]$ is minimally smooth.

Remark 2.2(c) For each object R in $\hat{\underline{C}}_\Lambda$, we set $h_R : \underline{C}_\Lambda \longrightarrow (\mathrm{Ens})$ to be the functor defined by $h_R(S) = \mathrm{Hom}_{\Lambda\text{-alg}}(R, S)$ for all S in $\underline{C}_\Lambda$. It associates a cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$, and a map $R \longrightarrow S$ in $\underline{C}_\Lambda$

induces a functor $\underline{h}_S \longrightarrow \underline{h}_R$ of cofibered groupoids. The functor $\underline{h}_S \longrightarrow \underline{h}_R$ induced by a map $R \longrightarrow S$ in $\underline{C}_\Lambda$ is smooth if and only if S is a formal power-series ring over R .

Remark 2.2(d) Given an object R in $\hat{\underline{C}}_\Lambda$, let us denote by $(R, \underline{C}_\Lambda)$ the category of arrows in $\hat{\underline{C}}_\Lambda$ with the source R and a target in $\underline{C}_\Lambda$. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Given $\xi \in \mathrm{ob}\,\hat{\underline{A}}$ with $P(\xi) = R$, let $\underline{A}' \longrightarrow (R, \underline{C}_\Lambda)$ be the pull-back of $\underline{A} \longrightarrow \underline{C}_\Lambda$ under the canonical functor $(R, \underline{C}_\Lambda) \longrightarrow \underline{C}_\Lambda$ and choose a section $\xi^{\#}$ such that $\xi^{\#}(1_R) = \xi$. (This simply means that we choose, for each map $R \xrightarrow{f} S$, a f-map $\xi \longrightarrow \xi(f)$). We then obtain a functor $\xi^{\#} : \underline{h}_R \longrightarrow \underline{A}$ over $\underline{C}_\Lambda$ by sitting $(R \xrightarrow{f} S) \longmapsto \xi(f)$. Of course this functor does not depend on the choice of $\xi^{\#}$ up to natural equivalence of functors. Now " $\xi^{\#} : \underline{h}_R \longrightarrow \underline{A}$ is (minimally) smooth" means that every surjective map $a' \xrightarrow{\alpha} \xi(f)$ in $\underline{A}$ can be completed into a commutative diagram

$$\begin{tikzcd} \xi(f') \arrow[r] \arrow[dr, "\xi(\ )"'] & a' \arrow[d, "\alpha"] \\ & \xi(f) \end{tikzcd}$$

(and $\underline{h}_R(k[\varepsilon]) \longrightarrow \underline{A}(k[\varepsilon])$ is bijective), which means precisely that ξ is a (minimally) versal formal object of $\underline{A}$ over $\underline{C}_\Lambda$. Thus we conclude that the existence of a (minimally) versal formal object of $\underline{A}$ is equivalent to the existence of a (minimally) smooth functor $\underline{h}_R \longrightarrow \underline{A}$ over $\underline{C}_\Lambda$ for some $R \in \mathrm{ob}(\hat{\underline{C}}_\Lambda)$.

The following general properties on smoothness are trivial to verify.

Proposition 2.3. Let $\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$ be functors of cofibered groupoids over $\underline{C}_\Lambda$.

(a) If φ, ψ are both smooth, then so is $\psi.\varphi$.

(b) If $\psi.\varphi$ is smooth and φ is surjective on isomorphism classes of objects, then ψ is smooth.

Remark 2.4. Let $\underline{A}$ be a cofibered groupoids over $\underline{C}_\Lambda$, admitting a versal formal object ξ . This means, by 2.2(d), that $\tilde{\xi} : \underline{h}_R \to \underline{A}$ is smooth, where $R = P(\xi)$. Therefore we have, " $\underline{A}$ is (formally) smooth over $\underline{C}_\Lambda$ (c.f. 1.13) i.e., R is a formal power-series algebra over Λ " $\Leftrightarrow$ "the composite functor $\underline{h}_R \xrightarrow{\tilde{\xi}} \underline{A} \xrightarrow{P} \underline{C}_\Lambda$ $(= \underline{h}_\Lambda)$ is smooth" $\Leftrightarrow$ " $\underline{A} \xrightarrow{P} \underline{C}_\Lambda$ is a smooth functor" by 2.3(b).

We now consider a stronger property than being semi-homogeneous.

Definition 2.5. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. It is called homogeneous if every diagram

$$\begin{tikzcd} & a' \arrow[d] \\ b' \arrow[r] & a \end{tikzcd}$$

where $a' \to a$ is surjective admits a fibre-product $a' \underset{a}{\times} b'$ in $\underline{A}$ such that

$$P(a' \underset{a}{\times} b') = P(a') \underset{P(a)}{\times} P(b') \quad .$$

Remark 2.6(a) Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and consider a diagram

(*) $$\begin{tikzcd} & a' \arrow[dl] \arrow[dr] & \\ a_1 \arrow[dr] & & a_2 \arrow[dl] \\ & a & \end{tikzcd}$$ in $\underline{A}$ above the diagram $$\begin{tikzcd} & R_1 \underset{R}{\times} R_2 \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ R_1 \arrow[dr] & & R_2 \arrow[dl] \\ & R & \end{tikzcd}$$

in $\underline{C}_\Lambda$ where π_i are the projections. If a fibre-product $a_1 \underset{a}{\times} a_2$ exists in $\underline{A}$, we obtain a unique map $\varphi : a' \to a_1 \underset{a}{\times} a_2$ compatible with the projections. If $P(a_1 \underset{a}{\times} a_2) = R_1 \underset{R}{\times} R_2$, then $P(\varphi)$ is the identity map on $R_1 \underset{R}{\times} R_2$ and therefore φ is a $\underline{C}_\Lambda$-isomorphism i.e., $a' = a_1 \underset{a}{\times} a_2$. In other words, if $\underline{A}$ is a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, every diagram (*) defines a fibre-product $a_1 \underset{a}{\times} a_2$ provided $R_1 \to R$ is surjective in $\underline{C}_\Lambda$. This fact entails immediately that a homogeneous cofibered groupoid is, a fortiori, semi-homogeneous.

Remark 2.6(b) Consider a diagram

$$\begin{tikzcd} X \arrow[dr, "\varphi"'] & & Y \arrow[dl, "\psi"] \\ & Z & \end{tikzcd}$$

of categories and functors. The ordinary fibre-product $X \underset{Z}{\times} Y$ is a category in which an object is a pair (x,y) such that $\varphi(x) = \psi(y)$, and a map $(x,y) \to (x', y')$ consists of a pair $\alpha : x \to x'$, $\beta : y \to y'$ such that $\varphi(\alpha) = \psi(\beta)$. The fibre-product $X \overset{2}{\underset{Z}{\times}} Y$ in the sense of 2-categories is a category in which an object is a triple $(x,y;\gamma)$, where $\varphi(x) \xrightarrow{\gamma} \psi(y)$ is an isomorphism, and a map $(x,y;\gamma) \to (x',y';\gamma')$ consists of a pair $x \xrightarrow{\alpha} x'$ $y \xrightarrow{\beta} y'$ with a commutative diagram

$$\begin{tikzcd} \varphi(x) \arrow[r, "\varphi(\alpha)"] \arrow[d, "\gamma"'] & \varphi(x') \arrow[d, "\gamma'"] \\ \psi(y) \arrow[r, "\psi(\beta)"'] & \psi(y') \end{tikzcd}$$

One can rephrase the notions of (semi)homogeneous cofibered groupoid or smooth functor in terms of fibre-product of categories (in the sense of 2-category). Now let $P : \underline{A} \to \underline{C}_\Lambda$ be a cofibered category, and consider an object $R_1 \underset{R}{\times} R_2$ in $\underline{C}_\Lambda$. If we choose a direct image functor $(\pi_i)_*: \underline{A}(R_1 \underset{R}{\times} R_2) \to \underline{A}(R_i)$, where $\pi_i : R_1 \underset{R}{\times} R_2 \to R_i$ is the projection for $i = 1,2$, we obtain a functor $\underline{A}(R_1 \underset{R}{\times} R_2) \to \underline{A}(R_1) \overset{2}{\underset{\underline{A}(R)}{\times}} \underline{A}(R_2)$. It is clear from the definition of cofibered category that this functor, up to canonical natural equivalence, does not depend on the choice of direct image functors. It is now clear from the definition that a cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is homogeneous over $\underline{C}_\Lambda$ if and only if $\underline{A}(R_1 \underset{R}{\times} R_2) \to \underline{A}(R_1) \overset{2}{\underset{\underline{A}(R)}{\times}} \underline{A}(R_2)$ is an equivalence of categories whenever $R_1 \to R$ is surjective in $\underline{C}_\Lambda$, and $\underline{A}$ is semi-homogeneous if and only if

$$[\underline{A}(R \underset{R}{\times} R_2)] \longrightarrow [\underline{A}(R_1) \overset{2}{\underset{\underline{A}(R)}{\times}} \underline{A}(R_2)]$$

is surjective whenever $R_1 \longrightarrow R$ is surjective in $\underline{C}_\Lambda$, and is bijective whenever $R = k$ and $R_1 = k[\epsilon]$. Now let $\varphi : \underline{A} \to \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. For each map $R' \to R$ in $\underline{C}$, φ induces a functor $\underline{A}(R') \to \underline{A}(R) \overset{2}{\underset{\underline{B}(R)}{\times}} \underline{B}(R')$ uniquely determined up to a (canonical) natural equivalence. It is again trivial to see that φ is smooth if and only if the functor $A(R') \to \underline{A}(R) \overset{2}{\underset{\underline{B}(R)}{\times}} \underline{B}(R')$ is surjective on isomorphism classes of objects, whenever $R' \to R$ is surjective.

Remark 2.6(c). If $\underline{A}$ is a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, the functor $A(k[V \times W]) \to \underline{A}(k[V]) \times \underline{A}(k[W])$ is an equivalence of categories for any finite-dimensional vector spaces V, W over k (We recall our convention $\underline{A}(k) = \{e\}$). In particular, we have

$$\mathrm{Aut}_{\underline{A}(k[V\times W])}(1_{V\times W}) \overset{\sim}{\to} \mathrm{Aut}_{\underline{A}(k[V])}(1_V) \times \mathrm{Aut}_{\underline{A}(k[W])}(1_W)$$

where 1_V is the direct image of e under the unique map $k \to k[V]$ in $\underline{C}_\Lambda$. It follows that if $\underline{A}$ is homogeneous over $\underline{C}_\Lambda$, $\mathrm{Aut}(1_\epsilon)$ as well as $[\underline{A}(k[\epsilon])]$ are canonically provided with vector space structures over k.

Remark 2.6(d). For any object R in $\hat{\underline{C}}_\Lambda$, the cofibered (discrete) groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$ is homogeneous.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. However, the homogeneity of $\underline{A}$ does not imply in general the homogeneity of $[\underline{A}]$. Indeed we have

Proposition 2.7. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$. The following statements are equivalent.

(i) $[\underline{A}]$ is homogeneous i.e. $[\underline{A}(R' \underset{R}{\times} S)] \longrightarrow [\underline{A}(R')] \underset{[\underline{A}(R)]}{\times} [\underline{A}(S)]$ is bijective for every small extension $R' \to R$.

(ii) For every small prolongation $a' \to a$ in $\underline{A}$, the map

$$\mathrm{Aut}_{\underline{A}(R')}(a') \longrightarrow \mathrm{Aut}_{\underline{A}(R)}(a)$$

<u>is surjective, where</u> $R' = P(a')$, $R = P(a)$.

Proof. (i) ⇒ (ii) Let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$ and let $a' \to a$ be a f-map in $\underline{A}$. Given $\tau \in \mathrm{Aut}_{\underline{A}(R)}(a)$, we consider $a' \times_a a'$ and $a' \times_a a'_\tau$ where $a'_\tau \longrightarrow a$ denotes the composite map $a' \longrightarrow a \xrightarrow{\tau} a$. Since $[A(R' \times_R R')] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(R')]$ is bijective, we conclude the existence of an isomorphism $a' \times_a a' \xrightarrow{u} a' \times_a a'_\tau$, and u induces, via the projection on the second factor, an isomorphism $a' \xrightarrow{\tau'} a'$ which is a lifting of τ .

(ii) ⇒ (i) Let a''_i $(i=1,2)$ be objects in $\underline{A}(R' \times_R S)$ which have the same image under $[\underline{A}(R' \times_R S)] \to [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$. We then obtain a diagram of solid arrows

$$
\begin{tikzcd}
 & & a''_2 \arrow[ddll] \arrow[ddrr] & & \\
 & & a''_1 \arrow[u, dashed] \arrow[dl] \arrow[dr] & & \\
a'_2 \arrow[ddrr] & a'_1 \arrow[l, "\sim"'] \arrow[dr] & & b_1 \arrow[r, "\tau"] \arrow[dl] & b_2 \arrow[ddll] \\
 & & a_1 \arrow[d, "\bar{\tau}"] & & \\
 & & a_2 & &
\end{tikzcd}
\quad \text{above} \quad
\begin{tikzcd}
 & R' \times_R S \arrow[dl] \arrow[dr] & \\
R' \arrow[dr] & & S \arrow[dl] \\
 & R &
\end{tikzcd}
$$

where $\bar{\tau} : a_1 \to a_2$ is the induced isomorphism from τ so that the square box on the lower right corner is commutative. If the condition (ii) is verified, one can always replace the given isomorphism $a'_1 \xrightarrow{\sim} a'_2$ so that the square box on the left corner is also commutative. Since $a''_i = a'_i \times_{a_i} b_i$ for $i = 1, 2$ by 2.6(a), we obtain the isomorphism $a''_1 \xrightarrow{\sim} a''_2$. This proves that $[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$ is injective, and hence is bijective since $\underline{A}$ is homogeneous by hypothesis.

Now let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$. We have a canonical map

$\tilde{f} : R'\times_R R' \longrightarrow k[\mathrm{Ker}\ f]$ given by $(r_1', r_2') \longrightarrow r_1'(0) + (r_1' - r_2')$, and $1\times\tilde{f} : R'\times_R R' \longrightarrow R'\times_k k[\mathrm{Ker}\ f]$ is an isomorphism. Given any two f-maps $a_i \to a$ $(i=1,2)$ in $\underline{A}$, $a_1 \times_a a_2$ is an object over $R'\times_R R'$ and in turn we may consider $\tau(a_1, a_2) = (\tilde{f})_*(a_1 \times_a a_2)$ which is an object in $\underline{A}(k[\mathrm{Ker}\ f])$. Since $1 \times \tilde{f} : R'\times_R R' \longrightarrow R'\times_k k[\mathrm{Ker}\ f]$ is an isomorphism, we have the isomorphism $a_1 \times_a a_2 \simeq a_1 \times \tau(a_1, a_2)$ compatible with the projections on the first factor. The statement 1.5(2) takes a more definite form for a homogeneous groupoid.

Lemma 2.8. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and $a_i \xrightarrow{\alpha_i} a$ $(i = 1, 2)$ f-maps in $\underline{A}$. Then

(1) There exists $\rho : a_1 \to a_2$ in $\underline{A}$ with $\alpha_2 \cdot \rho = \alpha_1$ if and only if there exists an arrow $a_1 \to \tau(a_1, a_2)$ in $\underline{A}$.

(2) There exists $\rho : a_1 \to a_2$ with $\alpha_2 \cdot \rho = \alpha_1$ such that $P(\rho) = id_{R'}$ if and only if $\tau(a_1, a_2) = 0$ in $[\underline{A}(k[\mathrm{Ker}\ f])]$.

Proof: We prove both at the same time. There exists $\rho : a_1 \to a_2$ in $\underline{A}$ with $\alpha_2 \cdot \rho = \alpha_1$ (such that $P(\rho) = id_{R'}$) $\Leftrightarrow$ there exists $h : a_1 \to a_1 \times_a a_2$ in $\underline{A}$ with $pr_1 \circ h = id_{a_1}$ (such that $P(h) =$ diagonal map) $\Leftrightarrow$ there exists $g : a_1 \to a_1 \times \tau(a_1, a_2)$ in $\underline{A}$ with $pr_1 \circ g = id_{a_1}$ (such that $P(g) : R' \to R'\times_k k[\mathrm{Ker}\ f]$ is given by $r' \to (r', r'(0))$ $\Leftrightarrow$ there exists an arrow $\gamma : a_1 \to \tau(a_1, a_2)$ in $\underline{A}$ (such that $P(\gamma) : R' \to k[\mathrm{Ker}\ f]$ is given by $r' \to r'(0)$). However, there exists an arrow $\gamma : a_1 \to \tau(a_1, a_2)$ in $\underline{A}$ such that $P(\gamma) : R' \to k[\mathrm{Ker}\ f]$ is factored through $R' \to k \to k[\mathrm{Ker}\ f]$ if and only if $\tau(a_1, a_2) = 0$.

Proposition 2.9. (a) Let $\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$ be functors of cofibered groupoids over $\underline{C}_\Lambda$. Assume that

(i) $\underline{B}$ is semi-homogeneous and $\underline{C}$ is homogeneous

(ii) $\psi(k[\epsilon]) : [\underline{B}(k[\epsilon])] \longrightarrow [\underline{C}(k[\epsilon])]$ is injective.

Then the composite functor $\psi \circ \varphi$ is smooth if and only if φ and ψ are both smooth.

(b) *Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. If $\underline{A}$ is semi-homogeneous and $\underline{B}$ is homogeneous, then $\hat{\varphi} : \hat{\underline{A}} \to \hat{\underline{B}}$ is smooth* (i.e., *for any surjective map* $b' \xrightarrow{\beta} \hat{\varphi}(a)$ *in* $\hat{\underline{B}}$, *there exists* $\alpha : a' \to a$ *in* $\hat{\underline{A}}$ *and a* $\hat{\underline{C}}_\Lambda$*-isomorphism* $\rho : \hat{\varphi}(a') \xrightarrow{\sim} b'$ *such that* $\beta \cdot \rho = \varphi(\alpha)$).

Proof: (a) Assume that the composite functor $\psi \circ \varphi$ is smooth. If φ is smooth, then so is ψ by 2.3(b), and therefore it suffices to show that φ is smooth. Let a small prolongation $b' \xrightarrow{\beta} \varphi(a)$ in $\underline{B}$ be given. Since the composite functor $\varphi \circ \psi$ is smooth, there exists an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ and a $\underline{C}_\Lambda$-isomorphism $\rho : \psi(b') \longrightarrow \psi\ \varphi(a')$ with a commutative diagram

$$\begin{tikzcd}
\psi(b') \arrow[rr, "\rho"] \arrow[dr, "\psi(\beta)"'] & & \psi\ \varphi(a') \arrow[dl, "\psi\ \varphi(\alpha)"] \\
& \psi\ \varphi(a) &
\end{tikzcd}$$

Since $\underline{B}$ is semi-homogeneous, we can choose a commutative diagram

$$\begin{tikzcd}
& b'' \arrow[dl] \arrow[dr] & \\
b' \arrow[dr, "\beta"'] & & \varphi(a') \arrow[dl, "\varphi(\alpha)"] \\
& \varphi(a) &
\end{tikzcd}
\quad \text{above} \quad
\begin{tikzcd}
& R' \times_R R' \arrow[dl] \arrow[dr] & \\
R' \arrow[dr] & & R' \arrow[dl] \\
& R &
\end{tikzcd}$$

where $R = P(a)$, $R' = P(b') = P(a')$. Since $\underline{C}$ is homogeneous, $\psi(b'') = \psi(b') \times_{\psi\ \varphi(a)} \psi\varphi(a')$ by 2.6(a), and it follows from 2.8(2) that $0 = \tau(\psi\ \varphi(a'), \psi(b')) = (\tilde{f})_*\psi(b'')$, and hence $(\tilde{f})_*(b'') = 0$ be the hypothesis (ii). It follows from 1.5(2) that there exists a map $\gamma : b' \longrightarrow \varphi(a')$ such that $\beta = \varphi(\alpha)\cdot\gamma$.

(b) Let a surjective map $\beta : b' \to \varphi(a)$ in $\hat{\underline{B}}$ be given. This means that we are given a sequence of commutative diagrams

$$\begin{tikzcd}
\cdots \arrow[r] & b_3' \arrow[r, "\beta_2'"] \arrow[d, "\beta_3"] & b_2' \arrow[r, "\beta_1'"] \arrow[d, "\beta_2"] & b_1' \arrow[d, "\beta_1"] \\
\cdots \arrow[r] & \varphi(a_3) \arrow[r, "\varphi(\alpha_2)"'] & \varphi(a_2) \arrow[r, "\varphi(\alpha_1)"'] & \varphi(a_1)
\end{tikzcd}$$

in $\underline{B}$, where α_v , β_v , β_v' are surjective for all v , such that $\beta = \varprojlim_v \beta_v$. To show that $\hat{\varphi}$ is smooth, it suffices to prove that a commutative diagram below of solid arrows in which α_n , β_n , β_{n+1} , β_n' are surjective can be completed into a commutative diagram including dotted arrows.

$$\begin{tikzcd}
\varphi(a_{n+1}') \arrow[rrr, dashed, "\varphi(\ )"] \arrow[dr, dashed, leftrightarrow] \arrow[ddr, dashed, "\varphi(\ )"'] & & & \varphi(a_n') \arrow[ddl, "\varphi(\alpha_n')"] \\
& b_{n+1}' \arrow[r, "\beta_n'"] \arrow[d, "\beta_{n+1}"] & b_n' \arrow[ur, leftrightarrow] \arrow[d, "\beta_n"] & \\
& \varphi(a_{n+1}) \arrow[r, "\varphi(\alpha_n)"'] & \varphi(a_n) &
\end{tikzcd}$$

Now $\underline{A}$ is, by hypotheses, semi-homogeneous and $\alpha_n : a_n' \to a_n$ is surjective, and therefore there exists a commutative diagram

$$\begin{tikzcd}
& a' \arrow[dl, "\gamma"'] \arrow[dr, "\gamma'"] & \\
a_{n+1} \arrow[dr, "\alpha_n"'] & & a_n' \arrow[dl, "\alpha_n'"] \\
& a_n &
\end{tikzcd} \quad \text{in } \underline{A} \text{ over } \quad
\begin{tikzcd}
& R_{n+1} \times_{R_n} R_n' \arrow[dl] \arrow[dr] & \\
R_{n+1} \arrow[dr] & & R_n' \arrow[dl] \\
& R_n &
\end{tikzcd} \quad \text{in } \underline{C}_\Lambda .$$

The hypothesis that $\underline{B}$ is homogeneous entails that

$$\begin{tikzcd}
& \varphi(a') \arrow[dl, "\varphi(\gamma)"'] \arrow[dr, "\varphi(\gamma')"] & \\
\varphi(a_{n+1}) \arrow[dr, "\varphi(\alpha_n)"'] & & \varphi(a_n') \arrow[dl, "\varphi(\alpha_n')"] \\
& \varphi(a_n) &
\end{tikzcd}$$

is a diagram of a fibre-product in $\underline{B}$, and therefore we obtain a unique map $\delta : b'_{n+1} \to \varphi(a')$ in $\underline{B}$ with an appropriate commutativity. Since β_{n+1} , β'_n are both surjective, so is δ and therefore, since φ is smooth, there exists $a'_{n+1} \xrightarrow{\alpha'} a'$ in $\underline{A}$ and a $\underline{C}_\Lambda$-isomorphism $\varphi(a'_{n+1}) \overset{\rho}{\underset{\sim}{\to}} b'_{n+1}$ such that $\delta \cdot \rho = \varphi(\alpha')$. We then obtain the commutative diagram

$$\begin{tikzcd}
\varphi(a'_{n+1}) \arrow[rrr, "\varphi(\gamma' \cdot \alpha')"] \arrow[dr, "\rho"] \arrow[ddr, "\varphi(\gamma \cdot \alpha')"'] & & & \varphi(a'_n) \arrow[dl, "\sim"'] \arrow[ddl, "\varphi(\alpha'_n)"] \\
& b'_{n+1} \arrow[r] \arrow[d] & b'_n \arrow[d] & \\
& \varphi(a_{n+1}) \arrow[r] & \varphi(a_n) &
\end{tikzcd}$$

Corollary 2.10. (a) Let $\varphi : \underline{A} \to \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. Let ξ be a versal formal object of $\underline{A}$, and η a minimally-versal formal object of $\underline{B}$. If $\underline{B}$ is homogeneous, then for any map $\beta : \eta \to \varphi(\xi)$ in $\hat{\underline{B}}$, $P(\xi)$ is a formal power-series ring over $P(\eta)$ through $P(\beta) : P(\eta) \to P(\varphi(\xi)) = P(\xi)$.

(b) Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let ξ , η be versal formal objects of $\underline{A}$. If η is minimal, then for any map $\alpha : \eta \to \xi$ in $\hat{\underline{A}}$, $P(\xi)$ is a formal power-series ring over $P(\eta)$ through $P(\alpha) : P(\eta) \to P(\xi)$.

Proof: (a) A map $\beta : \eta \to \varphi(\xi)$ induces a commutative diagram

$$\begin{tikzcd}
\underline{h}_{P(\xi)} \arrow[r, "\tilde{\beta}"] \arrow[d, "\tilde{\xi}"'] & \underline{h}_{P(\eta)} \arrow[d, "\tilde{\eta}"] \\
\underline{A} \arrow[r, "\varphi"'] & \underline{B}
\end{tikzcd}$$

Since $\tilde{\xi}$, φ are smooth, $\tilde{\eta} \cdot \tilde{\beta} = \varphi \cdot \tilde{\xi}$ is smooth. On the other hand,

$\eta(k[\epsilon]) : \underline{h}_{P(\eta)}(k[\epsilon]) \to [\underline{B}(k[\epsilon])]$ is an isomorphism and therefore β is smooth by 2.9(a), which means that $P(\xi)$ is a formal power-series ring over $P(\eta)$.

(b) follows from (a) by setting $\underline{A} = \underline{B}$.

Finally we consider the question of "representability" of a cofibered groupoid. Let $\underline{C}$ be a fixed category. Each object R in $\underline{C}$ defines a functor $h_R : \underline{C} \to (\mathrm{Ens})$ given by $h_R(S) = \mathrm{Hom}_{\underline{C}}(R,S)$, and in turn we obtain a functor $h : \underline{C}^o \to \underline{\mathrm{Hom}}(\underline{C},(\mathrm{Ens}))$ which is fully-faithful. A "cocartegory in $\underline{C}$ " is a diagram

$$h_{R_1} \underset{h_{R_0}}{\times} h_{R_1} \longrightarrow h_{R_1} \underset{\underset{s_2}{\longrightarrow}}{\overset{\overset{s_1}{\longrightarrow}}{\xleftarrow{\epsilon}}} h_{R_0} \quad \text{with } s_i \cdot \epsilon = \text{identity for } i = 1,2$$

(where $h_{R_1} \underset{h_{R_0}}{\times} h_{R_1} = (h_{R_1})_{s_1} \underset{h_{R_0}}{\times} {}_{s_2}(h_{R_1}))$ such that

(i) c is associative and is compatible with s_i for $i = 1,2$

(ii)
$$h_{R_1} \xrightarrow{\sim} h_{R_0} \underset{h_{R_0}}{\times} h_{R_1} \xrightarrow{\epsilon \times 1} h_{R_1} \underset{h_{R_0}}{\times} h_{R_1} \xrightarrow{c} h_{R_1}$$

$$h_{R_1} \xrightarrow{\sim} h_{R_1} \underset{h_{R_0}}{\times} h_{R_0} \xrightarrow{1 \times \epsilon} h_{R_1} \underset{h_{R_0}}{\times} h_{R_1} \xrightarrow{c} h_{R_1}$$

are both identities.

If $(R_0, R_1, s_1, s_2, \epsilon, c)$ is a cocategory in $\underline{C}$, then for an object S in $\underline{C}$, we can define a category $h_{(R_0, R_1, \ldots)}(S)$ by setting $\mathrm{ob}\, h_{(R_0, R_1,\ldots)}(S) = \mathrm{Hom}_{\underline{C}}(R_0, S)$, and $\mathrm{Fl}\, h_{(R_0, R_1,\ldots)}(S) = \mathrm{Hom}_{\underline{C}}(R_1,S)$ in which the composition rule of arrows is given by $c : h_{R_1} \underset{h_{R_0}}{\times} h_{R_1} \to h_{R_1}$. Then $h_{(R_0, R_1,\ldots)} : \underline{C} \to (\text{categories})$ is a functor and in turn defines a cofibered category $\underline{h}_{(R_0, R_1,\ldots)}$ over $\underline{C}$. For instance, any object R in $\underline{C}$ defines a discrete cocategory $\underline{h}_R = \underline{h}_{(R, R, s_1, s_2, \epsilon, c)}$ by setting s_1, s_2, ϵ, c to be all equal to the identity map. A homomorphism $(R_0, R_1,\ldots) \to (R_0', R_1',\ldots)$

of cocategories in $\underline{C}$ is a pair of maps $h_{R_0} \to h_{R_0'}$, $h_{R_1} \to h_{R'_1}$ in $\underline{C}$ such that the induced map

$$h_{(R_0,\ R_1,\dots)}(S) \to h_{(R_0',\ R_1',\dots)}(S)$$

is a functor for every object S in $\underline{C}$. For instance, each cocategory $(R_0, R_1,\dots)$ is provided with a homomorphism $\epsilon : \underline{h}_{R_0} \to \underline{h}_{(R_0,\ R_1,\dots)}$ of cocategories. A cogroupoid in $\underline{C}$ is by definition a cocategory $(R_0, R_1,\dots)$ in $\underline{C}$ such that $h_{(R_0,\ R_1,\dots)}(S)$ is a groupoid for every object S in $\underline{C}$. In other words, a cogroupoid in $\underline{C}$ is a cocategory $(R_0, R_1,\dots)$ in $\underline{C}$ such that $\underline{h}_{(R_0,\ R_1,\dots)}$ is a cofibered category in groupoids over $\underline{C}$.

In practice, a cocategory or cogroupoid occur in a following fashion: Let $P : \underline{F} \to \underline{C}$ be a cofibered category, and let ξ be a fixed object in $\underline{F}$. Set $R_0 = P(\xi)$ and let $(R_0, \underline{C})$ be the category of arrows in $\underline{C}$ with the source R_0 . Let $P : \underline{F}' \to (R_0, \underline{C})$ be the pull-back of $\underline{F}$ under the canonical functor $(R_0, \underline{C}) \to \underline{C}$, and choose a cocartesian section $\xi^{\#}$ of $\underline{F}'$ over $(R_0, \underline{C})$ such that $\xi^{\#}(id_{R_0}) = \xi$. (This simply means that we choose, for each arrow $R_0 \xrightarrow{f} S$ in $\underline{C}$, a cocartesian f-map $\xi \to \xi^{\#}(f)$). Define the functor $Hom_{\xi^{\#}} : (R_0 \amalg R_0, \underline{C}) \to (Ens)$ given by

$$X = (f_1, f_2) \longmapsto Hom_{\underline{F}(S)}(\xi^{\#}(f_1), \xi^{\#}(f_2)) \quad \text{where}$$

S = the target of f_1 (= the target of f_2). We note that this functor does not depend on the choice of $\xi^{\#}$ up to (canonical) natural equivalence. Henceforth, by abuse of notation, we shall simply write ξ for $\xi^{\#}$. If this functor is representable by an object $X_0 = (R_0 \overset{s_1}{\underset{s_2}{\rightrightarrows}} R_1)$ in $(R_0 \amalg R_0, \underline{C})$ i.e. if there exists R_0-map $\varphi : \xi(s_1) \to \xi(s_2)$ in $\underline{F}$ such that the induced map $\tilde{\varphi} : Hom_{(R_0 \amalg R_0,\ \underline{C})}(X_0, X) \to Hom_{\xi}(X)$ is bijective for all X in $(R_0 \amalg R_0, \underline{C})$, then the object X_0 defines a cocategory $(R_0, R_1,\dots)$ in $\underline{C}$ in a natural way. Indeed, a triple of arrows

$$R_0 \overset{u}{\underset{w}{\overset{\longrightarrow}{\underset{\longrightarrow}{\xrightarrow{v}}}}} S$$

in $\underline{C}$ defines a composition

$\mathrm{Hom}_{\underline{F}(S)}(\xi(v), \xi(w)) \times \mathrm{Hom}_{\underline{F}(S)}(\xi(u), \xi(v)) \to \mathrm{Hom}_{\underline{F}(S)}(\xi(u), \xi(w))$

and in turn yields the composition rule

$$h_{R_1}(S) \underset{h_{R_0}(S)}{\times} h_{R_1}(S) \to h_{R_1}(S)$$

depending functorially on S. Consequently the functor Hom_ξ defines the cocategory $(R_0, R_1, \ldots)$ in $\underline{C}$ and in turn defines a cofibered category $\underline{h}_{(R_0, R_1, \ldots)}$ in $\underline{C}$. By definition, $\mathrm{ob}\ \underline{h}_{(R_0, R_1, \ldots)} = \coprod_S \mathrm{Hom}_{\underline{C}}(R_0, S)$ (disjoint union for $S \in \mathrm{ob}(\underline{C})$), and an arrow $(R_0 \xrightarrow{f} S) \to (R_0 \xrightarrow{f'} S')$ is a pair $S \xrightarrow{h} S'$ and $R_1 \xrightarrow{u} S'$ in $\underline{C}$ such that $u \cdot s_1 = h \cdot f$ and $u \cdot s_2 = f'$, and the structural functor $P : \underline{h}_{(R_0, R_1, \ldots)} \to \underline{C}$ is given by $(R_0 \xrightarrow{f} S) \longmapsto S$.
Now the pair (ξ, φ) defines the functor

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \ldots)} \to \underline{F} \text{ over } \underline{C} \text{ by}$$

$$(R_0 \xrightarrow{f} S) \longmapsto \xi(f)$$

$$[(R_0 \xrightarrow{f} S) \xrightarrow{(h,u)} (R_0 \to S')] \longmapsto \text{the composit map } \xi(f) \xrightarrow{h_*} \xi(hf) \xrightarrow{u(\varphi)} \xi(f').$$

We note that, for each object S in $\underline{C}$, the functor $(\xi,\varphi)(S)$ $\underline{h}_{(R_0,R_1,\ldots)}(S) \to \underline{F}(S)$ is fully-faithful by virtue of the definition of $(R_0 \rightrightarrows R_1)$ coming from the representability of the functor Hom_ξ. Consequently the functor $(\xi,\varphi) : \underline{h}_{(R_0,R_1,\ldots)} \to \underline{F}$ is an $\underline{C}$-equivalence if and only if ξ is a quasi-initial object of $\underline{F}$ over $\underline{C}$ i.e., for every object a in $\underline{F}$ there exists a cocartesian arrow $\xi \to a$. Instead of this functor Hom_ξ one may also consider the functor

$$\mathrm{Isom}_\xi : (R_0 \amalg R_0, \underline{C}) \to (\mathrm{Ens})$$

given by $X = (f_1,f_2) \longmapsto \mathrm{Isom}_{\underline{F}(S)}(\xi(f_1),\xi(f_2))$ where S is the target of f_1 (= the target of f_2). If this functor is representable by an object $(R_0 \rightrightarrows R_1)$ in $(R_0 \amalg R_0, \underline{C})$, then it defines a cogroupoid $(R_0,R_1,\ldots)$ in $\underline{C}$ together with the functor $(\xi,\varphi) : \underline{h}_{(R_0,R_1,\ldots)} \to \underline{F}$, which is, when $\underline{F}$ is a cofibered groupoid, a $\underline{C}$-equivalence if and only if ξ is a quasi-initial object of $\underline{F}$ over $\underline{C}$.

Definition and proposition 2.11. A cogroupoid $(R_0,R_1,\ldots)$ in $\hat{\underline{C}}_\wedge$ is called smooth if any one of the following equivalent conditions holds:

(i) R_1 is a formal power-series ring over R_0 via s_1 (and s_2)

(ii) $\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0,R_1,\ldots)}$ is smooth over $\underline{C}_\wedge$.

A minimally smooth cogroupoid or a normalized smooth cogroupoid in $\hat{\underline{C}}_\wedge$ is a smooth cogroupoid $(R_0,R_1,\ldots)$ in $\hat{\underline{C}}_\wedge$ such that $\underline{h}_{(R_0,R_1,\ldots)}(k[\epsilon])$ is totally disconnected i.e., $s_1(k[\epsilon]) = s_2(k[\epsilon])$.

Proof: $\epsilon : \underline{h}_{R_0} \to \underline{h}_{(R_0,R_1,\ldots)}$ is smooth over $\underline{C}_\wedge \Leftrightarrow$ given any surjective map $(S',f') \xrightarrow{(h,u)} (S, f)$ in $\underline{h}_{(R_0,R_1,\ldots)}$, there exists a lifting f'' of f to S' together with an isomorphism $(S',f'') \xrightarrow{(1,u')} (S',f') \Leftrightarrow$ given $u \in h_{R_1}(S)$ and a lifting f' or $s_1(u)$ to S' , there exists a lifting u' of u to S' such that $s_1(u') = f'$, where $S' \to S$ is an arbitrary surjective map in $\underline{C}_\wedge \Leftrightarrow s_1 : \underline{h}_{R_1} \to \underline{h}_{R_0}$ is smooth over $\underline{C}_\wedge$.

Proposition 2.12. Let $(R_0,R_1,\ldots)$ be a cogroupoid in $\hat{\underline{C}}_\wedge$.

(a) If $(R_0,R_1,\ldots)$ is smooth, then $\underline{h}_{(R_0,R_1,\ldots)}$ is homogeneous.

(b) If $s_1(k[\epsilon]) = s_2(k[\epsilon])$, then the following statements are equivalent

(i) $(R_0,R_1,\ldots)$ is smooth

(ii) $\underline{h}_{(R_0,R_1,\ldots)}$ is homogeneous

(iii) $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous

(c) <u>If $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous, then it is equivalent to $\underline{h}_{(S_0,S_1,\ldots)}$ for some minimally smooth **cogroupoid**</u> S_* .

<u>Proof:</u> (a) Consider a diagram

$$\begin{tikzcd}
(S_1,f_1) \arrow[dr, "{(h_1,u_1)}"'] & & (S_2,f_2) \arrow[dl, "{(h_2,u_2)}"] \\
& (S,f) &
\end{tikzcd}$$

in $\underline{h}_{(R_0,R_1,\ldots)}$ where $h_2 : S_2 \longrightarrow S$ is surjective in $\underline{C}_\Lambda$. Since $s_1 : \underline{h}_{R_1} \longrightarrow \underline{h}_{R_0}$ is smooth by hypothesis, $u_1^{-1}\, u_2$ can be lifted to some $v \in h_{R_1}(S_2)$ such that $s_1(v) = f_2$. Since $h_2 \cdot s_2(v) = h_1 \cdot f_1$, $f_1 \times s_2(v) \in h_{R_0}(S_1 \underset{S}{\times} S_2)$ is defined and we obtain the commutative diagram

$$\begin{tikzcd}
& (S_1 \underset{S}{\times} S_2,\ f_1 \times s_2(v)) \arrow[dl, "\pi_1"'] \arrow[dr, "{(\pi_2,v^{-1})}"] & \\
(S_1,f_1) \arrow[dr, "{(h_1,u_1)}"'] & & (S_2,f_2) \arrow[dl, "{(h_2,u_2)}"] \\
& (S,f) &
\end{tikzcd}$$

We claim that the above diagram defines a fibre-product $(S_1,f_1) \underset{(S,f)}{\times} (S_2,f_2)$ in $\underline{h}_{(R_0,R_1,\ldots)}$. Indeed, for any object (T, g) in $\underline{h}_{(R_0,R_1,\ldots)}$ we have

$\mathrm{Hom}((T,g),(S_1 \underset{S}{\times} S_2, f_1 \times s_2(v))) = \{(h_1' \times h_2', w_1 \times w_2) \in h_T(S_1 \underset{S}{\times} S_2) \times h_{R_1}(S_1 \underset{S}{\times} S_2) \mid s_1(w_1 \times w_2) = (h_1' \times h_2')\, g ,$

$s_2(w_1 \times w_2) = f_1 \times s_2(v)\} = \{(h_1' \times h_2', w_1, w_2) \in h_T(S_1 \underset{S}{\times} S_2) \times h_{R_1}(S_1) \times h_{R_1}(S_2) \mid h_1 w_1 = h_2 w_2,$

$s_1(w_1) = h_1'\, g,\ s_1(w_2) = h_2'\, g,\ s_2(w_1) = f_1,\ s_2(w_2) = s_2(v)\} =$

$\{(h_1' \times h_2', w_1, w_2) \in h_T(S_1 \underset{S}{\times} S_2) \times h_{R_1}(S_1) \times h_{R_1}(S_2) \mid u_1^{-1}\, u_2\, h_2(v^{-1} w_2) = h_1\, w_1\ ,\ s_1(w_1) = h_1'\ g,$

$s_2(w_1)=f_1,\ s_1(v^{-1}w_2)=h_2'g,\ s_2(v^{-1}w_2)=f_2\} \xrightarrow{\sim} \{(h_1'\times h_2', w_1, w_2') \in h_T(S_1 \underset{S}{\times} S_2) \times h_{R_1}(S_1) \times h_{R_1}(S_2) \mid$

$u_2\, h_2\, w_2' = u_1\, h_1\, w_1,\ s_1(w_1) = h_1'g,\ s_2(w_1) = f_1,\ s_1(w_2') = h_2'g,\ s_2(w_2') = f_2\} =$

$\mathrm{Hom}((T,\ g),\ (S_1,\ f_1)) \underset{\mathrm{Hom}((T,g),\ (S,f))}{\times} \mathrm{Hom}((T,\ g),\ (S_2,\ f_2))$.

(b) It suffices to show (iii) $\Rightarrow$ (i) . Since $\underline{h}_{(R_0,R_1,\dots)}$ is semi-homogeneous and $\dim[\underline{h}_{(R_0,R_1,\dots)}(k[\epsilon])] = \dim h_{R_0}(k[\epsilon]) < \infty$, $\underline{h}_{(R_0,R_1,\dots)}$ admits a minimally versal formal object. On the other hand, $(R_0,\ 1_{R_0})$ is a quasi-initial minimal object of $\widehat{\underline{h}_{(R_0,R_1,\dots)}}$ since $h_{R_0}(k[\epsilon]) \to [\underline{h}_{(R_0,R_1,\dots)}(k[\epsilon])]$ is bijective, and therefore $(R_0,\ 1_{R_0})$ is a minimally versal formal object of $\underline{h}_{(R_0,R_1,\dots)}$ by 1.10(ii), which means that $\epsilon : h_{R_0} \longrightarrow \underline{h}_{(R_0,R_1,\dots)}$ is minimally smooth.

(c) Let $(S_0,\ \xi_0)$ be a minimally versal formal object for $\underline{h}_{(R_0,R_1,\dots)}$ (1.11). For any object $\xi : R_0 \to T$ in $\underline{h}_{(R_0,\ R_1,\dots)}(T)$, the functor $\mathbf{Isom}(\xi,\xi)$ from $C_\Lambda/T \otimes_\Lambda T$ to (sets) which to any R associates the sets of isomorphisms between the two images of ξ on $R : R_0 \xrightarrow{\xi} T \rightrightarrows T \otimes_\Lambda T \longrightarrow R$, is represented by $R_1 \otimes_{R_0 \hat{\otimes}_\Lambda R_0} (T \otimes_\Lambda T)$. By a passage to limit, one get that $\mathrm{Isom}(\xi_0,\ \xi_0)$ is pro-represented by some S_1 , and $\underline{h}_{(R_0,R_1,\dots)}$ is equivalent to $\underline{h}_{(S_0,S_1,\dots)}$. By (b), S_* is minimally smooth.

Definition 2.13. *A cofibered groupoid* $\underline{A}$ *over* $\underline{C}_\Lambda$ *is called (minimally) pro-representable if it is* $\underline{C}_\Lambda$*-equivalent to* $\underline{h}_{(R_0,R_1,\dots)}$ *for some (normalized) cogroupoid* $(R_0,R_1,\dots)$ *in* $\hat{\underline{C}}_\Lambda$.

Proposition 2.14. *If* $(R_0,\ R_1,\dots)$ *is a normalized cogroupoid in* $\hat{\underline{C}}_\Lambda$ *, then every homomorphism* $(\alpha_0,\alpha_1) : (R_0,R_1,\dots) \to (R_0,R_1,\dots)$ *which induces* $\underline{C}_\Lambda$*-equivalence* $\underline{h}_{(\alpha_0,\alpha_1)} : \underline{h}_{(R_0,R_1,\dots)} \longrightarrow \underline{h}_{(R_0,\ R_1,\dots)}$ *is an isomorphism. In* particular, for any cofibered groupoid A over $\underline{C}_\Lambda$, a normalized prorepresentation, if it exists, is unique up to an isomorphism.

Proof: Consider the commutative diagram

$$\begin{tikzcd} \underline{h}(R_0,R_1,\ldots) \arrow[r, "\underline{h}(\alpha_0,\alpha_1)"] & \underline{h}(R_0,R_1,\ldots) \\ \underline{h}_{R_0} \arrow[u, "\varepsilon"] \arrow[r, "\underline{h}_{\alpha_0}"'] & \underline{h}_{R_0} \arrow[u, "\varepsilon"'] \end{tikzcd}$$

where $\underline{h}_{(\alpha_0,\alpha_1)}$ is a $\underline{C}_\Lambda$-equivalence. Since $\underline{h}_{(R_0,R_1,\ldots)}(k[\varepsilon])$ is totally disconnected, $[\underline{h}_{(\alpha_0,\alpha_1)}(k[\varepsilon])$ and $[\varepsilon(k[\varepsilon])]$ are both bijections, and hence so is $\underline{h}_{\alpha_0}(k[\varepsilon])$.

Consequently, $\alpha_0 : R_0 \to R_0$ is a surjective endomorphism and hence is an automorphism. It then follows that α_1 is also an automorphism of R_1 .

Lemma 2.15. Let A, B be discrete groupoids over $\underline{C}_\Lambda$ and $\varphi : A \to B$ a minimally smooth functor. If A, B are homogeneous, then φ is a $\underline{C}_\Lambda$-equivalence.

Proof: We must show that $\varphi(S) : A(S) \longrightarrow B(S)$ is bijective for all $S \in \mathrm{ob}(\underline{C}_\Lambda)$. We note that $\varphi(S)$ is surjective for all $S \in \mathrm{ob}(\underline{C}_\Lambda)$ and bijective for $S = k[\varepsilon]$. We argue by induction on the length of S . Let $S' \xrightarrow{f} S$ be a small extension in $\underline{C}_\Lambda$. Since A, B are homogeneous discrete groupoids, the canonical isomorphism $1 \times \tilde{f} : S' \times_S S' \longrightarrow S' \times_k k[\mathrm{Ker}\ f]$ induces the commutative diagram

$$\begin{tikzcd} A(S') \times_{A(S)} A(S') \arrow[r, "\simeq"] \arrow[d] & A(S') \times A(k[\mathrm{Ker}\, f]) \arrow[d] \\ B(S') \times_{B(S)} B(S') \arrow[r, "\simeq"] & B(S') \times B(k[\mathrm{Ker}\, f]) \end{tikzcd}$$

Therefore $A(S')$, $B(S')$ are principal homogeneous sets over $A(S)$, $B(S)$ with the structural group $A(k[\mathrm{Ker}\, f]) \xrightarrow{\simeq} B(k[\mathrm{Ker}\, f])$. Consequently if $\varphi(S)$ is bijective, then $\varphi(S')$ is injective and hence is bijective.

Corollary 2.16. Let $\underline{A}$ be a homogeneous groupoid over $\underline{C}_\Lambda$ such that $\mathrm{Aut}_{\underline{A}(k[\varepsilon])}(1_\varepsilon)$ is of finite-dimensional. Then for any $\xi \in \mathrm{ob}(\hat{\underline{A}})$, the functor $\mathrm{Isom}_\xi : \underline{C}_{R\hat{\otimes}_\Lambda R} \longrightarrow (\mathrm{Ens})$ is prorepresentable, where $R = P(\xi)$.

Proof: Assume that Isom_ξ is homogeneous. We then have, by 1.11, a minimally smooth functor $\phi : \underline{h}_{R_1} \to \underline{\mathrm{Isom}}_\xi$, which is a $\underline{C}_{R\hat{\otimes}R}$-equivalence by 2.15 i.e., Isom_ξ is prorepresentable. Therefore it suffices to show that Isom_ξ is homogeneous. Consider a diagram

$$\begin{tikzcd} T_1 \arrow[dr] & & T_2 \arrow[dl] \\ & T & \end{tikzcd}$$

in $\underline{C}_{R\hat{\otimes}_\Lambda R}$ where $T_2 \longrightarrow T$ is surjective. Since $\underline{A}$ is homogeneous, $\xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2)$ exists in $\underline{A}$, and we have a canonical isomorphism

$\xi_i(T_1 \times_T T_2) \simeq \xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2)$ for $i = 1,2$ (cf. 2.6(a)). It follows that

$\mathrm{Isom}_\xi(T_1 \times_T T_2) \longrightarrow \mathrm{Isom}_\xi(T_1) \times_{\mathrm{Isom}_\xi(T)} \mathrm{Isom}_\xi(T_2)$ is an isomorphism.

Theorem 2.17. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then it is prorepresentable by a (normalized) smooth cogroupoid if and only if

(1) $\underline{A}$ is homogeneous

(2) $[A(k[\varepsilon])]$ and $[\mathrm{Aut}(1_\varepsilon)]$ are both finite-dimensional vector spaces over k .

Proof: Assume that $\underline{A}$ is prorepresentable by a smooth cogroupoid i.e., $\underline{A}$ is $\underline{C}_\Lambda$-equivalent to $\underline{h} = \underline{h}_{(R_0,R_1,\dots)}$ for some smooth cogroupoid $(R_0,R_1,\dots)$ in $\hat{\underline{C}}_\Lambda$. Since $\underline{h}$ is homogeneous by 2.12(a), and

$$[\underline{h}(k[\varepsilon])] = \mathrm{Coker}\,(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1,k[\varepsilon] \overset{s_1}{\underset{s_2}{\rightrightarrows}} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0,k[\varepsilon])) \text{ as well as}$$

$$\mathrm{Aut}_{\underline{h}}(1_\varepsilon) = \mathrm{Ker}(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1,k[\varepsilon]) \xrightarrow{s_1} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0,k[\varepsilon]))$$

$$= \mathrm{Ker}(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1,k[\varepsilon]) \xrightarrow{s_2} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0,k[\varepsilon]))$$ are both finite-dimensional vector spaces over k , we must have the same properties on $\underline{A}$. Conversely assume (1) and (2) on $\underline{A}$. We claim that $\underline{A}$ is prorepresentable by a normalized smooth cogroupoid. Since $\underline{A}$ is homogeneous and $[\underline{A}(k[\varepsilon])]$ is a finite-dimensional vector space over k , $\underline{A}$ admits a minimally versal formal object ξ . Set $P(\xi) = R_0$ and consider the functor $\mathrm{Isom}_\xi : \underline{C}_{R_0 \hat{\otimes}_\Lambda R_0} \to (\mathrm{Ens})$ given by

$$T \longmapsto \mathrm{Isom}_{\underline{A}(T)}(\xi_1(T),\xi_2(T))$$

where $\xi_i(T) = \xi(f_i)$ and $f_i : R_0 \to T$ denotes the composit maps

$$R_0 \overset{s_1}{\underset{s_2}{\rightrightarrows}} R_0 \hat{\otimes}_\Lambda R_0 \longrightarrow T \ .$$

This functor is pro-representable by 2.16, i.e. there exists $R_1 \in \mathrm{ob}(\widehat{\underline{C}_{R_0 \hat{\otimes}_\Lambda R_0}})$ and an isomorphism $\xi_1(R_1) \xrightarrow{\varphi} \xi_2(R_1)$ such that $\bar{\varphi} : h_{R_1} \to \mathrm{Isom}_\xi$ is an isomorphism over $\underline{C}_{R_0 \hat{\otimes}_\Lambda R_0}$. Then $(R_0,R_1,\dots)$ defines a cogroupoid in $\hat{\underline{C}}_\Lambda$ and the functor $(\xi,\varphi) : \underline{h}_{(R_0,R_1,\dots)} \to \underline{A}$ is a $\underline{C}_\Lambda$-equivalence. Since ξ is a minimally versal formal object of $\underline{A}$, the composit functor $\underline{h}_{(R_0,R_0,\dots)} \xrightarrow{\varepsilon} \underline{h}_{(R_0,R_1,\dots)} \xrightarrow{(\xi,\varphi)} \underline{A}$ is, by definition, minimally smooth and therefore so is ε since (ξ,φ) is a $\underline{C}_\Lambda$-equivalence i.e., $(R_0,R_1,\dots)$ is a normalized smooth cogroupoid in $\hat{\underline{C}}_\Lambda$.

Corollary 2.18. *A functor* $F : \underline{C}_\Lambda \longrightarrow (Ens)$ *is pro-representable if and only if it is homogeneous and* $\dim F(k[\epsilon]) < \infty$.

Corollary 2.19. *Every smooth cogroupoid in* $\hat{\underline{C}}_\Lambda$ *can be normalized up to isomorphism.*

Given a cogroupoid $(R_0,R_1,\dots)$ in $\hat{\underline{C}}_\Lambda$, we set $\bar{R}_1 = R_1/J$ where J is the ideal in R_1 generated by $\{s_1(r) - s_2(r) | r \in R_0\}$. Then $(R_0,\bar{R}_1,\dots)$ defines a formal group scheme over R_0 with respect to the induced composition rule. Now let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ prorepresented by a normalized smooth cogroupoid $(R_0,R_1,\dots)$ via (ξ,φ) . Let $\underline{C}_k \hookrightarrow \underline{C}_{R_0} \hookrightarrow \underline{C}_{R_0 \hat{\otimes}_\Lambda R_0}$ be the embeddings corresponding to the surjective maps $R_0 \hat{\otimes}_\Lambda R_0 \longrightarrow R_0 \longrightarrow k$, and let Aut_ξ , Aut^0 be the restriction functors of Isom_ξ on $\underline{C}_{R_0}$, $\underline{C}_k$ respectively. In other words, $\mathrm{Aut}_\xi(S)$ = the automorphism group of $\xi(S)$ for every $S \in ob(\underline{C}_{R_0})$, and $\mathrm{Aut}^0(S)$ = the automorphism group of 1_S for every $S \in ob(\underline{C}_k)$, where 1_S = the direct image of $1 \in \underline{A}(k)$ under the map $k \to S$. Since R_1 prorepresents the functor Isom_ξ on $\underline{C}_{R_0 \hat{\otimes}_\Lambda R_0}$, it follows that $\bar{R}_1$ prorepresents the functor Aut_ξ on $\underline{C}_{R_0}$, and $k \otimes_{R_0} \bar{R}_1$ prorepresents Aut^0 on $\underline{C}_k$. Though $\underline{A}$ is prorepresented by a smooth cogroupoid $(R_0,R_1,\dots)$, neither $(R_0,\bar{R}_1,\dots)$ nor $(k, k \otimes_{R_0} \bar{R}_1,\dots)$ are in general smooth. Indeed we have

Corollary 2.20. *Let* $\underline{A}$ *be a homogeneous cogroupoid over* $\underline{C}_\Lambda$ *prorepresented by a normalized smooth cogroupoid* $(R_0,R_1,\dots)$ *via* (ξ,φ) .

(a) $(R_0,\bar{R}_1,\dots)$ *and* $(k, k \otimes_{R_0} \bar{R}_1,\dots)$ *prorepresent the functors* Aut_ξ *and* Aut^0 *respectively.*

(b) *The following statements are equivalent*

(i) *the formal group scheme* $\underline{h}_{(R_0,\bar{R}_1,\dots)}$ *is formally smooth over* R_0 ;

(ii) *for every surjective arrow* $a' \to a$ *in* $\underline{A}$, $\mathrm{Aut}_{P(a')}(a') \longrightarrow \mathrm{Aut}_{P(a)}(a)$ *is surjective*,

(iii) R_0 *prorepresents the functor* $[\underline{A}]$ *over* $\underline{C}_\Lambda$.

Proof: We have already noted (a). As for (b), (i) $\Leftrightarrow$ (ii) is clear whereas (ii) $\Leftrightarrow$ (iii) follows from 2.7 and 2.15, since $\tilde{\xi} : h_{R_0} \longrightarrow [\underline{A}]$ is minimally smooth.

Remark 2.21. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and let $\underline{A}|\underline{C}_k$ be the pull-back of $p : \underline{A} \longrightarrow \underline{C}_\Lambda$ under $\underline{C}_k \hookrightarrow \underline{C}_\Lambda$. Then it is trivial to see that if $\underline{A}$ is prorepresented by $(R_0,R_1,\ldots)$ via (ξ,φ), then $\underline{A}|\underline{C}_k$ is prorepresented by $(k \underset{\Lambda}{\otimes} R_0 , k \underset{\Lambda}{\otimes} R_1,\ldots)$ via $(k \underset{\Lambda}{\otimes} \xi , k \underset{\Lambda}{\otimes} \varphi)$ where $k \underset{\Lambda}{\otimes} \xi$ = its direct image of ξ under the map $R_0 \longrightarrow k \underset{\Lambda}{\otimes} R_0$, $k \underset{\Lambda}{\otimes} \varphi$ = the image of φ under $\mathrm{Isom}_{\underline{A}(R_0)}(\xi(s_1), \xi(s_2)) \longrightarrow \mathrm{Isom}_{\underline{A}(k \underset{\Lambda}{\otimes} R_1)}(k \underset{\Lambda}{\otimes} \xi(s_1), k \underset{\Lambda}{\otimes} \xi(s_2))$. If $(R_0,R_1,\ldots)$ is normalized, then so is $(k \underset{\Lambda}{\otimes} R_0, k \underset{\Lambda}{\otimes} R_1,\ldots)$, and the relative dimension of R_1 over R_0 is equal to the relative dimension of $k \underset{\Lambda}{\otimes} R_1$ over $k \underset{\Lambda}{\otimes} R_0$ since R_1 is smooth over R_0.

Remark 2.22. Let $\underline{A}$ be prorepresented by smooth cogroupoid $(R_0,R_1,\ldots)$ and $(R_0', R_1',\ldots)$ where we assume that $(R_0,R_1,\ldots)$ is normalized. We then obtain a homomorphism $(i_0,i_1) : (R_0,R_1,\ldots) \longrightarrow (R_0', R_1',\ldots)$. Now R_0' is a formal power-series ring over R_0 by 2.10(b), and the homomorphism (i_0,i_1) induces an isomorphism $R_0' \underset{R_0}{\otimes} (R_0,R_1,\ldots) \xrightarrow{\sim} (R_0', R_1',\ldots)$.

3. The coherent sheaves $\underline{D}_X^i$

One of the nice functors which is usually not pro-representable but nevertheless admits a formal versal object is the deformation functor. To this end we recall a few basic facts on commutative ring extensions and its obstruction theory. This section contains nothing new (possibly except in its systematic approach via Grothendieck cohomology on site). A purpose of this section is to make this expose as much self-contained. (Indeed we prove here more than we need later). The basic facts are stated in theorem 3.12. Throughout rings are meant to be commutative rings with unit.

Given a ring R we denote by $\underline{C}_R$ the category of R-algebras. For each object A in $\underline{C}_R$ we set

Cov(A) = the set of all surjective arrows $\varphi : A' \to A$ in $\underline{C}_R$ with target A.

It clearly defines a pretopology and in turn we obtain a site $\underline{C}_R$. For each object A in $\underline{C}_R$, we shall denote by $\underline{C}_{A|R}$ the category of arrows in $\underline{C}_R$ with target A. We may and shall provide the category $\underline{C}_{A|R}$ with the topology induced from the forgetful functor $j_A : \underline{C}_{A|R} \to \underline{C}_R$. We note that the category $\underline{C}_{A|R}$ has a final object $(1_A : A \to A)$ and an initial object $(i_A : R \to A)$, and admits a fibre-product as well as coproduct. Henceforth we shall denote the object $(B \to A)$ in $\underline{C}_{A|R}$, by abuse of notation, simply by B in case when there is no possible confusion. Given an abelian category $\underline{C}$, a $\underline{C}$-valued sheaf on $\underline{C}_{A|R}$ is, by definition, a functor $F : \underline{C}^{\circ}_{A|R} \to \underline{C}$ such that, for any surjective arrow $B' \to B$ in $\underline{C}_{A|R}$,

$$0 \to F(B) \to F(B') \rightrightarrows F(B' \underset{B}{\times} B')$$

is exact. The additive presheaves are those which preserve coproduct i.e., $F(B_1 \underset{R}{\otimes} B_2) = F(B_1) \oplus F(B_2)$. For example, an A-module M defines the functor

$$\mathrm{Der}_R(-,M) : \underline{C}^{\circ}_{A|R} \to (A\text{-mod})$$

given by $B \to \mathrm{Der}_R(B,M)$ = R-linear derivations of B with values in M regarded as B-module via the structural map $B \to A$, and it is clearly an additive sheaf on $\underline{C}^{\circ}_{A|R}$, where (A-mod) denotes the category of A-modules. A presheaf F on $\underline{C}_{A|R}$ with values in (A-mod) is called "finite type" if $F(B)$ is an A-module of finite type whenever B is an R-algebra of finite type. For example, if A is noetherian and M is an A-module of finite type, then $\mathrm{Der}_R(-,M)$ is an additive sheaf of finite type on the site $\underline{C}_{A|R}$.

A few more words on notations and basic facts in homological algebra:

3.1(a) Throughout the rest of this section, we shall denote by $\underline{C}^{\sim}_{A|R}$ (or $\underline{C}^{\vee}_{A|R}$) the category of sheaves (or presheaves) with values in (A-mod). We have the (left exact) injection functor $i_{A|R} : \underline{C}^{\sim}_{A|R} \to \underline{C}^{\vee}_{A|R}$, and the associated-sheaf (exact) functor $G_{A|R}: \underline{C}^{\vee}_{A|R} \to \underline{C}^{\sim}_{A|R}$ which is adjoint to the functor $i_{A|R}$. An object $B \in ob(\underline{C}_{A|R})$ induces a continuous functor $j : \underline{C}_{B|R} \to \underline{C}_{A|R}$ and in turn a commutative diagram

(*)
$$\begin{tikzcd}
\underline{C}^{\sim}_{A|R} \arrow[r, "j^{\sim}"] \arrow[d, "i_{A|R}"'] & \underline{C}^{\sim}_{B|R} \arrow[d] \\
\underline{C}^{\vee}_{A|R} \arrow[r, "j^{\vee}"'] & \underline{C}^{\vee}_{B|R}
\end{tikzcd}$$

where the restriction functor $\tilde{j}$ (as well as $j^{\vee}$) is an exact functor, since the site $\underline{C}_{B|R}$ is nothing but the category of arrows in $\underline{C}_{A|R}$ having target B , with the induced topology. A ring homomorphism $S \to R$ induces a continuous functor $\beta : \underline{C}_{A|R} \to \underline{C}_{A|S}$, and in turn a commutative diagram

(**)
$$\begin{tikzcd}
\underline{C}^{\sim}_{A|S} \arrow[r, "\tilde{\beta}"] \arrow[d, "i_{A|S}"'] & \underline{C}^{\sim}_{A|R} \arrow[d] \\
\underline{C}^{\vee}_{A|S} \arrow[r, "\beta^{\vee}"'] & \underline{C}^{\vee}_{A|R}
\end{tikzcd}$$

Note that $\tilde{\beta}$ is not necessarily an exact functor.

3.1(b) The right derived functor of the injection functor $i_{A|R}: \underline{C}^{\sim}_{A|R} \to \underline{C}^{\vee}_{A|R}$ will be denoted by $\underline{H}^{\circ}_{A|R}$. For each sheaf F on $\underline{C}_{A|R}$, we set

$$H^{\circ}(A|R,F) = [H^{\circ}{}_{A|R}(F)](A)$$

For any $B' \to B$ in $Cov(B)$, let us define the functor $H^{\circ}((B' \to B),-) : \underline{C}^{\vee}_{A|R} \to (A\text{-mod})$ by $H^{\circ}((B' \to B),F) = Ker(F(B') \rightrightarrows F(B' \times_B B'))$. The right derived functors of $H^{\circ}((B' \to B),-)$ will be denoted by $H^{\circ}((B' \to B),-)$. One knows that the $H^{\circ}((B' \to B),F)$ are the cohomology groups of the (Čech) semi-simplicial module

$$F(B') \rightrightarrows F(B' \times_B B') \rightrightarrows F(B' \times_B B' \times_B B') \rightrightarrows \cdots$$

The right derived functor of $\underline{\check{H}}^0_{A|R} : \underline{C}_{A|R} \to \underline{C}_{A|R}$ will be denoted by $\underline{\check{H}}^{\bullet}_{A|R}$, where

$$[\underline{\check{H}}^0_{A|R}(F)](B) = \lim_{\{B' \to B\} \in \mathrm{Cov}(B)} H^0(\{B' \to B\}, F) \quad .$$

For each presheaf F on $\underline{C}_{A|R}$, we set

$$\check{H}^{\cdot}(A|R,F) = [\check{H}^{\cdot}_{A|R}(F)](A) = \lim_{\{A' \to A\} \in \mathrm{Cov}(A)} H^{\cdot}(\{A' \to A\}, F)$$

for any sheaf F on $\underline{C}_{A|R}$, we have the usual spectral sequence

$$\check{H}^p(A|R, \underline{H}^q_{A|R}(F)) \Rightarrow H^{p+q}(A|R,F)$$

where we note that $\check{H}^0(A|R, \underline{H}^q(F)) = 0$ for all $q > 0$.

In particular

$$\begin{cases} \check{H}^1(A|R,F) = H^1(A|R,F) \\ 0 \to \check{H}^2(A|R,F) \to H^2(A|R,F) \to \check{H}^1(A|R, \underline{H}^1_{A\ R}(F)) \to \check{H}^3(A|R,F) \quad \text{is exact.} \end{cases}$$

3.1(c) If B is an object in $\underline{C}_{A|R}$, the commutative diagram (*) in 2.1(a) together with the fact that $\tilde{j}$ as well as $j^{\vee}$ are exact functors yields the canonical isomorphism

$$H^{\cdot}(B|R, \tilde{j}F) = [\underline{H}^{\cdot}_{A|R}(F)](B)$$

On the other hand, a ring homomorphism $S \to R$ induces the commutative diagram (**) in 2.1(a). Now $\tilde{\beta}$ is not necessarily an exact functor. However, β preserves a fibre-product, and thus we obtain a spectral sequence

$$\underline{H}^p_{A|R}(R^q\tilde{\beta}F) \Rightarrow \check{\beta}\underline{H}^{p+q}_{A|S}(F) \qquad \text{i.e.,} \qquad H^p(A|R, R^q\tilde{\beta}F) \Rightarrow H^{p+q}(A|S,F)$$

Since we have the canonical isomorphism $\tilde{\beta} = a_{A|R} \circ \check{\beta} \circ i_{A|S}$ and $\beta^{\#} = a_{A|R} \circ \check{\beta}$ is an exact functor, we have $R^{\cdot}\widetilde{\beta}F = \beta^{\#}\underline{H}^{\cdot}_{A|S}(F)$. Thus the spectral sequence above can now be written as

$$H^p(A|R,\beta^{\#}\underline{H}^q_{A|S}(F)) \Rightarrow H^{p+q}(A|S,F) \quad .$$

Lemma 3.2. Given $B \in ob(\underline{C}_{A|R})$, choose a surjective arrow $P \to B$ in $\underline{C}_{B|R}$, where P is a polynomial algebra over R . Then for any $F \in ob(\underline{C}^{\vee}_{A|R})$, the natural map $H^{\cdot}(\{P \to B\},F) \to \underline{\check{H}}^{\cdot}(F)\ (B)$ is an isomorphism.

Proof: It suffice to see that the natural map

$$H^0(\{P \to B\},F) \to \underline{\check{H}}^0(F)\ (B) = \varinjlim_{\{B' \to B\} \in Cov(B)} H^0(\{B' \to B\},F)$$

is an isomorphism, which is clear since, for any $\{B' \to B\} \in Cov(B)$, there exists an arrow $\{P \to B\} \to \{B' \to B\}$.

Corollary 3.3.

(a) Assume that A is noetherian. If $F \in ob(\widetilde{\underline{C}}_{A|R})$ is of finite type, then $\underline{H}^n(F)$ is of finite type for all n .

(b) For any polynomial algebra P over R , we have $H^n(P|R,F) = 0$ for all $n > 0$ and for all $F \in ob(\widetilde{\underline{C}}_{P|R})$.

(c) A diagram $0 \to F' \to F \to F'' \to 0$ in $\widetilde{\underline{C}}_{A|R}$ is exact if and only if $0 \to F'(P) \to F(P) \to F''(P) \to 0$ is exact for every $P \in ob(\underline{C}_{A|R})$ which is a polynomial algebra over R .

Proof:

(a) In view of the spectral sequence

$$\check{H}^p(B|R,\underline{H}^q(F)) \Rightarrow H^{p+q}(B|R,F)$$

together with the fact that $\check{H}^0(\underline{H}^q(F)) = 0$ for all $q > 0$, it suffices to show that $\underline{H}^n(G)$ is of finite type for all n and for all presheaf G of finite type. However $\check{H}^{\cdot}(B|R,G) = H^{\cdot}(\{P \to B\},G)$, where P is a polynomial algebra over R and $\{P \to B\} \in Cov(B)$, and furthermore $H^{\cdot}(\{P \to B\},G)$ is simply the cohomology gotten from the complex

$$\cdots\cdots\cdots\cdots G(P \underset{B}{\times} P \underset{B}{\times} P) \leftleftarrows G(P \underset{B}{\times} P) \leftleftarrows G(P) \ .$$

If B is an R-algebra of finite type, then we may take P to be of finite type over R so that $P\underset{B}{\times} \cdots \underset{B}{\times}P$ are all R-algebras of finite type. If G is of finite type, then $G(P\underset{B}{\times} \cdots \underset{B}{\times}P)$ are all A-modules of finite type, so that $H^{\cdot}(\{P \to B\},G)$ is an A-module of finite type since A is noetherian.

(b) Let P be a polynomial algebra over R. If $B \to P$ is any surjective R-algebra map, then it amits a section and hence $H^i(\{B \to P\},G) = 0$ for all $i > 0$, so that $\check{H}^i(P|R, G) = 0$ for all $i > 0$ and for all presheaves G on $\underline{C}_{P|R}$. It follows that the spectral sequence $\check{H}^p(P|R, \underline{H}^q_{P|R}(F)) \Rightarrow H^{p+q}(P|R, F)$ degenerates and hence $\check{H}^0(P|R, \underline{H}^n(F)) \simeq H^n(P|R, F)$ for all n. However $\check{H}^0(P|R, \underline{H}^n(F)) = 0$ for all $n > 0$ whenever F is a sheaf, and hence $H^n(P|R, F) = 0$ for all $n > 0$ and for all $F \in ob(\underline{\tilde{C}}_{P|R})$.

(c) If $0 \to F' \to F \to F'' \to 0$ is an exact sequence in $\underline{\tilde{C}}_{A|R}$, then

$$0 \to F'(B) \to F(B) \to F''(B) \to H'(B|R, F') \to H'(B|R, F) \to \cdots$$

is exact. If $P \in ob(\underline{C}_{A|R})$ is a polynomial algebra over R, then $H^1(P|R, F') = 0$ by 3.3(b) and hence

$$0 \to F' \to F \to F'' \to 0$$

is exact. As for the other direction, let $\underline{P}_{A|R}$ be the full subcategory of $\underline{C}_{A|R}$ in which

$$ob(\underline{P}_{A|R}) = \{P \in ob(\underline{C}_{A|R}) \mid P \text{ is a polynomial algebra over } R\} ,$$

provided with the induced topology, and let $p : \underline{P}_{A|R} \to \underline{C}_{A|R}$ be the inclusion functor. Since every object in $\underline{C}_{A|R}$ can be covered by an object in $\underline{P}_{A|R}$, it follows from a comparison lemma (   ) that $\tilde{p} : \underline{\tilde{C}}_{A|R} \to \underline{\tilde{P}}_{A|R}$ is an equivalence of categories. Consequently, if $0 \to F'(P) \to F(P) \to F''(P) \to 0$ is exact for every $P \in ob(\underline{P}_{A|R})$, then $0 \to \tilde{p}F' \to \tilde{p}F \to \tilde{p}F'' \to 0$ is exact in $\underline{\tilde{P}}_{A|R}$ and hence $0 \to F' \to F \to F'' \to 0$ is exact in $\underline{\tilde{C}}_{A|R}$.

Remark 3.4. Every surjective arrow in $\underline{P}_{A|R}$ admits a section. It follows immediately that every presheaf on $\underline{P}_{A|R}$ is actually a sheaf i.e., $\underline{\tilde{P}}_{A|R} = \underline{P}_{A|R}$. Consequently we find that the restriction functor $\underline{\tilde{C}}_{A|R} \to \underline{\check{P}}_{A|R}$ is an equivalence of categories.

Lemma 3.5. *Consider a diagram of R-algebras*

$$\begin{tikzcd} & C \arrow[d] \\ B' \arrow[r, "\text{surjective}"'] & B \end{tikzcd}$$

and let S *be an R-algebra.*

(i) *If* $\operatorname{Tor}_1^R(B,S) = 0$, *then the canonical map*

$$(B' \underset{B}{\times} C) \underset{R}{\otimes} S \to (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S)$$

is an isomorphism.

(ii) *Assume that* B' *is R-flat. If* $\operatorname{Tor}_i^R(C,S) = 0$ *for all* $0 < i < n$ *and* $\operatorname{Tor}_i^R(B,S) = 0$ *for all* $0 < i < n$, *then* $\operatorname{Tor}_i^R(B' \underset{B}{\times} C, S) = 0$ *for all* $0 < i < n$.

Proof: Let $0 \to J \to B' \to B \to 0$ be exact.

(i) If $\operatorname{Tor}_i^R(B, S) = 0$, then

$$0 \to J \underset{R}{\otimes} S \to B' \underset{R}{\otimes} S \to B \underset{R}{\otimes} S \to 0$$

is exact so that

$$0 \to J \underset{R}{\otimes} S \to (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S) \to C \underset{R}{\otimes} S \to 0$$

is exact. We thus obtain an exact commutative diagram

$$\begin{tikzcd} & J \underset{R}{\otimes} S \arrow[r] \arrow[d, equal] & (B' \underset{B}{\times} C) \underset{R}{\otimes} S \arrow[r] \arrow[d] & C \underset{R}{\otimes} S \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & J \underset{R}{\otimes} S \arrow[r] & (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S) \arrow[r] & C \underset{R}{\otimes} S \arrow[r] & 0 \end{tikzcd}$$

and therefore $(B'\otimes_B C)\otimes_R S \to (B'\otimes_R S)\times_{(B\otimes_R S)}(C\otimes_R S)$ is an isomorphism.

(ii) Since B' is R-flat and $\mathrm{Tor}_i^R(B, S) = 0$ for all $0 < i \leq n$ by hypothesis, we have $\mathrm{Tor}_i^R(J, S) = 0$ for all $0 < i < n$. Then the exact sequence $0 \to J \to B'\times_B C \to C \to 0$ together with the hypothesis that $\mathrm{Tor}_i^R(C, S) = 0$ for all $0 < i < n$ entails that $\mathrm{Tor}_i^R(B'\times_B C, S) = 0$ for all $0 < i < n$.

Corollary 3.6.

(a) Let A, S be R-algebras. If $\mathrm{Tor}_1^R(A, S) = 0$, then the canonical map $\check{H}^{\cdot}(A\otimes_R S|S, G) \to \check{H}^{\cdot}(A|R, SG)$ is an isomorphism for any $G \in \mathrm{ob}(\underline{C}^{\vee}_{A\otimes_R S|S})$ where the functor $S : \underline{C}_{A|R} \to \underline{C}_{A\otimes_R S|S}$ is given by $X \longmapsto X\otimes_R S$. In particular if S is R-flat, then the canonical map $\check{S}\check{H}^{\cdot}(G) \to \check{H}(\check{S}G)$ is an isomorphism for every $G \in \mathrm{ob}(\underline{C}^{\vee}_{A\otimes_R S|S})$.

(b) Let $P \to B$ be a surjective R-algebra map, where P is a polynomial algebra over R. If $\mathrm{Tor}_i^R(B, S) = 0$ for all $0 < i \leq n$, then $\mathrm{Tor}_i^R(P\times_B P\times_B \ldots \times_B P, S) = 0$ for all $0 < i < n$.

Proof:

(a) Given $B \in \mathrm{ob}(\underline{C}_{A|R})$, choose $\{P \to B\} \in \mathrm{Cov}(B)$ where P is a polynomial algebra over R. If $\mathrm{Tor}_1^R(B, S) = 0$, the natural map $(\overbrace{P\times_B \ldots \times_B P}^{n})\otimes_R S \to (P\otimes_R S)\times_{(B\otimes_R S)} \ldots \times_{(B\otimes_R S)}(P\otimes_R S)$ is an isomorphism for all n by 3.5(i), and hence $\check{H}^{\cdot}(B|R, \check{S}G) = H^{\cdot}(\{P \to B\}, \check{S}G) = H^{\cdot}(\{P\otimes_R S \to B\otimes_R S\}, G) = \check{H}^{\cdot}(B\otimes_R S|S, G)$.
If S is R-flat, then $\mathrm{Tor}_1^R(X, S) = 0$ for every $X \in \mathrm{ob}(\underline{C}_{A|R})$ and hence $\check{H}^{\cdot}(X\otimes_R S|S, G) \to \check{H}^{\cdot}(X|R, \check{S}G)$ is an isomorphism for every $X \in \mathrm{ob}(\underline{C}_{A|R})$ i.e. $\check{S}\check{\underline{H}}^{\cdot}(G) \to \check{\underline{H}}^{\cdot}(\check{S}G)$ is an isomorphism.

(b) Set $P^{(m)} = \overbrace{P\times_B \ldots \times_B P}^{m+1}$. If $m = 0$ there is nothing to prove since P is R-flat. Assume that $\mathrm{Tor}_i^R(P^{(m-1)}, S) = 0$ for all $0 < i < n$. Since $P \to B$ is surjective and $P^{(m)} = P\times_B P^{(m-1)}$, it follows from 3.5(ii) that $\mathrm{Tor}_i^R(P^{(m)}, S) = 0$ for all $0 < i < n$.

Proposition 3.7.

(a) *Let* A, S *be* R-*algebras. If* $\mathrm{Tor}_i^R(A, S) = 0$ *for all* $0 < i \leq n$, *then for any sheaf* F *on* $\underline{C}_{A\otimes_R S|S}$, *the canonical map* $H^i(A\otimes_R S|S, F) \to H^i(A|R, \widetilde{S}F)$ *is an isomorphism for all* $i \leq n$, *where* $S : \underline{C}_{A|R} \to \underline{C}_{A\otimes_R S|S}$ *is given by* $X \longmapsto X\otimes_R S$.

(b) *Let* P *be a polynomial algebra over* R. *Then for any additive sheaf* F *on* $\underline{C}_{P\otimes_R A|R}$, *the canonical map* $H^n(P\otimes_R A|R, F) \to H^n(A|R, F)$ *is an isomorphism for all* $n > 0$.

(c) *Let* $S \to R \to A$ *be ring homomorphisms. Then for any additive sheaf* F *on* $\underline{C}_{A|S}$, *we have an exact sequence*

$$0 \to H^0(A|R, F_R) \to H^0(A|S, F) \to H^0(R|S, F) \to H^1(A|R, F_R) \to H^1(A|S, F) \to H^1(R|S, F) \to$$
$$\cdots\cdots \to H^n(A|R, F_R) \to H^n(A|S, F) \to H^n(R|S, F) \to H^{n+1}(A|R, F_R) \to \cdots$$

where F_R *is a sheaf on* $\underline{C}_{A|R}$ *defined by*

$$F_R(X) = \mathrm{Ker}(F(X) \to F(R))$$

for all $X \in \mathrm{ob}(\underline{C}_{A|R})$ *via the forgetful functor* $\underline{C}_{A|R} \to \underline{C}_{A|S}$.

Proof:

(a) Consider the continuous functor

$$S : \underline{C}_{A|R} \to \underline{C}_{A\otimes_R S|S}$$

where $S(X) = X\otimes_R S$. Since the functor S sends a polynomial algebras over R to polynomial algebras over S, $\widetilde{S} : \widetilde{\underline{C}}_{A\otimes_R S|S} \to \widetilde{\underline{C}}_{A|R}$ is an exact functor by 3.3(c) and in turn we obtain the canonical map $\check{S}\underline{H}^{\cdot}_{A\otimes_R S|S}(F) \to \underline{H}^{\cdot}_{A|R}(\widetilde{S}F)$, and in turn we obtain the commutative diagram of spectral sequences:

$$\begin{array}{ccc}
E_S^{p,q}(S(X)) = \check{H}^p(X\otimes_R S|S, \underline{H}^q_{A\otimes_R S|S}(F)) & \Longrightarrow & H^{p+q}(X\otimes_R S|S, F) \\
\downarrow & & \\
\check{H}^p(X|R, \check{S}\underline{H}^q_{A\otimes_R S|S}(F)) & & \downarrow \\
\downarrow & & \\
E_R^{p,q}(X) = \check{H}^p(X|R, \underline{H}^q_{A|R}(\widetilde{S}F)) & \Longrightarrow & H^{p+q}(X|R, \widetilde{S}F)
\end{array}$$

We note that $E_S^{p,q}(S(X)) \to \check{H}^p(X|R, \check{S}\underline{H}^q_{A\otimes_R S|S}(F))$ is an isomorphism whenever $\mathrm{Tor}_1^R(X, S) = 0$ by 3.6(a). Assume now that $[\check{S}\underline{H}^q_{A\otimes_R S|S}(F)](X) \to [\underline{H}^q_{A|R}(\tilde{S}F)](X)$ is an isomorphism for all $q < n$ and for all $X \in \mathrm{ob}(\underline{C}_{A|R})$ such that $\mathrm{Tor}_i^R(X, S) = 0$ for all $0 \; i \; n$. It follows immediately from 3.6(b) that $E_S^{p,q}(S(X)) \to E_R^{p,q}(X)$ is an isomorphism for all $q < n$ and for all p, provided $\mathrm{Tor}_i^R(X, S) = 0$ for all $0 < i \leq n$. Since $E_S^{0,n}(S(X)) \to E_R^{0,n}(X)$ is an isomorphism (since both sides are zero), we get that $H^n(X_R S|S, F) \to H^n(X|R, \tilde{S}F)$ is an isomorphism whenever $\mathrm{Tor}_i^R(X, S) = 0$ for all $0 < i \leq n$ i.e., $[\check{S}\underline{H}^i_{A\otimes_R S|S}(F)](X) \to [\underline{H}^i_{A|R}(\tilde{S}F)](X)$ is an isomorphism for all $i \leq n$ whenever $\mathrm{Tor}_i^R(X, S) = 0$ for all $0 < i \leq n$.

(b) Let P be a polynomial algebra over R and let $f : \underline{C}_{A|R} \to \underline{C}_{A\otimes_R P|R}$ be the continuous functor given by $f(X) = X \otimes_R P$. Let $P' \to X$ be a surjective arrow in $\underline{C}_{A|R}$ where P' is a polynomial algebra over R. Then $P' \otimes_R P \to X \otimes_R P$ is a surjective arrow in $\underline{C}_{A\otimes_R P|R}$, and $P' \otimes_R P$ is again a polynomial algebra over R. Since P is flat over R, it follows from 3.5(i) that $\check{H}^{\cdot}(X|R, \check{f}G) = H^{\cdot}(\{P' \to X\}, \check{f}G) = H(\{P' \otimes_R P \to X \otimes_R P\}, G) = \check{H}^{\cdot}(X \otimes_R P|R, G)$ for every $G \in \mathrm{ob}(\check{\underline{C}}_{A\otimes_R P|R})$. It follows that $\tilde{f} : \tilde{\underline{C}}_{A\otimes_R P|R} \to \underline{C}_{A|R}$ sends acyclic ones to acyclic ones. On the other hand, f sends polynomial algebras over R to polynomial algebras over R, and hence $\tilde{f} : \tilde{\underline{C}}_{A\otimes_R P|R} \to \tilde{\underline{C}}_{A|R}$ is an exact functor by 3.2(c). It follows that $H^{\cdot}(A \otimes_R P|R, F) \to H^{\cdot}(A|R, \tilde{f}F)$ is an isomorphism for every $F \in \mathrm{ob}(\tilde{\underline{C}}_{A\otimes_R P|R})$. Now for each $X \in \mathrm{ob}(\underline{C}_{A|R})$, we have canonical maps

$$(\tilde{f}F)\ (X) = F(X \otimes_R P) \to F(X) \oplus F(P) \quad ,$$

and hence a canonical map $\tilde{f}F \to R \oplus F(P)$ where $F(P)$ denotes a constant sheaf (and F on the right hand side denotes, strictly speaking, the restriction of F on $\underline{C}_{A|R}$). If F is an additive sheaf, then $(\tilde{f}F)\ (X) = F(X \otimes_R P) \xrightarrow{\sim} F(X) \oplus F(P)$ is an isomorphism for every $X \in \mathrm{ob}(\underline{C}_{A|R})$ and hence $\tilde{f}F \xrightarrow{\sim} R \oplus F(P)$ is an isomorphism. Therefore

$$H^n(A|R, \widetilde{f}F) \cong H^n(A|R, F\oplus F(P)) = H^n(A|R, F)$$

for all $n > 0$ since $F(P)$, being a constant sheaf, is flask. It follows that $H^n(A\underset{R}{\otimes}P|R, F) \to H^n(A|R, F)$ is an isomorphism for all $n > 0$, provided F is an additive sheaf on $\underline{C}_{A\underset{R}{\otimes}P|R}$.

(c) (communicated from Rinehart) Consider the forgetful functor $\beta : \underline{C}_{A|R} \to \underline{C}_{A|S}$. We have a spectral sequence

$$E^{p,q} = H^p(A|R, \beta^{\#}\underline{H}^q_{A|S}(F)) \Longrightarrow H^{p+q}(A|S, F)$$

where $\beta^{\#} = G_{A|R}\circ\check{\beta}$, $G_{A|R} : \check{\underline{C}}_{A|R} \to \widetilde{\underline{C}}_{A|R}$ is "associated-sheaf" functor. Now let F be an additive sheaf. We claim that $\beta^{\#}\underline{H}^q_{A|S}(F)$ is a constant sheaf for all $q > 0$. Indeed, if P is a polynomial algebra over R, then $P = R\underset{S}{\otimes}P'$ for a polynomial algebra P' over S and hence

$$[\beta^{\#}\underline{H}^q_{A|S}(F)](P) = H^q(P|S, F) = H^q(R\underset{S}{\otimes}P'|S, F) \to H^q(R|S, F)$$

is an isomorphism for all $q > 0$ by 3.7(b), and surjective for $q = 0$. It follows from 3.3(c) that the canonical map $\beta^{\#}\underline{H}^q_{A|S}(F) \to H^q(R|S, F) =$ constant sheaf is an isomorphism for all $q > 0$, and is surjective for $q = 0$. It follows that $E^{p,q} = H^p(A|R, \beta^{\#}\underline{H}^q_{A|S}(F)) = 0$ for all $p, q > 0$, and hence the above spectral sequence yields the exact sequence

$$(*) \quad 0 \to E^{1,0} \to H^1 \to E^{0,1} \to E^{2,0} \to H^2 \to E^{0,2} \to E^{3,0} \to H^3 \to \dots.$$

where we note that $E^{0,q} = [\beta^{\#}\underline{H}^q_{A|S}(F)](A) \cong H^q(R|S, F)$ for all $q > 0$ and $E^{p,0} = H^p(A|R, \widetilde{\beta}F)$. On the other hand, the exact sequence $0 \to F_R \to \widetilde{\beta}F \to F(R) \to 0$ gives us the exact sequence

$$\begin{array}{cccccccccccc}
(**) \quad 0 \to H^0(A|R,F_R) & \to & H^0(A|R,\widetilde{\beta}F) & \to & H^0(A|R,F(R)) & \to & H^1(A|R,F_R) & \to & H^1(A|R,\widetilde{\beta}F) \to 0 \\
& & \| & & \| & & & & \\
& & H^0(A|S,F) & & H^0(R|S,F) & & & &
\end{array}$$

The exact sequences (*) and (**) combined gives the desired exact sequence.

Corollary 3.8.

(a) If A, B are R-algebras such that $\mathrm{Tor}_i^R(A, B) = 0$ for all $0 < i \leq n$, then for any additive sheaf $F \in \mathrm{ob}(\widetilde{\underline{C}}_{A\otimes_R B|R})$, the canonical map

$$H^i(A\otimes_R B|R, F) \to H^i(A|R, F) \oplus H^i(B|R, F)$$

is an isomorphism for all $i \leq n$.

(b) Let A be an R-algebra such that $\mathrm{Tor}_i^R(A, A) = 0$ for all $i > 0$. If $A\otimes_R A \to A$ is an isomorphism, then for any additive sheaf $F \in \mathrm{ob}(\widetilde{\underline{C}}_{A|R})$, we have $H^i(A|R, F) = 0$ for all i .

Proof:

(a) Let $F \in \mathrm{ob}(\widetilde{\underline{C}}_{A\otimes_R B|R})$ be additive. By 3.7(c), the ring homomorphisms $R \to A \to A\otimes_R B$ gives us the exact sequence

$$(*) \quad \ldots \to H^i(A\otimes_R B\ A, F_A) \to H^i(A\otimes_R B\ R, F) \to H^i(A|R, F) \to \ldots.$$

consider now the commutative diagram

$$\begin{array}{ccc}
H^i(A\otimes_R B|A, F_A) & \longrightarrow & H^i(A\otimes_R B|R, F) \\
\downarrow & & \downarrow \\
H^i(B|R, \tilde{A}F_A) & \xrightarrow{\approx} & H^i(B|R, F)
\end{array}$$

where the functor $A : \underline{C}_{B|R} \to \underline{C}_{A\otimes_R B|A}$ is given by $X \longmapsto A\otimes_R X$, and the map $H^i(B|R, \tilde{A}F_A) \to H^i(B|R, F)$ is induced from the map $\tilde{A}F_A \to F$, which is an isomorphism since F is additive. Note that the commutativity of the above diagram, because of its functorial nature, is reduced to the case $i = 0$.
If $\mathrm{Tor}_i^R(A, B) = 0$ for all $0 < i \leq n$, then $H^i(A\otimes_R B|A, F_A) \to H^i(B|R, \tilde{A}F_A)$ is an isomorphism for all $i \leq n$ by 3.7(a), and hence the composite map

$$H^i(A\otimes_R B|A, F_A) \to H^i(A\otimes_R B|R, F) \to H^i(B|R, F)$$

and (because of the symmetry)

$$H^i(A\underset{R}{\otimes}B|B,\ F_B) \to H^i(A\underset{R}{\otimes}B|R,\ F) \to H^i(A|R,\ F)$$

are isomorphisms for all $i \leq n$. This implies that the exact sequence (*) breaks up to short split-exact sequences

$$0 \to H^i(A\underset{R}{\otimes}B|A,\ F_A) \to H^i(A\underset{R}{\otimes}B|R,\ F) \to H^i(A|R,\ F) \to 0$$

for all $i \leq n$, and that

$$H^i(A\underset{R}{\otimes}B|R,\ F) \to H^i(A|R,\ F) \oplus H^i(B|R,\ F)$$

is an isomorphism for all $i \leq n$.

(b) Since $\mathrm{Tor}_i^R(A,A) = 0$ for all $i > 0$, it follows from 3.8(a) that

$$H^i(A\underset{R}{\otimes}A|R,\ F) \to H^i(A|R,\ F) \oplus H^i(A|R,\ F)$$

is an isomorphism for all i . On the other hand, $A\underset{R}{\otimes}A \to A$ is an isomorphism by hypothesis and hence $H^i(A|R,\ F) \to H^i(A\underset{R}{\otimes}A|R,\ F)$ is an isomorphism for all i . This means that the diagonal map $H^i(A|R,\ F) \to H^i(A|R,\ F) \oplus H^i(A|R,\ F)$ is an isomorphism for all i i.e., $H^i(A|R,\ F) = 0$ for all i .

<u>Corollary</u> 3.9: Let S be a multiplicative subset of an R-algebra A . If $F \in \mathrm{ob}\left(\underline{C}^{\sim}_{S^{-1}A|R}\right)$ is additive such that F_A is also additive, then the canonical map

$$H^i(S^{-1}A|R,\ F) \to H^i(A|R,\ F)$$

is an isomorphism for all i , where $F_A \in \mathrm{ob}\left(\underline{C}^{\sim}_{S^{-1}A|A}\right)$ is defined by $F_A(X) = \mathrm{Ker}(F(X) \to F(A))$ for all $X \in \mathrm{ob}\left(\underline{C}_{S^{-1}A|A}\right)$.

<u>Proof</u>: Since F is additive, the ring homomorphisms $R \to A \to S^{-1}A$ gives us the exact sequence

$$\dots\dots \to H^n(S^{-1}A|A,\ F_A) \to H^n(S^{-1}A|R,\ F) \to H^n(A|R,\ F) \to \dots\dots$$

On the other hand, since $S^{-1}A\underset{A}{\otimes}S^{-1}A \to S^{-1}A$ is an isomorphism and F_A is additive by hypothesis, it follows from 3.8(b) that $H^n(S^{-1}A|A,\ F_A) = 0$ for all n , and hence $H^n(S^{-1}A|R,\ F) \to H^n(A|R,\ F)$ is an isomorphism for all n .

Now let A be an R-algebra. An A-module M defines an additive sheaf $\mathrm{Der}_R(-, M) \in \mathrm{ob}(\underline{\tilde{C}}_{A|R})$ by $X \to \mathrm{Der}_R(X, M)$. We now make the following definition:

Definition 3.10: Let A be an R-algebra. For each A-module M we set

$$D^i(A|R, M) = H^i(A|R, \mathrm{Der}_R(-, M)) \quad .$$

We want to make an explicit computation of low-dimensional ones i.e., $D^i(A|R, M)$ for $i = 1, 2$: Choose a surjective arrow $P \to A$ in $\underline{C}_{A|R}$ where P is a polynomial algebra over R , and let $0 \to I \to P \to A \to 0$ be exact. Then

Lemma 3.10:

$$D^1(A|R, M) = \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_P(I, M))$$
$$D^2(A|R, M) = H^1(A|R, \underline{H}^1_{A|R}(\mathrm{Der}_R(-, M)))$$

Proof: Consider the simplicial algebra

$$(*) \qquad \cdots\cdots\cdots \; P_A^{(2)} \rightrightarrows P_A^{(1)} \rightrightarrows P_A^{(0)}$$

where $P_A^{(n)} = \overbrace{P \underset{A}{\times} \cdots \underset{A}{\times} P}^{n+1}$. For each n we have the exact sequence

$$0 \to I^{(n)} \to P_A^{(n)} \to A \to 0$$

where $I^{(n)} = \overbrace{I \times I \times \cdots \times I}^{n+1}$, and it is easy to see that

(a):

$$0 \to \mathrm{Hom}_A(\bar{I}^{(n)}/\Delta(\bar{I}), M) \to \mathrm{Der}_R(P_A^{(n)}, M) \xrightarrow{\mathrm{Der}(\Delta,\, M)} \mathrm{Der}_R(P, M) \to 0$$

is exact for all n , where $\bar{I} = I/I^2$ and Δ stands for the diagonal map. We also note the exact sequence

(b):

$$0 \to \mathrm{Hom}_A(\bar{I}^{(n)}/\Delta(\bar{I}), M) \to \mathrm{Hom}_A(\bar{I}^{(n)}, M) \xrightarrow{\mathrm{Hom}(\Delta,\, M)} \mathrm{Hom}_A(I, M) \to 0$$

The exact sequences (a) and (b) yield the respective exact sequences of simplicial A-modules. Since $\{I \to 0\}$, $\{I \to I\}$ admits sections, we have

$$H^i(\{I \to 0\}, \mathrm{Hom}_A(-, M)) = H^i(\{I \to I\}, \mathrm{Hom}_A(-, M)) = 0$$

for all $i > 0$, and it follows readily that

$$\begin{cases} 0 \to \mathrm{Der}_R(A, M) \to \mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(\bar{I}, M) \to H^1(\{P \to A\}, \mathrm{Der}_R(-, M)) \to 0 \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{is exact} \\ H^i(\{P \to A\}, \mathrm{Der}_R(-, M)) = 0 \quad \text{for all } i > 1 \end{cases}$$

It follows from 3.1(b) that

$$\begin{aligned} D^1(A|R, M) &= \check{H}^1(A|R, \mathrm{Der}_R(-, M)) = H^1(\{P \to A\}, \mathrm{Der}_R(-, M)) \\ &= \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(\bar{I}, M)) \\ &= \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_P(I, M)) \quad , \\ D^2(A|R, M) &= H^1(A|R, \underline{H}^1_{A|R}(\mathrm{Der}_R(-, M))) \quad . \end{aligned}$$

Now choose a free P-module F and a P-linear map $F \xrightarrow{\gamma} P$ such that $\gamma(F) = I$. Associated to P-linear map $\gamma : F \to P$, there is a Koszul complex, which is simply denoted, by abuse of notation, by K_I .

<u>Lemma</u> 3.11: $D^2(A|R, M) \simeq \mathrm{Coker}(H^1(K_I, M) \to \mathrm{Hom}_A(H_1(K^I), M))$.

<u>Proof:</u> We have

$$\begin{aligned} D^2(A|R, M) &= H^1(A|R, H^1_{A|R}(\mathrm{Der}_R(-, M))) \\ &= \mathrm{Ker}(H^1(P_A^{(1)}|R, M) \rightrightarrows H^1(P_A^{(2)}|R, M)) \end{aligned}$$

since $[\underline{H}^1_{A|R}(\mathrm{Der}_R(-, M))](P_A^0) = [\underline{H}^1_{A|R}(\mathrm{Der}_R(-, M))](P) = 0$.

Let $0 \to K \to F \xrightarrow{\gamma} I \to 0$ be exact. If we set $P[F]$ to be the symmetric algebra over P generated by F , then $P[F]$ and $P[F] \otimes P[F]$ are polynomial algebras over R and we obtain the surjective arrows

$$P[F] \to P_A^{(1)} = P \times_A P \qquad \text{given by} \qquad X \to (0, \gamma(X))$$

$$P[F] \otimes_P P[F] \to P_A^{(2)} = P \times_A P \times_A P \qquad X \otimes 1 \to (0, \gamma(X), 0),\ 1 \otimes X \to (0, 0, \gamma(X))$$

where $X \in F$. If we set $J = \mathrm{Ker}(P[F] \to P_A^{(1)})$ and $J' = \mathrm{Ker}(P[F] \otimes_P P[F] \to P_A^{(2)})$, then

$$J = (F) \cap (F_\gamma) = (K) + (F\cdot F_\gamma)$$

$$J' = (F\otimes 1 + 1\otimes F) \cap (F_\gamma\otimes 1 + 1\otimes F) \cap (F\otimes 1 + 1\otimes F_\gamma)$$

$$= (K\otimes 1 + 1\otimes K) + (F\cdot F_\gamma\otimes 1 + 1\otimes F\cdot F_\gamma) + (F\otimes F)$$

where F_γ = the ideal generated by $\{X - \gamma(X) \mid X \in F\}$. Now $\pi_i : P_A^{(2)} \to P_A^{(1)}$ $i = 0, 1, 2$ can be lifted to $\hat{\pi}_i : P[F]\otimes_P P[F] \to P[F]$ where, for each $X \in F$,

$$\pi_0 = \begin{cases} X\otimes 1 \to \gamma(X) - X \\ 1\otimes X \to X \end{cases} \qquad \pi_1 = \begin{cases} X\otimes 1 \to 0 \\ 1\otimes X \to X \end{cases} \qquad \pi_2 = \begin{cases} X\otimes 1 \to X \\ 1\otimes X \to 0 \end{cases} .$$

A trivial computation shows that $\pi_0 - \pi_1 + \pi_2$ sends $K\otimes 1 + 1\otimes K + 1\otimes F\cdot F_\gamma$ to zero, whereas it sends $(F\cdot F_\gamma\otimes 1 + F\otimes F)$ onto $(F\cdot F_\gamma)$. Now for any $D \in \mathrm{Der}_R(P[F]\otimes_P P[F], M)$, $D(F\cdot F_\gamma\otimes 1 + F\otimes F) = 0$ since $I\cdot M = 0$, and consequently

$$D^2(A|R, M) = \mathrm{Ker}(D^1(P_A^{(1)}|R, M) \rightrightarrows D^1(P_A^{(2)}|R, M))$$

$$= \{[f] \in D^1(P_A^{(1)}|R, M) \mid f(F\cdot F_\gamma) = 0\}$$

$$= \mathrm{Coker}(\mathrm{Der}_R(P[F], M) \to \mathrm{Hom}_{P[F]}((F) \cap (F_\gamma)/(F\cdot F_\gamma), M))$$

$$= \mathrm{Coker}(\mathrm{Der}_P(P[F], M) \to \mathrm{Hom}_{P[F]}((F) \cap (F_\gamma)/(F)\cdot(F_\gamma), M)) .$$

Now
$$0 \to (F) \to P[F] \xrightarrow{P[0]} P \to 0$$
$$0 \to (F_\gamma) \to P[F] \xrightarrow{P[\gamma]} P \to 0 .$$

are exact and hence $(F) \cap (F_\gamma)/(F)\cdot(F_\gamma) = \mathrm{Tor}_1^{P[F]}(P, P_\gamma) = H_1(K_I)$ where K_I is the Koszul complex associated to the P-linear map $F \xrightarrow{\gamma} I(\subset P)$. We thus have

$$D^2(A|R, M) = \mathrm{Coker}(\mathrm{Hom}_P(F, M) \to \mathrm{Hom}_P(H_1(K_I), M))$$

$$= \mathrm{Coker}(H^1(K_I, M) \to \mathrm{Hom}_A(H^1(K_I), M)) .$$

We can now state all the basic facts needed for our purpose, summarizing the various results we have already established.

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Theorem 3.12.

(1) *If* $0 \to M' \to M \to M'' \to 0$ *is an exact sequence of* A-*modules, then*

$$0 \to D^0(A|R, M') \to D^0(A|R, M) \to D^0(A|R, M'') \to D^1(A|R, M') \to$$
$$D^1(A|R, M) \to D^1(A|R, M'') \to \ldots\ldots\ldots \to D^n(A|R, M') \to$$
$$D^n(A|R, M) \to D^n(A|R, M'') \to \ldots$$

is exact.

(2) *Let* $S \to R \to A$ *be ring homomorphisms. Then for any* A-*module* M *we have the exact sequence*

$$0 \to D^0(A|R, M) \to D^0(A|S, M) \to D^0(R|S, M) \to D^1(A|R, M) \to$$
$$D^1(A|S, M) \to D^1(R|S, M) \to \ldots\ldots\ldots \to D^n(A|R, M) \to$$
$$D^n(A|S, M) \to D^n(R|S, M) \to \ldots.$$

(3) *Let* A, S *be* R-*algebras such that* $\mathrm{Tor}_i^R(A, S) = 0$ *for all* $i > 0$. *Then for any* $A\otimes_R S$-*module* M, *the canonical map* $D^i(A\otimes_R S|S, M) \to D^i(A|R, M)$ *is an isomorphism for all* i. *In particular, if* A *is* R-*flat we have*

$$D^i(A\otimes S|S, M) \cong D^i(A|R, M)$$

for all i *and for all* R-*algebras* S.

(4) *Let* S *be a multiplicative subset of* A. *Then for any* $S^{-1}A$-*module* N, *the canonical map*

$$D^i(S^{-1}A|R, N) \to D^i(A|R, N)$$

is an isomorphism for all i. *Consequently, for any* A-*module* M *we have the canonical isomorphism*

$$D^i(S^{-1}A|R, S^{-1}M) \cong S^{-1}D^i(A|R, M)$$

for all i.

(5) *If* A, B *are* R-*algebras such that* $\mathrm{Tor}_i^R(A, B) = 0$ *for all* $0 < i \leq n$, *then for any* $A\otimes_R B$-*module* M, *the canonical map*

$$D^i(A\otimes_R B|R, M) \to D^i(A|R, M) \oplus D^i(B|R, M)$$

is an isomorphism for all $i \leq n$. *In particular, if* $\mathrm{Tor}_i^R(A, A) = 0$ *for all*

$i > 0$ and

$$\Delta : \mathrm{Spec}(A) \to \mathrm{Spec}(A) \underset{\mathrm{Spec}(R)}{\times} \mathrm{Spec}(A)$$

is an open immersion (for instance A/R etale), then $D^i(A|R, M) = 0$ for all $i > 0$ and for all A-modules M.

(6) Let A be an R-algebra of essentially finite type over R, where R is noetherian, and let E be a flat A-module. Then we have a canonical isomorphism

$$E \otimes_A D^i(A|R, M) \xrightarrow{\sim} D^i(A|R, E \otimes_A M)$$

for all $i \geq 0$ and for all A-modules M.

(7) If R is noetherian and A is an R-algebra of essentially finite type, then $D^i(A|R, M)$ is A-module of finite type for all i, provided M is an A-module of finite type.

(8) Let P be a polynomial algebra over R and let $0 \to I \to P \to A \to 0$ be exact. Then $D^0(A|R, M) = \mathrm{Der}_R(A, M)$,

$$\begin{aligned} D^1(A|R, M) &= \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) \\ &= \text{the set of isomorphism classes of} \\ &\quad \text{R-algebra extensions of } A \text{ by } M, \\ D^2(A|R, M) &= \mathrm{Coker}(H^1(K_I, M) \to \mathrm{Hom}_A(H_1(K_I), M)) . \end{aligned}$$

Proof:

(1) If $0 \to M' \to M \to M'' \to 0$ is an exact sequence of A-modules, then for any polynomial algebra P over R, $0 \to \mathrm{Der}_R(P, M') \to \mathrm{Der}_R(P, M) \to \mathrm{Der}_R(P, M'') \to 0$ is exact and hence $0 \to \mathrm{Der}_R(-, M') \to \mathrm{Der}_R(-, M) \to \mathrm{Der}_R(-, M'') \to 0$ is an exact sequence of sheaves by 3.3(c).

(2) $\mathrm{Der}_S(-, M)$ is an additive sheaf on $\underline{C}_{A|S}$, and $\mathrm{Der}_R(B, M) = \mathrm{Ker}(\mathrm{Der}_S(B, M) \to \mathrm{Der}_S(R, M))$ for every $B \in \mathrm{ob}(\underline{C}_{A|R})$. Therefore our assertion follows from 3.7(c).

(3) Let $S : \underline{C}_{A|R} \to \underline{C}_{A \otimes_R S|S}$ be the functor given by $X \longmapsto X \otimes_R S$. Then $\tilde{S}\mathrm{Der}_S(-, M) = \mathrm{Der}_R(-, M)$ and hence our assertion follows from 3.7(a).

(4) Follows from 3.9. Indeed, $F = \mathrm{Der}_R(-, N)$ is an additive sheaf on $\underline{C}_{S^{-1}A|R}$, and $F_A \in \mathrm{ob}\left(\underline{C}^{\sim}_{S^{-1}A|A}\right)$ is also additive since $F_A(X) = \mathrm{Ker}(F(X) \to F(A)) = \mathrm{Ker}(\mathrm{Der}_R(X, N) \to \mathrm{Der}_R(A, N)) = \mathrm{Der}_A(X, N)$ i.e., $F_A = \mathrm{Der}_A(-, N)$.

(5) The first assertion follows from 3.8(a). Now assume that $\mathrm{Tor}_i^R(A, A) = 0$ for all $i > 0$ and that

$$\Delta : \mathrm{Spec}(A) \to \mathrm{Spec}(A) \underset{\mathrm{Spec}(R)}{\times} \mathrm{Spec}(A)$$

is an open immersion i.e., $(A \otimes_R A)_{\varphi^{-1}(\underline{m})} \to A_{\underline{m}}$ is an isomorphism for all maximal ideals $\underline{m}$ of A, where $\varphi : A \otimes_R A \to A$ is the multiplication map. Then the first assertion combined with 3.12(4) entails that the diagonal map

$$\Delta_{\underline{m}} : D^i(A|R, M)_{\underline{m}} \to D^i(A|R, M)_{\underline{m}} \oplus D^i(A|R, M)_{\underline{m}}$$

is an isomorphism for every maximal ideal $\underline{m}$ of A and hence

$$\Delta : D^i(A|R, M) \to D^i(A|R, M) \oplus D^i(A|R, M)$$

is an isomorphism, which entails that $D^i(A|R, M) = 0$ for all i.

(6) In view of 3.12(4), we may assume that A is an R-algebra of finite type. Now consider the map

$$E \otimes_A \mathrm{Der}_R(-, M) \to \mathrm{Der}_R(-, E \otimes_A M)$$

in $\underline{C}^{\sim}_{A|R}$. Since E is A-flat, we have an isomorphism

$$E \otimes_A \mathrm{Der}_R(B, M) \xrightarrow{\sim} \mathrm{Der}_R(B, E \otimes_A M)$$

whenever $B \in \mathrm{ob}(\underline{C}_{A|R})$ is of finite type over R. It follows from 3.3(b) that

$$D^i(A|R, E \otimes_A \mathrm{Der}_R(-, M)) \to D^i(A|R, E \otimes_A M)$$

is an isomorphism. On the other hand, the functor $E \otimes_A - : (A\text{-mod}) \to (A\text{-mod})$ is exact, and therefore we have

$$E \otimes_A D^i(A|R, M) = D^i(A|R, E \otimes_A \mathrm{Der}_R(-, M)) \quad ,$$

and hence our assertion.

(7) follows from 3.3(a).

(8) In view of 3.10 and 3.11, it suffices to show that $\mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) = \mathrm{Exalcomm}(A|R, M)$, where $\mathrm{Exalcomm}(A|R, M)$ denotes the set of isomorphism classes of R-algebra extensions of A by M . Let $0 \to M \to A' \to A \to 0$ be an R-algebra extension of A by M . Since P is a polynomial algebra over R , we obtain a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & I \arrow[r] \arrow[d] & P \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & M \arrow[r] & A' \arrow[r] & A \arrow[r] & 0 \end{tikzcd}$$

where $I \to M$ is uniquely determined by the extensions A' up to $\mathrm{Der}_R(P, M)$. We thus obtain a map

$$\Phi : \mathrm{Exalcomm}(A|R, M) \to \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) .$$

Conversely a P-linear map $I \to M$ yields an extension by push-out

$$\begin{tikzcd} 0 \arrow[r] & I \arrow[r] \arrow[d] & P \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & M \arrow[r] & A' \arrow[r] & A \arrow[r] & 0 \end{tikzcd}$$

Two P-linear maps $I \rightrightarrows M$ which differ by $\mathrm{Der}_R(P, M)$ is trivially seen to yield isomorphic extensions and therefore we obtain a map

$$\Psi : \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) \to \mathrm{Exalcomm}(A|R, M) .$$

It is straightforward to verify that Φ, Ψ are inverse process to each other.

Corollary 3.13. (a) R-algebra A is formally smooth over R if and only if $D^1(A|R, M) = 0$ for every A-module M .

(b) Let R be noetherian and A an R-algebra of finite type. Then A is a relative complete-intersection over R if and only if $D^2(A|R, M) = 0$ for every A-module M .

(c) Let R be noetherian and A an R-algebra of finite type. Then for any multiplicative subset S in A , we have a canonical isomorphism $D^i(S^{-1}A|R, S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R, M)$ for all $i \geq 0$.

(d) Let R be noetherian and A a local R-algebra of essentially finite type. We denote by $\hat{A}$ the $\underline{m}$-adic completion of A where $\underline{m}$ = the maximal ideal of A . Then for any A-module E of finite type we have a canonical isomorphism

$$D^1(\hat{A}|R, \hat{A} \otimes_A E) \xrightarrow{\sim} \hat{A} \otimes_A D^1(A|R, E) .$$

Proof: (a) follows immediately from $D^1(A|R, M) = \text{Exalcomm}(A|R, M)$.

(b) Let P be a polynomial algebra over R in a finite number of variables and let $0 \to I \to P \to A \to 0$ be exact. Localizing if necessary, we may assume that I is generated by a P-regular sequence which would imply that $H_1(K_I) = 0$ and therefore $D^2(A|R, M) = \text{Coker}(H^1(K_I, M) \to \text{Hom}_A(H_1(K_I), M)) = 0$. Conversely assume that $D^2(A|R, M) = 0$ for every A-module M .

Let $0 \to I \to P \to A \to 0$ be exact where P is a polynomial algebra over R in a finite number of variables. We have

$$0 = D^2(A|R, M) = \text{Coker}(H^1(K_I, M) \to \text{Hom}_A(H_1(K_I), M))$$

i.e., $H^1(K_I, M) \to \text{Hom}_A(H_1(K_I), M)$ is surjective for all A-module M . Choose an exact sequence $0 \to K \to F \to I \to 0$ where F is a free P-module of finite type. We note that $H_1(K_I) = K/K_0$ where $K_0 = \text{Im}(\overset{2}{\wedge} F \to \overset{1}{\wedge} F)$. The fact that $D^2(A|R, H_1(K_I)) = 0$ i.e.,

$$H^1(K_I, H_1(K_I)) \to \text{Hom}_A(H_1(K_I), H_1(K_I)) \text{ is surjective}$$

entails that $0 \to K/K_0 \to F/IF \to I/I^2 \to 0$ is exact and splits as A-modules and in particular I/I^2 is projective as A-module. Therefore, localizing A if necessary, we may assume that I/I^2 is A-free, and that $F/IF \to I/I^2$ is an isomorphism. Then the fact that $0 \to K/K_0 \to F/IF \to I/I^2 \to 0$ is exact and splits as A-modules entails that $H_1(K_I) = K/K_0 = 0$. This means that I is generated by a P-regular sequence i.e., A is a relative complete intersection over R .

(c) follows immediately from 3.12(4) and (6).

(d) Firstly it follows from 3.12(6) that

$$D^i(A R, \hat{A}\otimes_A E) \xrightarrow{\sim} \hat{A}\otimes_A D^i(A|R, E)$$

is an isomorphism for all i , and therefore it suffices to show that

$D^1(\hat{A}|R, \hat{A}\otimes_A E) \to D^1(A|R, \hat{A}\otimes_A E)$ is an isomorphism. Our proof is based on the description of D^1 in terms of algebra extensions. Let $0 \to \hat{A}\otimes_A E \to A' \xrightarrow{\varphi} A \to 0$ be an R-algebra extension. The fact that $\hat{A}\otimes_A E$ is an $\hat{A}$-module of finite type entails readily that A' is a local R-algebra, complete with respect to $\underline{m}'$-adic topology where $\underline{m}'$ = the maximal ideal of A'. Therefore if there is an R-algebra map $\psi : A \to A'$ such that $\varphi \circ \psi = id_A$, then ψ extends uniquely to $\hat{\psi} : \hat{A} \to A'$ such that $\varphi \circ \hat{\psi} = id_{\hat{A}}$. This proves that

$$D^1(\hat{A}|R, \hat{A}\otimes_A E) \to D^1(A|R, \hat{A}\otimes_A E)$$

is injective. Now let $0 \to \hat{A}\otimes_A E \to B \xrightarrow{\varphi} A \to 0$ be an R-algebra extension. Firstly we claim that the topology on $\hat{E} = \hat{A}\otimes_A E$ induced from $\underline{m}_B$-adic topology on B coincides with the topology on $\hat{E}$ as $\hat{A}$-module. Indeed, since A is essentially of finite type over R, one can find a subalgebra B_0 of B such that B_0 is a local R-algebra of essentially finite type over R and that $\varphi(B_0) = A$. Set $E_0 = E \cap B_0$ and denote by $\underline{m}_B$, $\underline{m}_0$, $\underline{m}$ the maximal ideals of B, B_0, A respectively. Then $\underline{m}_B = \underline{m}_0 + \hat{E}$ and hence $\underline{m}_B^{n+1} = \underline{m}_0^{n+1} + \underline{m}^n\hat{E}$ for all n. Since B_0 is noetherian, there exists an integer $r > 0$ by Artin-Rees Theorem such that

$$\hat{E} \cap \underline{m}_B^{n+1} = \underline{m}^n\hat{E} + \hat{E} \cap \underline{m}_0^{n+1} = \underline{m}^n\hat{E} + E_0 \cap \underline{m}_0^{n+1}$$
$$= \underline{m}^n\hat{E} + \underline{m}^{n+1-r}(E_0 \cap \underline{m}_0^r) \subset \underline{m}^{n+1-r}\hat{E}$$

for all $n \geq r$, which proves our assertion about the two topologies on E. In particular $\hat{E} = \hat{A}\otimes_A E$ is complete with respect to the topology induced from $\underline{m}_B$-adic topology on B, and hence we obtain the exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \hat{E} & \longrightarrow & B & \longrightarrow & A & \longrightarrow & 0 \\ & & \| & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \hat{E} & \longrightarrow & \hat{B} & \longrightarrow & \hat{A} & \longrightarrow & 0 \end{array}$$

where $\hat{B}$ stands for $\underline{m}_B$-adic completion of B. This proves that D^1 $D^1(\hat{A}|R, \hat{E}) \to D^1(A|R, \hat{E})$ is surjective and hence we are done.

Corollary 3.14. Let R be noetherian and A an R-algebra of finite type, smooth everywhere except at one closed point $x \in \mathrm{Spec}(A)$. We then have isomorphisms

$$D^1(A|R,\ A) \simeq D^1(A_x|R,\ A_x) \simeq D^1(\hat{A}_x|R,\ \hat{A}_x)\ .$$

Proof: Since A is smooth over R except at $x \in \mathrm{Spec}(A)$, $D^1(A|R,\ A)$ is an A-module of finite type with its support on the set $\{x\}$, and therefore we have

$$D^1(A|R,\ A) = D^1(A|R,\ A)_x = \widehat{D^1(A|R,\ A)_x}\ .$$

Therefore our assertion follows from 3.13.

Remark 3.15. Let R be topological ring and A a topological R-algebra. We then define, for any A-module M,

$$D^i_{top}(A|R,\ E) = \varinjlim_{\alpha} D^i(A_\alpha|R_\alpha,\ A_\alpha \otimes_\alpha E)$$

where $A_\alpha = A/J_\alpha$, $R_\alpha = R/I_\alpha$, and J_α, I_α are both open in A, R respectively. Then A is formally smooth topological R-algebra means that $D^1_{top}(A|R,\ E) = 0$ for every A-module E.

Now let X be a scheme over a ring R and $\underline{E}$ a quasi-coherent X-module. We then define the sheaves $\underline{D}^i_{X|R}(\underline{E}) = \underline{D}^i(X|R,\ \underline{E})$ on X via presheaves on X given by

$$[\underline{D}^i_{X|R}(\underline{E})](U) = D^i(\Gamma(U,\ \underline{O}_X)|R,\ \Gamma(U,\ \underline{E}))$$

for every affine open set $U \subset X$. If R is noetherian and X is of finite type over R, then the property 3.13(a) shows that $\underline{D}^i_{X|R}(\underline{E})$ can actually be constructed canonically, as a quasi-coherent X-module.

In case when there is no possible confusion we shall abreviate by setting $\underline{D}^i_{X|R} = \underline{D}^i_{X|R}(\underline{O}_X)$. All the statements of 3.13 can now be translated in terms of these quasi-coherent sheaves $\underline{D}^i_{X|R}(\underline{E})$, which we leave to the readers. We list below a few properties which is most relevant to our purpose.

Corollary 3.15. Let R be noetherian and X an R-scheme of finite type. Then

(a) For any coherent X-module $\underline{E}$, $\underline{D}^i_{X|R}(\underline{E})$ is again a coherent X-module for all i, and furthermore $\mathrm{Supp}\ \underline{D}^i_{X|R}(\underline{E}) \subset X^{\mathrm{Sing}}$ for all $i > 0$ where

$X^{\mathrm{Sing}} = \{x \in X \mid X$ is non-smooth over R at the point $x \in X\}$.

(b) If X is flat over R, then for any R-algebra S, we have the canonical isomorphism

$$\underline{D}^i_{X\otimes_R S|S}(\underline{E}) \xrightarrow{\sim} \underline{D}^i_{X|R}(\underline{E})$$

for all i .

(c) If X is a relative complete intersection over R, then for every surjective ring homomorphism $S \to R$ we have the exact sequence

$$0 \to \underline{D}^1_{X|R}(\underline{E}) \to \underline{D}^1_{X|S}(\underline{E}) \to \underline{\mathrm{Hom}}_{\underline{O}_X}(I/I^2\otimes_R \underline{O}_X,\ \underline{E}) \to 0$$

where $I = \mathrm{Ker}(S \to R)$. If, furthermore, R is local with the residue field k and X is R-flat, then we have the exact sequence

$$0 \to \underline{D}^1_{X_0|h}(\underline{O}_{X_0}) \to \underline{D}^1_{X|S}(\underline{O}_{X_0}) \to D^1(R|S,\ k)\otimes_k \underline{O}_{X_0} \to 0$$

where $X_0 = X\otimes_R k$.

Proof: (a) Follows from 3.12(6) and (8) whereas (b) follows from 3.12(3).

(c) Since X is a relative complete-intersection over R, we have $\underline{D}^i_{X|R}(\underline{E}) = 0$ for all $i \geq 2$ by 3.12(10) and hence

$$0 \to \underline{D}^0_{X|R}(\underline{E}) \to \underline{D}^0_{X|S}(\underline{E}) \to \underline{D}^0_{R|S}(\underline{E}) \to \underline{D}^1_{X|R}(\underline{E}) \to \underline{D}^1_{X|S}(\underline{E}) \to \underline{D}^1_{R|S}(\underline{E}) \to 0$$

is exact by 3.12(2), where $\underline{D}^i_{R|S}(\underline{E})$ are quasi-coherent sheaves of X defined by

$$[\underline{D}^i_{R|S}(\underline{E})](U) = D^i(R|S,\ \underline{E}(U))$$

for every affine open subset $U \subset X$. In particular, $\underline{D}^0_{R|S}(\underline{E}) = 0$ since $S \to R$ is surjective, and

$$\underline{D}^1_{R|S}(\underline{E}) = \underline{\mathrm{Hom}}_{\underline{O}_X}(I/I^2\otimes_R \underline{O}_X,\ \underline{E})$$

by 3.12(7), and hence the first exact sequence. If X is R-flat, we have

$\underline{D}^i_{X_0|k}(\underline{E}) \xrightarrow{\sim} \underline{D}^i_{X|R}(\underline{E})$ for all i and for all quasi-coherent X_0-module $\underline{E}$ and in particular

$$\underline{D}^1_{X_0|k}(\underline{O}_{X_0}) \xrightarrow{\sim} \underline{D}^1_{X|R}(\underline{O}_{X_0})$$

from which the second exact sequence follows.

Remark 3.16: The quasi-coherent sheaves $\underline{D}^i_{X|R}(\underline{E})$ we have defined should appear as $E_2^{0,i}$ of a spectral sequence relating the local ones and the global ones. L. Illusie has obtained such a theory (Lecture Notes in Mathematics 239). His method is via "homotopical algebra" generalizing the methods of M. Andre and D. Quillen rather than "cohomological algebra". It seems that the approach we have taken here can be globalized via Grothendieck cohomology on the category of schemes with an appropriate topology generalizing the one on the affine category.

4. Formal moduli of deformations

One amongst the nice cofibered categories in which the isomorphism classes of objects are usually not prorepresentable, but nevertheless admit a formal versal object is the cofibered category of deformations. We now come back to the situations and notations of paragraph 1. Thus, throughout the section, Λ is a fixed complete local noetherian ring with the residue field k, and $\underline{C}_\Lambda$ is the category of artinian local Λ-algebras with the same residue field k.

Let X_0 be a fixed scheme over the field k. By an (infinitesimal) deformation of X_0, we mean a pair (R, X) where $R \in ob(\underline{C}_\Lambda)$ and X is a flat R-scheme together with a commutative diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow \text{flat} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that the induced morphism $X_0 \to X \times_{\mathrm{spec}(R)} \mathrm{Spec}(k)$ is an isomorphism. We note that, since R is an artinian local ring, $i_X : X_0 \to X$ is necessarily a closed immersion which is a homeomorphism on underlying topological spaces. If (R, X), (R', X') are deformations of X_0 to R, R' respectively, we define an arrow $(R', X') \to (R, X)$ to consist of a map $R' \xrightarrow{\varphi} R$ in $\underline{C}_\Lambda$ and a morphism $\Phi : X \to X'$ of schemes with a commutative diagram

$$\begin{tikzcd}
X \arrow[rr, "\Phi"] \arrow[ddd] & & X' \arrow[ddd] \\
& X_0 \arrow[ul, "i_X"'] \arrow[ur, "i_{X'}"] \arrow[d] & \\
& \mathrm{Spec}(k) \arrow[dl] \arrow[dr] & \\
\mathrm{Spec}(R) \arrow[rr, "\mathrm{Spec}(\varphi)"'] & & \mathrm{Spec}(R')
\end{tikzcd}$$

We thus obtain the category $\underline{M}_{X_0}$ of (infinitesimal) deformations of X_0 in which the objects and the arrows are defined as above. The category $\underline{M}_{X_0}$ is provided with a functor $p : \underline{M}_{X_0} \to \underline{C}_\Lambda$ defined by $(R, X) \to R$. Now let $(R, X) \in \mathrm{ob}(\underline{M}_{X_0})$ and a map $R \xrightarrow{\varphi} S$ in $\underline{C}_\Lambda$ be given. We then obtain a commutative diagram

$$\begin{tikzcd}
X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(S) \arrow[r, "p_1"] \arrow[d, "\text{flat}"'] & X \arrow[d, "\text{flat}"] \\
\mathrm{Spec}(S) \arrow[r, "\mathrm{Spec}(\varphi)"'] & \mathrm{Spec}(R)
\end{tikzcd}$$

and φ-arrow $(\varphi, p_1) : (R, X) \to (S, X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(S))$ is obviously cocartesian in $\underline{M}_{X_0}$. It follows that the category $\underline{M}_{X_0}$ is a cofibered category over $\underline{C}_\Lambda$ via $P : \underline{M}_{X_0} \to \underline{C}_\Lambda$. We observe the following facts:

4.1(a): Let $(1_R, \Phi) : (R, X_1) \to (R, X_2)$ be an arrow in $\underline{M}_{X_0}$ i.e., $\Phi : X_2 \to X_1$ is a morphism over R such that $X_2 \otimes_R k \to X_1 \otimes_R k$ is an isomorphism. Since R is artinian and X_i is R-flat, it follows immediately that Φ is an isomorphism so that $(1_R, \Phi) : (R, X_1) \to (R, X_2)$ is an isomorphism in $\underline{M}_{X_0}$. Consequently, $\underline{M}_{X_0}$ is a cofibered category in groupoids over $\underline{C}_\Lambda$.

4.1(b): Suppose that X_0 is affine, say $X_0 = \mathrm{Spec}(A_0)$. Then for any object $(R, X) \in \mathrm{ob}(\underline{M}_{X_0})$, X is also affine. Consequently, for any $R \in \mathrm{ob}(\underline{C}_\Lambda)$, the category $\underline{M}_{X_0}(R)$ is canonically equivalent, via the sections, to the category in which an object is a flat algebra A over R together with an R-algebra map $j_A : A \to A_0$ inducing an isomorphism $A \otimes_R k \xrightarrow{\sim} A_0$, and a map $\Phi : A_1 \to A_2$ is an R-algebra map with a commutative diagram

$$\begin{tikzcd} A_1 \arrow[r, "\Phi"] \arrow[d, "j_{A_1}"'] & A_2 \arrow[d, "j_{A_2}"] \\ A_0 \arrow[r, equal] & A_0 \end{tikzcd}$$

4.1(c): Consider the cofibered category $\hat{\underline{M}}_{X_0}$ in groupoids over $\hat{\underline{C}}_\Lambda$. By definition, the category $\hat{\underline{M}}_{X_0}(R)$ for any $R \in \mathrm{ob}(\hat{\underline{C}}_\Lambda)$ is canonically equivalent to the category of formal schemes $\mathfrak{X}$, adic over the formal spectrum $\mathrm{Spf}(R)$, such that $\mathfrak{X}_n = \mathfrak{X} \otimes_R R/\underline{m}^{n+1}$ is flat over $R/\underline{m}^{n+1}$ for every $n \geq 0$ (or what amounts to the same, if X_0 is noetherian, such that $\mathfrak{X}$ is flat over $\mathrm{Spf}(R)$). Consequently, if X_0 is noetherian, $\hat{\underline{M}}_{X_0}$ is canonically $\hat{\underline{C}}_\Lambda$-equivalent to the category of deformations of X_0 over the objects in $\hat{\underline{C}}_\Lambda$, in the sense of formal schemes.

4.1(d): Let $(R, X) \in \mathrm{ob}(\underline{M}_{X_0})$ i.e., we have a commutative diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow \text{flat} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that $X_0 \to X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k)$ is an isomorphism. Now let $U_0 \subset X_0$ be a fixed open subscheme of X_0. Since $i_X : X_0 \to X$ is a closed immersion which is a homeomorphism on the underlying topological spaces, we obtain a commutative diagram

$$\begin{array}{ccc} U_0 & \xrightarrow{i_X|U_0} & X|U_0 \\ \downarrow & & \downarrow \text{flat} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that $U_0 \to (X|U_0) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k) = X \otimes_R k|U_0$ is an isomorphism, so that $(R, X|U_0)$ is a deformation of U_0 to R. Furthermore, for any arrow $(\varphi, \Phi) : (R, X) \to (R', X')$ in $\underline{M}_{X_0}$, $(\varphi, \Phi|U_0) : (R, X|U_0) \to (R', X'|U_0)$ is an arrow in $\underline{M}_{U_0}$. Consequently we have the restriction functor $\underline{M}_{X_0} \to \underline{M}_{U_0}$ over $\underline{C}_\Lambda$. One may also consider a localization as well as a formalization at a point. Namely let x_0 be a fixed point in X_0. If $(R, X) \in \mathrm{ob}(\underline{M}_{X_0})$, then $\mathrm{Spec}(\underline{O}_{X,x_0})$ is a deformation of $\mathrm{Spec}(\underline{O}_{X_0,x_0})$ to R and $\mathrm{Spec}(\hat{\underline{O}}_{X,x_0})$ is a deformation of $\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_0})$ over R, where ^ stands for the completion with respect to the linear topology given by the powers of the maximal ideal. We thus obtain functors $\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0,x_0})} \to \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_0})}$.

Firstly we establish that $\underline{M}_{X_0}$ is a homogeneous cofibered category over $\underline{C}_\Lambda$.

Lemma 4.2. Consider a diagram

$$\begin{array}{ccc} & & R' \\ & & \downarrow \\ R & \xrightarrow[\text{surjective}]{} & \bar{R} \end{array}$$

in $\underline{C}_\Lambda$, and let E, E' be flat modules over R, R' respectively. Then for any R'-homomorphism $E' \to \bar{E} = E \otimes_R \bar{R}$, the fibre-product $E \times_{\bar{E}} E'$ is flat as $R \times_{\bar{R}} R'$-module

Proof: Let $0 \to I \to R \to \bar{R} \to 0$ be exact so that $0 \to I \to R \times_{\bar{R}} R' \to R' \to 0$ is exact. Since E is R-flat, $0 \to I \otimes_R E \to E \to \bar{E} \to 0$ is exact and hence we obtain the exact sequence

$$(*) \qquad 0 \to I \otimes_R E \to E \times_{\bar{E}} E' \to E' \to 0 \ .$$

Set $S = R \times_{\bar{R}} R$. Since I is nilpotent, $E \times_{\bar{E}} E'$ is flat as an S-module if and only if $(E \times_{\bar{E}} E') \otimes_S R'$ is flat as an R'-module and $\mathrm{Tor}_1^S (E \times_{\bar{E}} E', R') = 0$. However, we note that $I \otimes_R E = I \otimes_S (E \times_{\bar{E}} E')$ and $E' = R' \otimes_S (E \times_{\bar{E}} E')$ and therefore the exact sequence (*) can be rewritten as

$$0 \to I \otimes_S (E \times_{\bar{E}} E') \to E \times_{\bar{E}} E' \to R' \otimes_S (E \times_{\bar{E}} E') \to 0 \ .$$

This means that $\mathrm{Tor}_1^S (R', E \times_{\bar{E}} E') = 0$ and $R' \otimes_S (E \times_{\bar{E}} E') = E'$ is flat as R'-module, and therefore $E \times_{\bar{E}} E'$ is S-flat.

Corollary 4.3. Given a diagram

$$\begin{tikzcd}
(R, X) \arrow[dr, "{(\varphi, \Phi)}"'] & & (R', X') \arrow[dl, "{(\varphi', \Phi')}"] \\
& (\bar{R}, \bar{X}) &
\end{tikzcd}$$

in $\underline{M}_{X_0}$, the amalgamated sum $X \coprod_{\bar{X}} X'$ (in the sense of schemes) is a flat scheme over $R \times_{\bar{R}} R'$, provided $\varphi : R \to \bar{R}$ is surjective. Consequently, $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Proof: For each affine open $U_0 \subset X_0$, we have by definition that $X \coprod_{\bar{X}} X' | U_0 \simeq \mathrm{Spec}(\Gamma(U_0, X) \times_{\Gamma(U_0, \bar{X})} \Gamma(U_0, X'))$. Consequently, if $\varphi : R \to \bar{R}$ is surjective, $X \coprod_{\bar{X}} X'$ is a flat scheme over $R \times_{\bar{R}} R'$ by 4.2, and the morphism $i_X \coprod_{i_{\bar{X}}} i_{X'} : X_0 \to X \coprod_{\bar{X}} X'$ induces an isomorphism $X_0 \to (X \coprod_{\bar{X}} X') \otimes_S k$, where $S = R \times_{\bar{R}} R'$. Therefore $(R \times_{\bar{R}} R', X \coprod_{\bar{X}} X') \in \mathrm{ob}(\underline{M}_{X_0})$ and we obviously have that $(R \times_{\bar{R}} R', X \coprod_{\bar{X}} X') = (R, X) \times_{(\bar{R}, \bar{X})} (R', X')$ in the category $\underline{M}_{X_0}$, which means that $\underline{M}_{X_0}$ is homogeneous over $\underline{C}_\Lambda$.

Lemma 4.4. For any scheme X_0 over k , we have a canonical exact sequence

$$0 \to H^1(X_0, \underline{D}^0_{X_0}) \to [\underline{M}_{X_0}(k[\epsilon])] \to H^0(X_0, \underline{D}^1_{X_0})$$

where $\underline{D}^i_{X_0}$ are as defined in paragraph 3.

Proof: (Special case) Assume that X_0 is affine and let $\Gamma(X_0, \underline{O}_{X_0}) = A_0$. Then as it was noted in 4.1(b), the category $\underline{M}_{X_0}(k[\epsilon])$ is canonically equivalent to the category $\underline{E}$, where

ob($\underline{E}$) : object in $\underline{E}$ is a flat algebra A over $k[\epsilon]$

together with a $k[\epsilon]$ -algebra map $j_A : A \to A_0$ inducing the isomorphism $k \otimes_{k[\epsilon]} j_A : k \otimes_{k[\epsilon]} A \xrightarrow{\sim} A_0$. Alternatively, an object in $\underline{E}$ is a k-algebra A together with an exact sequence $0 \to A_0 \xrightarrow{\epsilon} A \xrightarrow{j_A} A_0 \to 0$ in which $\epsilon(A_0)^2 = 0$ as an ideal in A .

Fl($\underline{E}$) : An arrow $A_1 \to A_2$ in $\underline{E}$ with respect to first description of the object in $\underline{E}$ is a $k[\epsilon]$-algebra map $\Phi : A_1 \to A_2$ such that $j_{A_1} = j_{A_2} \circ \Phi$. Therefore, an arrow $A_1 \to A_2$ in $\underline{E}$ with respect to the alternative description of objects in $\underline{E}$ is a k-algebra map $\Phi : A_1 \to A_2$ yielding a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & A_0 \arrow[r, "\epsilon"] \arrow[d, equal] & A_1 \arrow[r, "j_{A_1}"] \arrow[d, "\Phi"] & A_0 \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & A_0 \arrow[r] & A_2 \arrow[r, "j_{A_2}"] & A_0 \arrow[r] & 0
\end{tikzcd}$$

It follows that $[\underline{M}_{X_0}(k[\epsilon])]$ is canonically isomorphic, via $\Gamma(X_0, -)$, to the set of isomorphism classes of k-algebra extensions of A_0 by A_0 . In other words, we have a canonical bijection $[\underline{M}_{X_0}(k[\epsilon])] \xrightarrow{\sim} H^0(X_0, \underline{D}^1_{X_0})$ via $\Gamma(X_0, -)$, which is trivially seen to respect the vector space structures on both sides.

(General case) For each affine open $U_0 \subset X_0$, we get a canonical map $[\underline{M}_{X_0}(k[\epsilon])] \to H^0(U_0, \underline{D}^1_{X_0})$ defined as the composite map $[\underline{M}_{X_0}(k[\epsilon])] \to [\underline{M}_{U_0}(k[\epsilon])] \xleftarrow{\sim} H^0(U_0, \underline{D}^1_{X_0})$ and hence we obtain a canonical map $[\underline{M}_{X_0}(k[\epsilon])] \to H^0(X_0, \underline{D}^1_{X_0})$. Its kernel is the set of isomorphism classes of locally-trivial deformations of X_0 to $k[\epsilon]$, i.e., the set of isomorphism classes of deformations of X_0 to $k[\epsilon]$, locally (on X) isomorphic to $X_0 \otimes_k k[\epsilon]$. Since $\underline{D}^0_{X_0} = \underline{\mathrm{Der}}_k(\underline{O}_{X_0}, \underline{O}_{X_0}) = \underline{\mathrm{Aut}}(X_0 \otimes_k k[\epsilon]|X_0)$ = the sheaf of germs of automorphisms of $X_0 \otimes_k k[\epsilon]$ inducing the identity on the subscheme X_0 , it is canonically isomorphic to $H^1(X_0, \underline{D}^0_{X_0})$, so that we obtain the desired exact sequence.

Theorem 4.5. Let X_0 be a scheme of finite type over k for which either (i) X_0 is proper over k or (ii) X_0 is affine but $X_0^{\mathrm{sing}} = \{x \in X_0 | X_0$ is non-smooth over k at the point $x\}$ is a finite set. Then

(a) $\underline{M}_{X_0}$ admits a formal versal object (called a formal versal deformation of X_0) over $\underline{C}_\Lambda$. If $(R, \mathfrak{X})$ is a formal versal deformation of X_0 , then for any $(S, \mathfrak{Y}) \in \mathrm{ob}(\hat{\underline{M}}_{X_0})$, there is a map $(\varphi, \Phi) : (R, \mathfrak{X}) \to (S, \mathfrak{Y})$ in $\hat{\underline{M}}_{X_0}$ such that the diagram below is cartesian

$$\begin{tikzcd} \mathfrak{Y} \arrow[r, "\Phi"] \arrow[d] & \mathfrak{X} \arrow[d] \\ \mathrm{Spf}(S) \arrow[r, "\mathrm{Spf}(\varphi)"'] & \mathrm{Spf}(R) \end{tikzcd} .$$

Furthermore a formal versal deformation $(R, \mathfrak{X})$ of X_0 prorepresents the functor $[\underline{M}_{X_0}]$ if and only if, for every surjective map $R' \to R$ in $\underline{C}_\Lambda$ and a deformation (R', X') of X_0 to R' , the homomorphism

$$\mathrm{Aut}_{R'}(X'|X_0) \to \mathrm{Aut}_R(X' \otimes_{R'} R|X_0)$$

is surjective.

(b) If X_0 is proper over k , then $\underline{M}_{X_0}$ is smoothly pseudo-representable over $\underline{C}_\Lambda$.

Proof: (a) Since $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$ by 4.3, it suffices to see that $[\underline{M}(k[\epsilon])]$ is a finite-dimensional vector space over k by 1.11 and 2.9(b).

Now consider the exact sequence

$$(*) \qquad 0 \to H^1(X_0, \underline{D}^0_{X_0}) \to [\underline{M}_{X_0}(k[\epsilon])] \to H^0(X_0, \underline{D}^1_{X_0}) \ .$$

If X_0 is proper over k , then $H^1(X_0, \underline{D}^0_{X_0})$ and $H^0(X_0, \underline{D}^1_{X_0})$ are both finite dimensional since $\underline{D}^i_{X_0}$ are coherent for all $i \geq 0$ by 2.13(a). Now assume that X_0 is affine such that X_0^{sing} is a finite set. Since $\underline{D}^1_{X_0}$ is a coherent sheaf on X_0 such that $\operatorname{Supp} \underline{D}^1_{X_0} \subset X_0^{sing}$ by 2.13(a), we find that $H^0(X_0, \underline{D}^1_{X_0})$ is a finite-dimensional vector space over k , whereas $H^1(X_0, \underline{D}^0_{X_0}) = 0$ since X_0 is affine. Therefore, in either case, it follows from the exact sequence (*) that $[\underline{M}_{X_0}(k[\epsilon])]$ is a finite dimensional vector space over k . The second statement of (a) follows from 2.7.

(b) If X_0 is proper, $\operatorname{Aut}(X_0 \otimes_k k[\epsilon] | X_0) = H^0(X_0, \underline{D}^0_{X_0})$ is a finite dimensional vector space and hence our statement follows from 2.13.

Remark 4.6(a): Let X_0 be affine. Then a formal versal deformation of X_0 exists if and only if $\underline{D}^1_{X_0}$ has its support on a finite number of points. However this does not imply that non-smooth points of X_0 are all isolated. There are affine schemes X_0 over k having a non-isolated non-smooth point but $\operatorname{supp} \underline{D}^1_{X_0}$ is a finite set.

Remark 4.6(b): Henceforth a deformation of X_0 is meant by an object in $\underline{M}_{X_0}$. In case when there is a possible confusion, an object in $\underline{M}_{X_0}$ will be referred as an infinitesimal deformation of X_0 .

Remark 4.6(c): Let X_0 be a scheme over k . We have fixed a complete local noetherian ring Λ with the residue field k , and considered the deformations of X_0 over the category $\underline{C}_\Lambda$. (In practice one usually takes $\Lambda = k$ or $W(k)$ if k is a perfect field of positive characteristic). Now $\underline{C}_k \subset \underline{C}_\Lambda$ is a fully faithful imbedding, and that $\underline{M}_{X_0} | \underline{C}_k$ is simply the category of deformations of X_0 over S where $S \in \mathrm{ob}(\underline{C}_k)$. Therefore if $(R, \mathfrak{X})$ is a formal versal deformation of X_0 over $\underline{C}_\Lambda$, then $(k \otimes_\Lambda R, k \otimes_\Lambda \mathfrak{X})$ is a formal versal deformation of X_0 over $\underline{C}_k$. It follows, for example, that $\dim_{\underline{C}_\Lambda} \underline{M}_{X_0} = \dim k \otimes_\Lambda R$ does not depend on the choice of Λ but depends only on X_0 .

One can generalize the above theorem in many ways. Let us for instance consider a pair $Y_0 \to X_0$ where X_0 is a scheme of finite type over k and Y_0 a fixed closed subscheme of X_0 . For each $R \in \mathrm{ob}(\underline{C}_\Lambda)$, a deformation of (X_0, Y_0) to R is meant to be a commutative diagram

$$\begin{tikzcd}
Y_0 \arrow[rrr] \arrow[dr, hook] \arrow[ddr] & & & Y \arrow[dl, hook] \arrow[dd, "\text{flat}"] \\
& X_0 \arrow[r] \arrow[d] & X \arrow[d, "\text{flat}"] & \\
& \mathrm{Spec}(k) \arrow[r] & \mathrm{Spec}(R) &
\end{tikzcd}$$

such that $(X_0, Y_0) \to (X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k), Y \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k))$ is an isomorphism. Defining a morphism $(R'; X', Y') \to (R: X, Y)$ of deformations of (X_0, Y_0) to be the obvious ones, we obtain the cofibered category $\underline{M}_{X_0, Y_0}$ in groupoids over $\underline{C}_\Lambda$. (This is a process often used in rigidification of automorphisms in moduli problems of deformations.) One sees immediately, by the same argument as in 4.3 based on 4.2, that $\underline{M}_{X_0, Y_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Lemma 4.7. $0 \to H^0(Y_0, \underline{\mathrm{Hom}}_{\underline{O}_{Y_0}}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0})) \to [\underline{M}_{X_0, Y_0}(k[\varepsilon])] \to [\underline{M}_{X_0}(k[\varepsilon])]$ is exact, where $\underline{I}$ is the ideal sheaf of $\underline{O}_{X_0}$ defining Y_0 i.e.

$0 \to \underline{I} \to \underline{O}_{X_0} \to \underline{O}_{Y_0} \to 0$ <u>is exact.</u>

<u>Proof</u>: Let $\underline{B}$ be the full subcategory of $\underline{M}_{X_0,Y_0}(k[\varepsilon])$ whose objects are of the form $(k[\varepsilon];\ X \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon]),\ Y)$. We then have $[B] = \mathrm{Ker}\ ([\underline{M}_{X_0,Y_0}(k[\varepsilon])] \to [\underline{M}_{X_0}(k[\varepsilon])]\ )$. The category $\underline{B}$ is the groupoid of deformations of Y_0 to $k[\varepsilon]$ "inside $X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon])$ ", i.e., an object in $\underline{B}$ is a commutative diagram

$$\begin{tikzcd} Y \arrow[rr, "i_Y"] \arrow[dr, "\text{flat}"'] & & X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon]) \arrow[dl] \\ & \mathrm{Spec}(k[\varepsilon]) & \end{tikzcd}$$

such that $\mathrm{Spec}(k) \underset{\mathrm{Spec}(k[\varepsilon])}{\times} i_Y = i_{Y_0}$, and an arrow $(Y_1, i_{Y_1}) \to (Y_2, i_{Y_2})$ is a $k[\varepsilon]$-morphism $\Phi : Y_1 \to Y_2$ (necessarily an isomorphism) such that $\mathrm{Spec}(k) \underset{\mathrm{Spec}(k[\varepsilon])}{\times} \Phi = \mathrm{id}_{Y_0}$ and $i_{Y_2} \circ \Phi = i_{Y_1}$. Alternatively, an object in $\underline{B}$ is a pair (Y, j_Y) , where Y is a deformation of Y_0 to $k[\varepsilon]$ and $j_Y : Y \to X_0$ is a morphism over k such that the composite morphism $Y_0 \to Y \xrightarrow{j_Y} X_0$ is i_{Y_0} , and an arrow $(Y_1, j_{Y_1}) \to (Y_2, j_{Y_2})$ is a morphism $\Phi : Y_1 \to Y_2$ such that $k \underset{k[\varepsilon]}{\otimes} \Phi = \mathrm{id}_{Y_0}$ and $j_{Y_2} \circ \Phi = j_{Y_1}$. Therefore the association

$$(Y, j_Y) \longmapsto \begin{tikzcd} 0 \arrow[r] & \underline{I} \arrow[r] \arrow[d] & \underline{O}_{X_0} \arrow[r] \arrow[d, "\mathfrak{J}_Y"] & \underline{O}_{Y_0} \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & \underline{O}_{Y_0} \arrow[r, "\varepsilon"] & \underline{O}_Y \arrow[r] & \underline{O}_{Y_0} \arrow[r] & 0 \end{tikzcd}$$

gives rise to an isomorphism

$$[B] \xrightarrow{j|\underline{I}} \mathrm{Hom}_{\underline{O}_{X_0}}(\underline{I}, \underline{O}_{Y_0}) = \mathrm{Hom}_{\underline{O}_{Y_0}}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0}) = H^0(Y_0, \underline{\mathrm{Hom}}_{\underline{O}_{Y_0}}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0})) \ .$$

Remark 4.8. For a more general formula implying the above lemma, see SGA3 III 4.4. One may also note the following fact. One can define for any S-scheme X and a quasi-coherent X-module $\underline{E}$ a quasi-coherent X-module $\underline{D}^i_{X|S}(\underline{E})$ for all $i \geq 0$, generalizing the case when S is affine. As usual we simply write $\underline{D}^i_{X|S}(\underline{O}_X) = \underline{D}^i_{X|S}$. We then have

$$H^0(Y_0, \underline{\mathrm{Hom}}_{Y_0}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0})) = H^0(Y_0, \underline{D}^1_{Y_0|X})$$

where $0 \to \underline{I} \to \underline{O}_{X_0} \to \underline{O}_{Y_0} \to 0$ is exact.

Theorem 4.9. Let X_0 be k-scheme of finite type and Y_0 a closed subscheme of X_0. Assume that either (i) X_0 is proper over k or (ii) X_0 is affine and smooth over k outside a finite number of points and Y_0 is proper over k. We then have

(a) $\underline{M}_{X_0, Y_0}$ admits a formal versal deformation over $\underline{C}_\wedge$, and it pro-represents the functor $[\underline{M}_{X_0, Y_0}]$ if and only if, for every surjective map $R' \to R$ in $\underline{C}_\wedge$ and a deformation $(R'; X', Y')$,

$$\mathrm{Aut}_{R'}((X', Y')|X_0) \to \mathrm{Aut}_R(X' \otimes_{R'} R, Y' \otimes_{R'} R)|X_0)$$

is surjective.

(b) If furthermore $H^0(X_0, \underline{D}^0_{Y_0|k} \times_{\underline{D}^0_{X_0|k}(\underline{O}_{Y_0})} \underline{D}^0_{X_0|k})$ is finite dimensional, then $\underline{M}_{X_0, Y_0}$ is smoothly pseudo-representable.

Proof: It is obvious from 4.4 and 4.6 together with 1.11, 2.7, and 2.13.

We now consider the passage to localization and completion in the deformation theory of schemes. Firstly on a notation: Let X_0 be a k-scheme of finite type. Given a fixed point $x_0 \in X_0$, we have natural functors

$$\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0, x_0})} \to \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0, x_0})}$$

and in turn we obtain the functors

$$\underline{\hat{M}}_{X_0} \to \underline{\hat{M}}_{\mathrm{Spec}(\underline{O}_{X_0},\, x_0)} \to \underline{\hat{M}}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0},\, x_0)} \ .$$

As it was remarked in 4.1(c), $\underline{M}_{X_0}$ is canonically equivalent to the category in which an object is a pair $(R, \mathfrak{X})$ where $R \in \mathrm{ob}(\underline{C}_\Lambda)$ and $\mathfrak{X}$ is a flat R-adic formal scheme together with a fixed isomorphism $\mathfrak{X}_0 \longleftrightarrow X_0$. Henceforth we shall write, when there is no possible confusion, the images of $(R, \mathfrak{X})$ under $\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0},\, x_0)}$ and $\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0},\, x_0)}$ by $(R, \mathfrak{X}_{(x_0)})$ and $(R, \mathfrak{X}_{(x_0)})$ respectively. Therefore if $\mathfrak{X} = \varinjlim_v \mathfrak{X}_v$ where $\mathfrak{X}_v = \mathfrak{X} \times_{\mathrm{Spf}(R)} \mathrm{Spec}(R/\underline{m}^{v+1})$ then

$$\mathfrak{X}_{(x_0)} = \varinjlim_v \mathrm{Spec}(\underline{O}_{\mathfrak{X}_v},\, x_0)$$

$$\mathfrak{X}_{(\hat{x}_0)} = \varinjlim_v \mathrm{Spec}(\hat{\underline{O}}_{\mathfrak{X}_v},\, x_0) \ .$$

<u>Theorem</u> 4.10. <u>Let</u> X_0 <u>be an affine scheme of finite type over</u> k .

(a) <u>If</u> X_0 <u>is locally a complete-intersection, then</u> $\underline{M}_{X_0}$ <u>is smooth over</u> $\underline{C}_\Lambda$.

(b) <u>Assume that, for a fixed closed point</u> $x \in X_0$, $X_0 - \{x\}$ <u>is smooth over</u> k . <u>Then</u>

$$\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0},\, x)} \to \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0},\, x)}$$

<u>are both minimally smooth functors. In particular, if</u> $(R, \mathfrak{X})$ <u>is a formal versal deformation of</u> X_0 , <u>then</u> $(R, \mathfrak{X}_{(x)})$ <u>and</u> $(R, \mathfrak{X}_{(\hat{x})})$ <u>are formal versal deformations of</u> $\mathrm{Spec}(\underline{O}_{X_0},\, x)$ <u>and</u> $\mathrm{Spec}(\hat{\underline{O}}_{X_0},\, x)$ <u>respectively</u>.

<u>Proof</u>: In this proof we go back to the notations of the cohomology of commutative algebras defined in the paragraph 3. We set $X_0 = \mathrm{Spec}(A_0)$

(a) It suffices to show that $P : [\underline{M}_{X_0}] \to \underline{C}_\Lambda$ is smooth.

Let $(R, \mathrm{Spec}(A))$ be a deformation of $\mathrm{Spec}(A_0)$ to R, where $R \in ob(\underline{C}_\Lambda)$. Given a small extension $R' \to R$ in $\underline{C}_\Lambda$ (i.e., $0 \to k \to R' \to R \to 0$ is exact), we must show that $(R, \mathrm{Spec}(A))$ can be lifted to R'. The exact sequence of D^i with coefficient A_0 applied to the ring homomorphisms $R' \to R \to A$ gives us the exact sequence

$$\dots \to D^1(A|R, A_0) \to D^1(A|R', A_0) \to D^1(R|R', k) \otimes_k A_0 \to D^2(A|R, A_0) \to \dots$$

Since $R' \to R$ is a small extension, $D^1(R'|R, k)$ is 1-dimensional vector space over k generated by the element (denoted by $[R']$) represented by the extension $0 \to k \to R' \to R \to 0$. On the other hand, A is R-flat and hence $D^i(A|R, A_0) = D^i(A_0|k, A_0)$ for all $i > 0$ and in particular $D^2(A|R, A_0) = 0$ by 3.12(11) since A_0 is locally a complete-intersection. Consequently we get the exact sequence

$$\to D^1(A_0|k, A_0) \to D^1(A|R', A_0) \to D^1(R|R', k) \otimes_k A_0 \to 0$$

This means that there exists an R'-algebra extension of A by A_0, denoted by A', such that $[A'] \to [R'] \otimes 1$ under the map $D^1(A|R', A_0) \to D^1(R|R', k) \otimes_k A_0$, which means that the deformation $(R, \mathrm{Spec}(A))$ can be lifted to a deformation $(R', \mathrm{Spec}(A'))$.

(b) By 3.14 we have the isomorphisms

$$D^1(A_0|k, A_0) \simeq D^1((A_0)_x|k, (A_0)_x) \simeq D^1(\widehat{(A_0)_x}|k, \widehat{(A_0)_x})$$

i.e., $[\underline{M}_{X_0}(k[\epsilon])] \simeq [\underline{M}_{\mathrm{Spec}(\underline{O}_{X_0, x})}(k[\epsilon])] \simeq [\underline{M}_{\mathrm{Spec}(\widehat{\underline{O}_{X_0, x}})}(k[\epsilon])]$.

It suffices to show, by 2.7(a), that

$$[\underline{M}_{X_0}] \to [\underline{M}_{\mathrm{Spec}(\underline{O}_{X_0, x})}] \to [\underline{M}_{\mathrm{Spec}(\widehat{\underline{O}_{X_0, x}})}]$$

are both smooth functors. Let $(R, \mathrm{Spec}(A))$ be a deformation of $X_0 = \mathrm{Spec}(A_0)$, and let $0 \to k \to R' \to R \to 0$ be a small extension in $\underline{C}_\Lambda$. As above, the ring homomorphisms $R' \to R \to A$ gives us the exact commutative diagram

$$\begin{array}{ccccccc}
\to D^1(A_0|k,A_0) & \to & D^1(A|R',A_0) & \xrightarrow{p} & D^1(R|R',k)\otimes_k A_0 & \to & D^2(A_0|k,A_0) \to \cdots \\
\downarrow\approx & & \downarrow & & \downarrow & & \downarrow\approx \\
\to D^1(A_0|k,(A_0)_x) & \to & D^1(A|R',(A_0)_x) & \to & D^1(R|R',k)\otimes_k (A_0)_x & \to & D^2(A_0|k,(A_0)_x) \to \cdots \\
\uparrow & & \uparrow & & \uparrow & & \uparrow \\
\to D^1((A_0)_x|k,(A_0)_x) & \to & D^1(A_x|R',(A_0)_x) & \xrightarrow{p_x} & D^1(R|R',k)\otimes_k (A_0)_x & \to & D^2((A_0)_x|k,(A_0)_x) \to \cdots
\end{array}$$

where the indicated isomorphism comes from the fact that A_0 is smooth over k except at one closed point x so that

$$D^i(A_0|k,\ (A_0)_x) = D^i(A_0|k,\ A_0)_x = D^i(A_0|k,\ A_0)$$

for all $i > 0$. Now suppose that we are given an arrow $(R',\ \mathrm{Spec}(B)) \to (R,\ \mathrm{Spec}(A_x))$ in $[\underline{M}_{\mathrm{Spec}((A_0)_x)}]$. It means that we are given a class $[B] \in D^1(A_x|R',\ (A_0)_x)$ such that $p_x([B]) = [R']$. Then a trivial diagram chasing shows that there exists $[A'] \in D^1(A|R',\ A_0)$ such that $p([A']) = [R']$ and $A'_x = B$ i.e., $(R',\ \mathrm{Spec}(A'))$ is a deformation of $\mathrm{Spec}(A_0)$ to R' such that $\mathrm{Spec}(A')_x = \mathrm{Spec}(B)$. This proves that $\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0},\ x)}$ is smooth. The smoothness of the functor $\underline{M}_{\mathrm{Spec}((A_0)_x)} \to \underline{M}_{\mathrm{Spec}(\widehat{(A_0)_x})}$ comes from the similar diagram together with the fact that

$$D^i((A_0)_x|k,\ \widehat{(A_0)_x}) = \widehat{D^i((A_0)_x|k,\ (A_0)_x)} = D^i((A_0)_x|k,\ (A_0)_x)$$

for all $i > 0$ since x is an only possible non-smooth point.

Theorem 4.11. *Let X_0 be a (separated) k-scheme of finite type, smooth outside a finite closed set $X_0^{\mathrm{Sing}} = \{x_1,\dots,x_r\}$ of X_0 . Let $U_0^1,\dots,U_0^r$ be affine open neighbourhoods of $x_1,\dots,x_r$ such that $U_0^i \cap X_0^{\mathrm{Sing}} = \{x_i\}$. If $H^2(X_0,\ \underline{D}^0_{X_0}) = 0$, then the natural functor $\underline{M}_{X_0} \to \underline{M}_{U_0^1} \times \dots \times \underline{M}_{U_0^r}$ is smooth.*

Proof: We can complete $U_0^1, \dots ,U_0^r$ into an affine open covering of X_0 by choosing $U_0^{r+1},\dots,U_0^N$ such that $U_0^j \cap X_0^{\mathrm{Sing}} = \emptyset$ for all $j > r$. Now let $R' \to R$ be a small extension in $\underline{C}_\Lambda$, and let $(R',\ U'^1),\dots,(R',\ U'^r)$ be deformations of $U_0^1,\dots,U_0^r$ to R' such that $U'^i \otimes_{R'} R = X|U_0^i$, $1 \le i \le r$, for some deformation (R, X) of X_0 to R. We must show that there exists a deformation (R', X') of X_0 to R' such that $X' \otimes_{R'} R = X$ and $X'|U_0^i = U'^i$ for $1 \leq i \leq r$. Since U_0^j for $j > r$ is smooth over k, $X|U_0^j$ can be lifted, by 4.9(a), to a deformation (R', U'^j) i.e., there exist deformations (R', U'^j) of U_0^j to R' such that $U'^j \otimes_{R'} R = X|U_0^j$ for all $j > r$. Since, for any pair $i \neq j$, $U_0^i \cap U_0^j$ is affine and smooth over k, we obtain an isomorphism

$$\tau_{ij} : U'^i|U_0^i \cap U_0^j \xrightarrow{\sim} U'^j|U_0^i \cap U_0^j$$

such that $\tau_{ij} \otimes_{R'} R = \mathrm{id}_{X|U_0^i \cap U_0^j}$, and in turn defines a cohomology class in $H^2(X_0\ \underline{D}^0_{X_0})$ given by $U_0^i \cap U_0^j \cap U_0^k \to \tau_{ij}\tau_{jk}\tau_{ik}^{-1}$. Since $H^2(X_0, \underline{D}^0_{X_0}) = 0$ by hypothesis, the obstruction for glueing these to obtain a global deformation vanishes, and hence we get a deformation (R', X') of X_0 to R' such that $X' \otimes_{R'} R = X$ and $X'|U_0^j \simeq U'^j$. This completes our proof.

Remark 4.12. The previous proof assumes X_0 to be separated, which is unnecessary. In fact, for a fixed deformation (R, X) of X_0 and deformations (R', U'^j) of U_0^i, the deformations (R', V') of open set V of X to R' compatible with the deformation (R', U'^j) already given, form a "gerbe" (in Giraud's terminology) whose structure sheaf is $\underline{D}^0_{X_0}$. Hence the obstruction to the existence of a section of this "gerbe" (i.e., a global deformation (R', X') of X compatible with the given datum (R', U'^j)) is in $H^2(X_0, \underline{D}^0_{X_0})$ which is zero by assumption.

Corollary 4.13. *Let X_0 be a k-scheme of finite type, smooth outside a finite subset $X_0^{Sing} = \{x_1, \dots, x_r\}$ of X_0 such that $H^2(X_0, \underline{D}^0_{X_0}) = 0$. Then*

(a) $\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\widehat{\underline{O}}_{X_0,x_1})} \times \dots \times \underline{M}_{\mathrm{Spec}(\widehat{\underline{O}}_{X_0,x_r})}$ *is smooth.*

In particular, $\dim \underline{M}_{X_0} = \dim H^1(X_0, \underline{D}^0_{X_0}) + \sum_{c=1}^{r} \dim \underline{M}_{\mathrm{Spec}(\widehat{\underline{O}}_{X_0,x_0})}$.

(b) *If furthermore X_0 is locally a complete-intersection, then $\underline{M}_{X_0}$ is smooth.*

<u>Proof</u>:

(a) The first statement follows from 4.10 and 4.9(b). Now let $(R, \mathfrak{X})$ and $(R_1, \mathfrak{X}_1), \dots, (R_r, \mathfrak{X}_r)$ be formal versal deformations of X_0 and $\mathrm{Spec}(\underline{O}_{X_0,x_1}), \dots, \mathrm{Spec}(\underline{O}_{X_0,x_r})$ respectively. We then obtain a commutative diagram

$$\begin{tikzcd}
\underline{h}_R \arrow[r] \arrow[d] & \underline{h}_{R_1 \hat{\otimes}_\Lambda \cdots \hat{\otimes}_\Lambda R_r} \arrow[d] \\
\underline{M}_{X_0} \arrow[r] & \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_1})} \times \dots \times \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_r})}
\end{tikzcd}$$

Since the composite functor

$$\underline{h}_R \to \underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_1})} \times \dots \times \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_r})}$$

is smooth whereas

$$\underline{h}_{R_1 \hat{\otimes}_\Lambda \cdots \hat{\otimes}_\Lambda R_r} \to \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_1})} \times \dots \times \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_r})}$$

is minimally smooth, it follows from 2.7(a) that

$\underline{h}_R \to \underline{h}_{R_1 \hat{\otimes}_\Lambda \cdots \hat{\otimes}_\Lambda R_r}$ is smooth i.e., R is a formal power-series ring over $R_1 \hat{\otimes}_\Lambda \cdots \hat{\otimes}_\Lambda R_r$. Since

$$0 \to H^1(X_0, \underline{D}^0_{X_0}) \to [\underline{M}_{X_0}(k[\epsilon])] \to [\underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_1})}(k[\epsilon])] \dots [\underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_r})}(k[\epsilon])]$$

is exact by 4.4 i.e.,

$$0 \to H^1(X_0, \underline{D}^0_{X_0}) \to \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\epsilon]) \to \mathrm{Hom}_{\Lambda\text{-alg}}(R_1 \hat{\otimes}_\Lambda \cdots \hat{\otimes}_\Lambda R_r, k[\epsilon])$$

is exact, it follows that R is a formal power-series ring over $R_1 \hat{\otimes}_\Lambda \cdots \hat{\otimes}_\Lambda R_r$ in d-variables where $d = \dim_k H^1(X_0, \underline{D}^0_{X_0})$.

Therefore

$$\begin{aligned}
\dim \underline{M}_{X_0} &= \dim k \otimes_\Lambda R = d + \dim k \otimes_\Lambda (R_1 \hat{\otimes}_\Lambda \cdots \hat{\otimes}_\Lambda R_r) \\
&= d + \dim(k \otimes_\Lambda R_1) \hat{\otimes}_k \cdots \hat{\otimes}_k (k \otimes_\Lambda R_r) \\
&= d + \sum_{i=1}^{r} \dim k \otimes_\Lambda R_i = d + \sum_{i=1}^{r} \dim \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_i})}
\end{aligned}$$

(b) follows from 4.10 and 4.9(a).

4.14. We close this paragraph with an explicit construction of a formal versal deformation in a simple situation. In fact, if X_0 is affine and is a complete-intersection, it is trivial to write down explicitely. Thus let

$$A_0 = k[x_1,\dots,x_n]/(f_1,\dots,f_r) \quad S/J$$

where $S = k[x_1,\dots,x_n]$, $J = (f_1,\dots,f_r)$, and $f_1,\dots,f_r$ is an S-regular sequence. We then obtain the exact sequence

$$\dots \to \mathrm{Der}_k(S, A_0) \to \mathrm{Hom}_{A_0}(J/J^2, A_0) \to D^1(A_0|k, A_0) \to 0$$

We note that J/J^2 is A_0-free module of rank r generated by $f_1 (\mathrm{mod}\ J^2),\dots,f_r(\mathrm{mod}\ J^2)$, since $f_1,\dots,f_r$ is an S-regular sequence. We now assume that $D^1(A_0|k, A_0)$ is a finite-dimensional vector space over k , say A has isolated non-smooth points only. Let $d = \dim_k D^1(A_0|k, A_0)$ and choose $p_j : J/J^2 \to A_0$ $(j = 1, 2,\dots,d)$ which form a k-basis of $D^1(A_0|k, A_0)$. Now choose $P_{jk} \in \Lambda[x_1,\dots,x_n]$ such that $p_j(f_i) = P_{ji}$ (mod $\underline{m}_\Lambda + J$) where $\underline{m}_\Lambda$ is the maximal ideal of Λ . We then consider

$$R = \Lambda[[t_1,\dots,t_d]]$$
$$B = R[x_1,\dots,x_n]/(G_1,\dots,G_r)$$

where $G_j = F_j - \sum_{i=1}^{d} t_i P_{ji}$ and F_j is a polynomial with coefficients in Λ gotten from f_j by lifting the coefficients from k to Λ . Since $R[x_1,\dots,x_n]$ is R-flat and $G_1,\dots,G_r$ modulo $\underline{m}_R$ (which becomes $f_1,\dots,f_r$) is a regular sequence in $k \otimes_R R[x_1,\dots,x_n]$, it follows that B is R-flat, and $k \otimes_R B = A_0$. Furthermore,

$$\mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\epsilon]) \to D^1(A_0|k, A_0) = [\underline{M}_{\mathrm{Spec}(A_0)}(k[\epsilon])]$$

is bijective by our construction of B . Therefore, if we set

$$\mathrm{Spf}(B) = \varinjlim_{v} \mathrm{Spec}(R/\underline{m}^{v+1} \otimes_R B)$$

then $(R, \mathrm{Spf}(B))$ is an object in $\widehat{\underline{M}_{\mathrm{Spec}(A_0)}}$ and the functor $\underline{h}_R \to \underline{M}_{\mathrm{Spec}(A_0)}$ induced by $(R, \mathrm{Spf}(B))$ is smooth by 1.6, since R is a formal power-series ring over Λ. In other words, $(R, \mathrm{Spf}(B))$ is a formal versal object of $\underline{M}_{\mathrm{Spec}(A_0)}$.

We note, by 4.10(b), that if we set

$$B = R[[X_1,\dots,X_n]]/(G_1,\dots,G_r) \quad ,$$

then $(R, \mathrm{Spf}(B))$ is a formal versal deformation of $\mathrm{Spec}(A_0)$ where $A_0 = k[[X_1,\dots,X_n]]/(f_1,\dots,f_r)$.

Remark 4.15. Let X_0 be a k-scheme of finite type with isolated non-smooth points only, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0 . The above construction shows that if X_0 is affine and locally a complete-intersection, then $\mathfrak{X}$ is a formalization of an R-scheme of finite type. This fact can easily be generalized to the case when X_0 is not necessarily affine but $H^2(X_0, \underline{D}^0_{X_0}) = 0$.

Remark 4.16. The global theory of the relative cotangent complex (the latter theory being reduced to the truncated complex of length 1 , which is sufficient however for many purposes of deformation theory) allows to generalize and simplify certain results, such as 4.10. Namely let S be a scheme, $S_0 \subset S' \subset S$ two subschemes of S defined by ideals $J \supset I$ such that $JI = 0$, X' a scheme flat over S' , $L. = L^{X_0/S_0}$ the relative cotangent complex of X_0 over S_0 , which is an element of the derived category $D(X_0, \underline{O}_X)$. We consider the category $\underline{F}$ of all flat schemes X over S which extend X'/S' , i.e. together with an S'-isomorphism $X \times_S S' \simeq X'$. Then

(a) For an object X' of $\underline{F}$, the group of automorphisms is canonically isomorphic to $\mathrm{Ext}^0(X_0; L., I_{X_0})$ (where I_{X_0} is the inverse image of I , viewed as a module on S_0 , on X_0) .

(b) The set of isomorphism classes of objects of $\underline{F}$ is empty or a torsor (= a principal homogeneous space) under the group $\mathrm{Ext}^1(X_0; L.,I_{X_0})$.

(c) One can define a canonical element $c \in \mathrm{Ext}^2(X_0; L.,I_{X_0})$ whose vanishing is necessary and sufficient for $\underline{F}$ to be non empty.

In order to study the $\mathrm{Ext}^i(X_0; L.,I_{X_0})$, it is often convenient to write

$$\mathrm{Ext}^i(X_0;L.,I_{X_0}) = H^i(X_0, \underline{\mathrm{RHom}}(L.,I_{X_0})) \quad .$$

and to notice that the complex $\underline{\mathrm{RHom}}(L.,I_{X_0})$ has as cohomology sheaves the sheaves $\underline{D}^i_{X_0}(I_{X_0})$ introduced in § 3. For $i = 0, 1, 2,$ these sheaves may be viewed respectively as the sheaf of automorphisms of any object X of $\underline{F}$ (understood: automorphisms inducing the identity on X') , the sheaf of indeterminacy of isomorphism classes of local solutions to the flat deformation problem, and the sheaf of obstructions to the existence of local solutions. On the other hand, the standart spectral sequence gives us

$$\mathrm{Ext}^*(X_0 \; L.,I_{X_0}) \leftarrow E_2^{p,q} = H^p(X_0, \underline{D}^q_{X_0}(I_{X_0})) \quad ,$$

which yields the well-known isomorphism

$$\mathrm{Aut}(X/X') \simeq H^0(X_0,\underline{D}^0_{X_0}(I_{X_0})) \simeq H^0(X_0,\underline{\mathrm{Hom}}(\underline{\Omega}^1_{X_0/S_0},I_{X_0})) = \mathrm{Hom}(\underline{\Omega}^1_{X_0/S_0},I_{X_0}) \quad ,$$

and the exact sequence in low degrees

$$0 \to H^1(X_0,\underline{D}^0_{X_0}) \to \mathrm{Ext}^1 \to H^0(X_0,\underline{D}^1_{X_0}) \to H^2(X_0,\underline{D}^0_{X_0}) \to \mathrm{Ext}^2$$

This shows in particular that if $H^2(X_0,\underline{D}^0_{X_0}) = 0$, and if there exists any local flat deformation X of X'/S' , then there exists a flat global deformation which corresponds to given local deformations. More precisely and generally, for a given local deformations (i.e. a section of the "sheaf of isomorphism classes of local flat deformations"), the argument 4.10 shows that the obstruction to finding a global flat deformation compatible with these is in $H^2(X_0, \underline{D}^0_{X_0})$, hence such a global flat deformation exists if $H^2(X_0, \underline{D}^0_{X_0}) = 0$: this is essentially 4.10.

On the other hand, the spectral sequence shows that the Ext^2 is zero whenever we have the relations

$$(*) \quad H^0(X_0, \underline{D}^2_{X_0}) = 0 \ , \ H^1(X_0, \underline{D}^1_{X_0}) = 0 \ , \ H^2(X_0, \underline{D}^0_{X_0}) = 0 \ .$$

The first relation is always satisfied if X_0 is a relative complete intersection over S_0 (as then $\underline{D}^2_{X_0} = 0$ for any module M on X_0), the second is satisfied provided the support of $\underline{D}^1_{X_0}$ is discrete, as is the case if X_0 is smooth over S_0 except on a discrete subset of X_0 over S_0 . When the previous three relations (*) are all satisfied, then by (c) above, $\underline{F}$ is non empty, which generalizes 4.12(b).

More geometrically, and without using the global theory of the relative cotangentcomplex (but only the globalizing of the functors $\underline{D}^i_{X_0}$), one can get a simple geometric interpretation of the three successive obstructions to deforming flatly X' to X , lying in the three left hand terms of (*). Namely by the local theory, the first obstruction c_1 in $H^0(X_0, \underline{D}^2_{X_0})$ can be defined as the obstruction to finding local solutions, when this obstruction vanishes, the sheaf $\underline{F}$ of isomorphism classes of local solutions of the problem is (by the local theory) in a canonical way a torsor under $\underline{D}^1_{X_0}$, and the second obstruction c_2 , lying in $H^1(X_0, \underline{D}^1_{X_0})$, is the obstruction to finding a section of this torsor, i.e. of finding a "coherent system of isomorphism classes of local solutions" ; if this second obstruction also vanishes, then, choosing arbitrarily a section of $\underline{F}$, one gets, as noted above, an obstruction c_3 in $H^2(X_0, \underline{D}^0_{X_0})$ to finding a global solution corresponding to the given system of (isomorphism classes of) local solutions. Using the fact that the section of F is determined up to adding a section of $\underline{D}^1_{X_0}$, we get a map $d : H^0(X_0, \underline{D}^1_{X_0}) \to H^2(X_0, \underline{D}^0_{X_0})$, and a solution to the deformation problem exists if and only if $c_1 = 0$, $c_2 = 0$, and c_3 is zero as an element of Coker d (each c_i being defined when the previous ones are zero). Of course, the groups $H^0(X_0, \underline{D}^2_{X_0})$, $H^1(X_0, \underline{D}^1_{X_0})$ and

$H^2(X_0, \underline{D}^0_{X_0})/\mathrm{Im}\, d$ are just the initial terms $E_2^{0,2}$, $E_2^{1,1}$, and $E_3^{2,0}$ of the spectral sequence above, d being the differential (up to sign ?) $d_2 : E_2^{0,1} \to E_2^{2,0}$.

5. Formal Jacobian subschemes

Let X_0 be a scheme of finite type over a field k , and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0 . Then $\mathfrak{X}$ is not an ordinary scheme over R but an R-adic formal scheme. The purpose of this section is to clarify "the non-smooth locus of $\mathfrak{X}$ over R ".

We fistly recall the definition of Fitting ideals associated to a module of finite type: Let A be a commutative ring and E an A-module of finite type. Choose a presentation

$$K \to F \to E \to 0$$

where F is locally-free of constant rank n . We define

$$I_A^p(E) = \mathrm{Im}(\textstyle\bigwedge^{n-p} K \otimes \bigwedge^{n-p} F^* \to A) \qquad \text{for } p = 0, 1, 2, \ldots$$

The ideals $I_A^p(E)$ thus defined does not depend on the choice of a presentation and therefore are invariants of A-module E , called the Fitting ideals of A-module E . The following properties readily follow from the definition.

5.1. (a) For any ring homomorphism $A \to B$, the canonical map

$$B \otimes_A A/I_A^p(E) \xrightarrow{\sim} B/I_B^p(B \otimes_A E)$$

is an isomorphism for all P . In particular we have

$$I^p_{S^{-1}A}(S^{-1}E) = S^{-1}I_A^p(E)$$

for any multiplicative subset S in A .

(b) We have $I_A^0(E) \subset I_A^1(E) \subset I_A^2(E) \subset \ldots$, and, for each point $x \in \operatorname{Spec}(A)$, $I_A^p(E)_x = A_x$ for all $p \geq \dim_{\varkappa(x)} E(x)$, where $E(x) = \varkappa(x) \otimes_A E$.

(c) $x \in \operatorname{Spec}(A) - \operatorname{Supp} A/I_A^p(E)$ if and only if, for any (or for some) presentation

$$K \to F \to E \to 0$$

where $F \approx A^n$, K_x contains A_x^{n-p} which is isomorphic to a direct summand of F_x . In particular, we have that

$$\operatorname{Supp} A/I_A^0(E) = \operatorname{Supp} E \ .$$

(d)

$$I_A^r(E_1 \oplus E_2) = \sum_{p+q=r} I_A^p(E_1) \cdot I_A^q(E_2)$$

(e) If A is a noetherian adic ring, then

$$A/I_A^p(E) = \varprojlim_{v} A_v/I_{A_v}^p(E_v)$$

where $A = \varprojlim_{v} A_v$ and $E_v = E \otimes_A A_v$.

In view of the property 5.1(a), our definition of Fitting ideals is readily globalized. Thus let X be any scheme, and $\underline{E}$ a quasi-coherent X-module of finite presentation. We then obtain a sequence of quasi-coherent ideal sheaves $\underline{I}_X^p(\underline{E})$ for $p \geq 0$ defined by

$$\Gamma(U, \underline{I}_X^p(\underline{E})) \Rightarrow I^p_{\Gamma(U, \underline{O}_X)}(\Gamma(U, \underline{E}))$$

for every affine open subset $U \subset X$.

Now let $X \to Y$ be a morphism of schemes. The relative Kähler-differentials $\underline{\Omega}_{X|Y} (= \underline{\Omega}^1_{X|Y})$ is a quasi-coherent X-module. If the morphism $X \to Y$ is of essentially finite presentation, then $\underline{\Omega}_{X|Y}$ is a quasi-coherent X-module of finite presentation, and in turn we may consider the ideal sheaves $\underline{I}_X^p(\underline{\Omega}_{X|Y})$ ..

Definition 5.2. <u>Let</u> $X \to Y$ <u>be a morphism of essentially finite presentation.</u> <u>For each integer</u> $p \geq 0$, <u>we define</u> $J_X^p(X|Y)$ <u>to be the closed subscheme of</u> X

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defined by the ideal sheaf $\underline{I}^p_X(X|Y) = \underline{I}^p_X(\underline{\Omega}_{X|Y})$, *called the Jacobian subschemes of* X *over* Y . *For each point* $x \in X$, *we set* $X_x = \mathrm{Spec}(\underline{O}_{X,x})$.

Theorem 5.3. *Let* $X \to Y$ *be a morphism of essentially finite presentation. Then*

(1) *For any base change* $Y \leftarrow Y'$, *the canonical map* $J^p_{X'}(X'|Y') \xrightarrow{\sim} J^p_X(X|Y) \underset{Y}{\times} Y'$ *is an isomorphism for all* p *where* $X' = X \underset{Y}{\times} Y'$.

(2) *For each point* $x \in X$, *we have the canonical isomorphism* $J^p_{X_x}(X_x|Y) \xrightarrow{\sim} J^p_X(X|Y)_x$.

(3) $J^0_X(X|Y) \supset J^1_X(X|Y) \supset J^2_X(X|Y) \supset \ldots,$ *and* $x \notin J^p_X(X|Y)$ *for all* $p \geq \dim_{\kappa(x)} \Omega_{X|Y}(x)$.

(4) $x \notin J^p_X(X|Y)$ *if and only if there exists an open neighbourhood* U *of* x *which admits a closed immersion into a* Y*-scheme* U' *such that* U' *is smooth over* Y *at* x' *and* $\dim_{x'} U'_{f(x)} = p$ *where* x' *is the image of* x *under* $U \to U'$.

(5) $$\underline{I}^r_{X_1 \underset{Y}{\times} X_2}(X_1 \underset{Y}{\times} X_2|Y) = \sum_{p+q=r} \pi_1^* \underline{I}^p_{X_1}(X_1|Y) \cdot \pi_2^* \underline{I}^q_{X_2}(X_2|Y)$$

where $\pi_i : X_1 \underset{Y}{\times} X_2 \to X_i$ *is the projection.*

Proof: (3) follows from 5.1(b) whereas (1) follows from 5.1(a) together with the fact that $P^* \underline{\Omega}_{X|Y} \xrightarrow{\sim} \underline{\Omega}_{X'|Y'}$ is an isomorphism where $P : X \underset{Y}{\times} Y' \to X$ is the projection. (2) also follows from 5.1(a) since $\underline{O}_{X,x} \underset{\underline{O}_X}{\otimes} \underline{\Omega}_{X|Y} \xrightarrow{\sim} \underline{\Omega}_{X_x|Y}$ is an isomorphism (5) follows from 5.1(d) and the isomorphism

$$\pi_1^* \underline{\Omega}_{X_1|Y} \oplus \pi_2^* \underline{\Omega}_{X_2|Y} \xrightarrow{\sim} \underline{\Omega}_{X_1 \underset{Y}{\times} X_2|Y}$$

(4) Assume the existence of a diagram

$$\begin{tikzcd} U \arrow[d, "f"'] & U' \arrow[l, "i"'] \arrow[dl] \\ Y & \end{tikzcd}$$

where U' is smooth over Y at $x' = i(x)$ and $\dim_{x'} U'_{f(x)} = p$. Then $\dim_{\kappa(x')} \underline{\Omega}_{U'|Y}(x') = p$ and $i^*\underline{\Omega}_{U'|Y} \to \underline{\Omega}_{U|Y} \to 0$ is exact, so that $\dim_{\kappa(x)} \underline{\Omega}_{U|Y}(x) \leq p$ which implies that $x \notin J^p_U(U|Y)$ by 5.3(3). Since $J^p_U(U|Y) = J^p_X(X|Y)|U$, this means that $x \notin J^p_X(X|Y)$. Now assume conversely that $x \notin J^p_X(X|Y)$.

The question being local, we may assume that X, Y are both affine, say $X = \mathrm{Spec}(B)$, $Y = \mathrm{Spec}(A)$. Choose a presentation over A

$$0 \to J \to P \to B \to 0$$

where $P = A[x_1,\ldots,x_n]$. Then $J/J^2 \xrightarrow{d} B \otimes_P \Omega_{P|A} \to \Omega_{B|A} \to 0$ is exact and $B \otimes_P \Omega_{P|A}$ is B-free of rank n, and therefore

$I^p_B(\Omega_{B|A}) = \mathrm{Im}(^{n-p}\!\wedge J/J^2 \otimes_B (B \otimes_P {}^{n-p}\!\wedge \Omega_{P|A})^* \to B)$. Therefore, $x \notin J^p_X(X|Y)$ $\Leftrightarrow$ there exists $f_1,\ldots,f_{n-p}$ in J such that $[\det\left(\frac{f_i}{x_j}\right)](x) \neq 0$ for some choice of $n-p$ variables among $x_1, x_2,\ldots,x_n$ $\Leftrightarrow$ there exists $f_1, f_2,\ldots,f_{n-p}$ in J such that $\mathrm{Spec}(P/(f_1,\ldots,f_{n-p})) \to \mathrm{Spec}(A)$ is smooth at x' where x' is the image of x under $\mathrm{Spec}(B) \to \mathrm{Spec}(P/(f_1,\ldots,f_{n-p}))$. Thus it suffices to take $U' = \mathrm{Spec}(P/(f_1,\ldots,f_{n-p}))$.

The following is a generalization of the usual Jacobian criteria for relative situation.

<u>Theorem</u> 5.4. <u>Let</u> $X \xrightarrow{f} Y$ <u>be a morphism of essentially finite presentation. Assume that there exists a fixed integer</u> $n \geq 0$ <u>with either conditions:</u>

(1) f <u>is flat and</u> $\dim_x X_{f(x)} = n$ <u>for every point</u> $x \in X$.

(2) f <u>is a relative complete-intersection with virtual dimension</u> n (SGA6 VIII 1.9).

<u>Then</u> $x \notin J^n_X(X|Y)$ <u>if and only if</u> f <u>is smooth at</u> x.

<u>Proof:</u> We have, by 5.3(4) that $x \notin J^n_X(X|Y)$ if and only if there exists a diagram

$$\begin{tikzcd} U \arrow[r, hook, "i"] \arrow[d, "f"'] & U' \arrow[dl, "f'"] \\ Y & \end{tikzcd}$$

such that f' is smooth at x' and $\dim_{x'} U'_{f(x)} = n$, where i is a closed immersion and $x' = i(x)$. Thus it suffices to show that either conditions (1) or (2) entails the isomorphism of i at the point x . Set $A = \underline{O}_{Y,f(x)}$, $B' = O_{U',x'}$, $B = O_{U,x}$ and let

$$0 \to I \to B' \to B \to 0$$

be exact. We must show that $I = 0$. Assume (1). Since B is A-flat,

$$0 \to I/\underline{m}I \to B'/\underline{m}B' \to B/\underline{m}B \to 0$$

is exact where $\underline{m}$ is the maximal ideal of A . By hypothesis, $\dim_x U_{f(x)} = \dim_{x'} U'_{f(x)}$ so that $\dim B'/\underline{m}B' = \dim B/\underline{m}B$. However $B'/\underline{m}B'$ is a regular local ring and therefore it entails that $I/\underline{m}I = 0$ so that $I = 0$. Now assume (2), and consider the exact sequence

$$I/I^2 \xrightarrow{d} B \otimes_{B'} \Omega_{B'|A} \to \Omega_{B|A} \to 0$$

Since X is a relative complete-intersection over Y and $U' \to Y$ is smooth at x , it follows that $d : I/I^2 \to B \otimes_{B'} \Omega_{B'|A}$ is injective at x and I/I^2 and $B \otimes_{B'} \Omega_{B'|A}$ are both free at x . Therefore by the definition of virtual dimension of relative complete-intersection we must have that

$$n = \dim_{\kappa(x)} \Omega_{B'|A}(x') - \dim_{\kappa(x)} I/I^2(x) = n - \dim_{\kappa(x)} I/I^2(x)$$

i.e., $I/I^2 = 0$ i.e., $I = 0$.

Definition 5.5. Let $X \xrightarrow{f} Y$ be a morphism of essentially finite presentation. In case when there exists a fixed integer n with the property (1) or (2) in 5.4,

<u>we set</u> $\underline{I}_X(X|Y) = \underline{I}_X^n(X|Y)$ <u>and</u> $J_X(X|Y) = J_X^n(X|Y)$.

We can extend this notion to formal schemes in an obvious manner.

Thus let $\mathfrak{X}$ be a noetherian formal scheme and $\underline{E}$ a coherent $\mathfrak{X}$-module. Then we can define the ideal sheaf $\underline{I}_{\mathfrak{X}}^P(\underline{E})$ by setting

$$\underline{I}_{\mathfrak{X}}^P(\underline{E})(U) = I_{\mathfrak{X}(U)}^P(\underline{E}(U))$$

for every affine open $U \subset \mathfrak{X}$. If $\mathfrak{X} = \varinjlim_{v} X_v$, then $\underline{E} = \varinjlim_{v} \underline{E}_v$ where $\underline{E}_v = \underline{E} \otimes_{\underline{O}_{\mathfrak{X}}} \underline{O}_{X_v}$ and it follows from the fact that $\mathfrak{X}$ is noetherian that

$$\underline{I}_{\mathfrak{X}}^P(\underline{E}) = \varprojlim_{v} \underline{I}_{X_v}^P(\underline{E}_v)$$

Now let S be a noetherian formal scheme and $\mathfrak{X}$ an S-adic formal scheme of essentially finite type. Let $\underline{I}_{\mathfrak{X}}$ be the ideal sheaf defining the closed immersion $\Delta : \mathfrak{X} \to \mathfrak{X} \times_S \mathfrak{X}$ (= the fibre-product in the category of formal schemes) i.e.,

$$0 \to \underline{I}_{\mathfrak{X}} \to \underline{O}_{\mathfrak{X} \times_S \mathfrak{X}} \to \underline{O}_{\mathfrak{X}} \to 0$$

is exact. We define $\underline{\Omega}_{\mathfrak{X}|S} = \underline{I}_{\mathfrak{X}}/\underline{I}_{\mathfrak{X}}^2$. Since $\mathfrak{X}$ is an S-adic formal scheme of essentially finite type, it follows that $\mathfrak{X} = \varinjlim_{v} X_v$ where $X_v = \mathfrak{X} \times_S S_v$, and hence $\underline{\Omega}_{\mathfrak{X}|S} \simeq \varprojlim_{v} \underline{\Omega}_{X_v|S_v}$ is a coherent $\mathfrak{X}$-module. We note that for any formal S-scheme S' of finite type, we have a canonical isomorphism

$$\underline{\Omega}_{\mathfrak{X} \times_S S'|S'} \simeq \underline{O}_{S'} \otimes_{\underline{O}_S} \underline{\Omega}_{\mathfrak{X}|S} \quad .$$

<u>Definition 5.6. Let S be a noetherian formal scheme and $\mathfrak{X}$ an S-adic formal scheme of essentially finite type. We define the ideal sheaf $\underline{I}_{\mathfrak{X}}^p(\mathfrak{X}|S)$ by</u>

$$\underline{I}_{\mathfrak{X}}^p(\mathfrak{X}|S) = \underline{I}_{\mathfrak{X}}^p(\underline{\Omega}_{\mathfrak{X}|S})$$

<u>and we set $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ the formal closed subscheme of $\mathfrak{X}$ defined by $\underline{I}_{\mathfrak{X}}^p(\mathfrak{X}|S)$</u> .

Since $\Omega_{\mathfrak{X}|S} = \varprojlim_{n} \Omega_{X_n|S_n}$ we have

$$I^p_{\mathfrak{X}}(\mathfrak{X}|S) = \varprojlim_n I^p_{X_n}(X_n|S_n)$$

$$J^p_{\mathfrak{X}}(\mathfrak{X}|S) = \varinjlim_n J^p_{X_n}(X_n|S_n) \quad .$$

$\mathfrak{X}_{(x)}$ is an S-adic formal scheme of essentially finite type and we have a commutative diagram

$$\begin{tikzcd} \mathfrak{X}_{(\hat{x})} \arrow[rr, "i_{(\hat{x})}"] \arrow[dr] & & \mathfrak{X} \\ & \mathfrak{X}_{(x)} \arrow[ur, "i_{(x)}"'] & \end{tikzcd}$$

We note that if x is an isolated point of $\mathfrak{X}$, then $i_{(x)}$, $i_{(\hat{x})}$ are both isomorphisms to the connected component of $\mathfrak{X}$ containing the point x .

In particular, if $|\mathfrak{X}| = \{x_1,\dots,x_m\}$ is a finite discrete space, then we have a canonical identification $\mathfrak{X} = \coprod_{1 \le i \le m} \mathfrak{X}_{(x_i)}$ (disjoint union).

Corollary 5.7. *Let* S *be a noetherian formal scheme, and* $\mathfrak{X}$ *an* S*-adic formal scheme of essentially finite type. We then have*

(1) $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ *commutes with base change* i.e., $J^p_{\mathfrak{X}'}(\mathfrak{X}'|S') = J^p_{\mathfrak{X}}(\mathfrak{X}|S) \times_S S'$ *where* $\mathfrak{X} = \mathfrak{X} \times_S S'$.

(2) *The connected components of* $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ *is in bijective correspondence with that of* $J^p_{X_0}(X_0|S_0)$. *For each point* $x \in |J^p_{X_0}(X_0|S_0)|$ *we have a canonical isomorphism* $J^p_{\mathfrak{X}}(\mathfrak{X}|S)_{(x)} \simeq J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)$. *In particular if*

$$|J^p_{X_0}(X_0|S_0)| = \{x_1,\dots,x_m\}$$

is discrete, then

$$J^p_{\mathfrak{X}}(\mathfrak{X}|S) = \coprod_{1 \le i \le m} J^p_{\mathfrak{X}_{(x_i)}}(\mathfrak{X}_{(x_i)}|S)$$

(disjoint union).

(3) Let S be an ordinary noetherian scheme and X an S-scheme of essentially finite type. Given a closed subscheme $S_0 \subset S$, let $\hat{S}$ be the formal completion of S along S_0.

If we set

$\hat{X} = X \underset{S}{\times} \hat{S}$ = the formal completion of X along $X_0 = X \underset{S}{\times} S_0$, then

$J^p_{\hat{X}}(\hat{X}|\hat{S}) = J^p_X(X|S) \underset{S}{\times} \hat{S}$ = the formal completion of $J^p_X(X|S)$ along $J^p_{X_0}(X_0|S_0)$.

(4) If there exists a fixed integer $n \geq 0$ with the property 5.4(1) or (2) for $X_\nu \to S_\nu$ for all ν, then $x \notin J^n_{\mathfrak{X}}(\mathfrak{X}|S)$ if and only if $\mathfrak{X}$ is formally smooth over S at x.

Proof: (1) follows from $\underline{\Omega}_{\mathfrak{X}'|S'} \simeq \underline{O}_{S'} \hat{\otimes}_{\underline{O}_S} \underline{\Omega}_{\mathfrak{X}|S}$ whereas (2) follows from $\underline{O}_{\mathfrak{X}_{(x)}} \hat{\otimes}_{\underline{O}_{\mathfrak{X}}} \underline{\Omega}_{\mathfrak{X}|S} \simeq \underline{\Omega}_{\mathfrak{X}_{(x)}|S}$.

(3) $J^p_{\hat{X}}(\hat{X}|\hat{S}) = \varinjlim_{v} J^p_{X \times_S S_v}(X \times_S S_v|S_v) = \varinjlim_{v} J^p_X(X|S) \times_S S_v = J^p_X(X|S) \times_S \hat{S}$.

(4) Since $\mathfrak{X}$ is an S-adic formal scheme, $\mathfrak{X} \to S$ is formally smooth at $x \Leftrightarrow X_\nu \to S_\nu$ is formally smooth at x for all $\nu \Leftrightarrow x \notin J^n_{X_\nu}(X_\nu|S_\nu)$ for all ν

$\Leftrightarrow x \notin \varinjlim_{\nu} J^n_{X_\nu}(X_\nu|S_\nu) = J^n_{\mathfrak{X}}(\mathfrak{X}|S)$.

Remark 5.8. Let $\mathfrak{X}$ be an S-adic formal scheme of essentially finite type. Given a point $x \in \mathfrak{X}$ we also consider the S-adic formal scheme $\mathfrak{X}_{(\hat{x})} = \varinjlim_{v} \operatorname{Spec}(\widehat{\underline{O}_{X_v,x}})$. We thus have the natural morphisms $\mathfrak{X}_{(\hat{x})} \to \mathfrak{X}_{(x)} \to \mathfrak{X}$. Now $\widehat{\underline{O}_{X_0,x}}$ is in general not essentially finite type over $S_{(y)}$ where y = the image of x under $\mathfrak{X} \to S$. For simplicity, let us set $A_v = \underline{O}_{X_v,x}$, $R_v = \underline{O}_{S_v,y}$. Now the natural map $\Omega_{A_v|R_v} \to \Omega_{\hat{A}_v|R_v}$ induces an isomorphism $\widehat{\Omega_{A_v|R_v}} \to \widehat{\Omega_{\hat{A}_v|R_v}}$, and therefore $\varprojlim_{v} \widehat{\Omega_{A_v|R_v}} \simeq \varprojlim_{v} \widehat{\Omega_{\hat{A}_v|R_v}}$ where $\widehat{\ }$ stands for the separable completion of a module with respect to the topology induced from the ring. Therefore if we define $\widehat{\Omega}_{\mathfrak{X}_{(\hat{x})}|S} = \varprojlim_{v} \widehat{\Omega_{\underline{O}_{X_v,x}|\underline{O}_{S_v}}}$, then $\widehat{\Omega}_{\mathfrak{X}_{(\hat{x})}|S}$ is a coherent $\mathfrak{X}_{(\hat{x})}$-module. Thus if we set $J^p_{\mathfrak{X}_{(\hat{x})}}(\mathfrak{X}_{(\hat{x})}|S)$ to be the formal closed subscheme of $\mathfrak{X}_{(\hat{x})}$ defined by the ideal sheaf $I^p_{\mathfrak{X}_{(\hat{x})}}(\mathfrak{X}_{(\hat{x})}|S) = I^p_{\mathfrak{X}_{(\hat{x})}}(\Omega_{\mathfrak{X}_{(\hat{x})}}|S)$, then we have

$$J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)_{(\hat{x})} \simeq J^p_{\mathfrak{X}_{(\hat{x})}}(\mathfrak{X}_{(\hat{x})}|S) .$$

We note that if x is an isolated point of $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$, then $J^p_{\mathfrak{X}_{(\hat{x})}}(\mathfrak{X}_{(\hat{x})}|S) \to J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)$ is an isomorphism.

To consider the "image" of $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ under the morphism $\mathfrak{X} \xrightarrow{f} S$, one may simply take the formal subscheme of S defined by the ideal sheaves $\mathrm{Ker}(\underline{O}_S \to \underline{O}_{\mathfrak{X}}/I^p_{\mathfrak{X}}(\mathfrak{X}|S))$. However, it is more convenient for our purpose to use the Fitting ideal again.

<u>Definition</u> 5.9. <u>Let</u> S <u>be a noetherian formal scheme and</u> $f : \mathfrak{X} \to S$ <u>an</u> S-<u>adic formal scheme of essentially finite type. Assume that</u> $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ <u>is proper over</u> S , <u>so that</u> $f_*\underline{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)}$ <u>is a coherent</u> $\underline{O}_S$-<u>module</u>. <u>We then define the ideal sheaf</u> $\underline{I}^p_S(\mathfrak{X}|S)$ <u>of</u> $\underline{O}_S$ <u>by</u>

$$\underline{I}^p_S(\mathfrak{X}|S) = \underline{I}^0_S(f_*\underline{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)})$$

<u>and set</u> $J^p_S(\mathfrak{X}|S)$ <u>to be the closed formal subscheme of</u> S <u>defined by the ideal sheaf</u> $\underline{I}^p_S(\mathfrak{X}|S)$. <u>In case when there exists a fixed integer</u> n <u>having the property</u> (1) <u>or</u> (2) <u>in</u> 5.4 <u>for</u> $X_v \to S_v$ <u>for all</u> v , <u>we set</u>

$$\underline{I}_{\mathfrak{X}}(\mathfrak{X}|S) = \underline{I}^n_{\mathfrak{X}}(\mathfrak{X}|S) , \qquad J_{\mathfrak{X}}(\mathfrak{X}|S) = J^n_{\mathfrak{X}}(\mathfrak{X}|S)$$

$$\underline{I}_S(\mathfrak{X}|S) = \underline{I}^n_S(\mathfrak{X}|S) , \qquad J_S(\mathfrak{X}|S) = J^n_S(\mathfrak{X}|S)$$

<u>Corollary</u> 5.10. <u>Let</u> S <u>be a noetherian formal scheme and</u> $\mathfrak{X} \xrightarrow{f} S$ <u>an</u> S-<u>adic formal scheme, essentially of finite type such that</u> $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ <u>is proper over</u> S . <u>Then</u>

(1) $|J^p_S(\mathfrak{X}|S)|$ = <u>the image of</u> $|J^p_{\mathfrak{X}}(\mathfrak{X}|S)|$ <u>under</u> $f : \mathfrak{X} \to S$.

(2) $J^p_S(\mathfrak{X}|S)$ <u>commute with base change</u> i.e., $J^p_{S'}(\mathfrak{X}'|S') = J^p_S(\mathfrak{X}|S) \times_S S'$.

(3) If $J^p_{\mathfrak{X}}(\mathfrak{X}|S) = \coprod_{1\leq i\leq m} \mathcal{Y}^{(i)}$ <u>(disjoint union), then</u>

$\underline{I}^p_S(\mathfrak{X}|S) = \prod_{1\leq i\leq m} \underline{I}^0_S(f_* \underline{O}_{\mathcal{Y}(i)})$. <u>In particular if</u>

$$|J^p_{X_0}(X_0|S_0)| = \{x_1,\ldots,x_m\}$$

<u>is discrete, then</u>

$$\underline{I}^p_S(\mathfrak{X}|S) = \prod_{1\leq i\leq m} \underline{I}^p_S(\mathfrak{X}_{(x_i)}|S) = \prod_{1\leq i\leq m} \underline{I}^p_S(\mathfrak{X}_{(\mathfrak{x}_i)}|S)$$

(4) <u>Let</u> S <u>be an ordinary noetherian scheme and</u> X <u>an</u> S-<u>scheme of essentially finite type. Given a closed subscheme</u> $S_0 \to S$, <u>let</u> $\hat{S}$ <u>be the formal completion of</u> S <u>along</u> S_0 . <u>If we set</u> $\hat{X} = X \times_S \hat{S}$ = <u>the formal completion of</u> X <u>along</u> $X_0 = X \times_S S_0$, <u>then</u> $J^p_{\hat{S}}(\hat{X}|\hat{S}) = J^p_S(X|S) \times_S \hat{S}$ = <u>the formal completion of</u> $J^p_S(X|S)$ <u>along</u> $J^p_{S_0}(X_0|S_0)$.

<u>Proof</u>: Supp $f_*\underline{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)} = |J^p_S(\mathfrak{X}|S)|$ by 5.1(c), and hence (1) follows. (2) and (4) are clear from 5.1(a), and (3) follows from 5.1(d), and the isomorphism $J^p_{\mathfrak{X}_{(\mathfrak{x})}}(\mathfrak{X}_{(\mathfrak{x})}|S) \xrightarrow{\sim} J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)$ for an isolated point x in $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$.

<u>Remark</u> 5.11. Our definition of Jacobian ideal, adopted from the classical one over base field, commutes with base change and gives a smoothness criteria under a mild condition of 5.4. However one can define, probably in many different ways, a canonical subscheme which gives the non-smooth locus in a general setting. We shall outline one here: Given an A-algebra B , define $I_{B|A} = \bigcap_E \operatorname{Ann} D^1(B|A,E)$ where E runs through all B-modules of finite type. This ideal $I_{B|A}$ commutes with localizations by 3.13(c), and therefore, given a morphism $X \xrightarrow{f} Y$, we obtain the ideal sheaf $\underline{I}_{X|Y}$. If f is a morphism of essentially finite type, then f is smooth at x if and only if $x \notin \operatorname{Supp} \underline{O}_X/\underline{O}_{X|Y}$ by 3.13(a). This notion is in an obvious way carried over to adic formal schemes of essentially

finite type and thus it is equally suitable for deformation theory.

6. Non-degenerate quadratic singularities

The objective in this section is to investigate the deformations of a k-scheme having non-degenerate quadratic singularities only. In view of 4.13, it becomes purely a local problem in an appropriate situation.

We go back to the notations of §4. Thus Λ is a fixed complete noetherian local ring with the residue field k, and $\underline{C}_\Lambda$ stands for the category of local artinian Λ-algebras with the same residue field k.

Let X be a k-scheme of finite type, and let x be a closed point of X. We recall that the point x is said to be a non-degenerate quadratic singularity rational over k if $\hat{O}_{X,x} - k[[x_1,\dots,x_n]]/(f)$ with $(\partial f/\partial x_1,\dots,\partial f/\partial x_n) = (x_1,\dots,x_n)$. One can always assume f to be a quadratic form. Indeed we have

__Lemma__ 6.1. __Let__ $f \in k[[x_1,\dots,x_n]]$. __If__ $(\partial f/\partial x_1,\dots,\partial f/\partial x_n) = M$ $(= (x_1,\dots,x_n))$, __then one can find__ $x_1',\dots,x_n'$ __with__ $x_i = x_i'$ $(\mathrm{mod}\ M^2)$ __such that__ $f = q(x_1',\dots,x_n')$ __where__ q __is a quadratic form.__

__Proof__: We do this by a successive approximation. Write $f = f_2 + h_3$ where f_2 is a quadratic form and $h_3 \in M^3$. Since $(\partial f/\partial x_1,\dots,\partial f/\partial x_n) = M$, there exists $\epsilon_1^{(1)},\dots,\epsilon_n^{(1)}$ in M^2 such that $\Sigma\, \epsilon_i^{(1)}\, \partial f/\partial x_i \equiv h_3 \pmod{M^4}$. If we set $x_i^{(1)} = x_i - \epsilon_i^{(1)}$, then $f(x_1^{(1)},\dots,x_n^{(1)}) =$
$f(x_1,\dots,x_n) - \Sigma\, \epsilon_i^{(1)}\, \partial f/\partial x_i + \dots \equiv f_2(x_1,\dots,x_n) \pmod{M^4}$ i.e.,
$f(x_1^{(1)},\dots,x_n^{(1)}) \equiv f_2(x_1,\dots,x_n) + h_4$ with $h_4 \in M^4$. Now
$(\partial f/\partial x_1^{(1)},\dots,\partial f/\partial x_n^{(1)}) = (\partial f/\partial x_1,\dots,\partial f/\partial x_n) = M$ and therefore we can choose

$\epsilon_1^{(2)},\dots,\epsilon_n^{(2)}$ in M^3 such that $\Sigma\ \epsilon_i^{(2)}\ \partial f/\partial x_u^{(1)} \equiv h_4 \pmod{M^5}$. Then $f(x_1^{(1)} - \epsilon_1^{(2)},\dots,x_n^{(1)} - \epsilon_n^{(2)}) = f(x_1^{(1)},\dots,x_n^{(1)}) - \Sigma\ \epsilon_i^{(2)}\partial f/\partial x_i^{(1)} + \dots$ $\equiv f_2(x_1,\dots,x_n) \pmod{M^5}$. Continuing this process we can find $\epsilon_i^{(1)},\ \epsilon_i^{(2)},\ \epsilon_i^{(3)},\dots$ $(i = 1,2,\dots,n)$ with $\epsilon_i^{(v)} \in M^{v+1}$ such that if we set $x_i^{(v)} = x_i - (\epsilon_i^{(1)} + \dots + \epsilon_i^{(v)})$ then $f(x_1^{(v)},\dots,x_n^{(v)}) \equiv f_2(x_1,\dots,x_n)$ $\pmod{M^{v+3}}$. Thus if we set $\epsilon_i = \sum_v \epsilon_i^{(v)}$ then $f(x_1-\epsilon_1,\dots,x_n-\epsilon_n) =$ $= f_2(x_1,\dots,x_n)$ i.e., $f(x_1,\dots,x_n) = f_2(x_1+\epsilon_1,\dots,x_n+\epsilon_n)$.

Lemma 6.2. Let A_0 be a local k-algebra, essentially of finite type such that

$$\hat{A}_0 \simeq k[[x_1,\dots,x_n]]/(f)$$

where f is a non-degenerate quadratic form, and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \mathrm{Spec}(A_0)$. Then

(1) $J_{\mathfrak{X}}(\mathfrak{X}|R) \to J_R(\mathfrak{X}|R) \to \mathrm{Spec}(\Lambda)$ are both isomorphisms.

(2) $J_R(\mathfrak{X}|R)$ is a principal ideal in R generated by a free formal generator of R over Λ i.e. $I_R(\mathfrak{X}|R) = (t)$ and $R = \Lambda[[t]]$.

(3) $\mathfrak{X} \to \mathrm{Spec}(\Lambda)$ is formally smooth.

Proof: We set $\mathfrak{X} = \mathrm{Spf}(A)$. By 4.10(b), $(R, \mathrm{Spf}(\hat{A}))$ is a formal versal deformation of $\hat{X}_0 = \mathrm{Spec}(\hat{A}_0)$ where $\hat{A}_0 = \varprojlim_v A \otimes_R R_v$.
Now $\hat{A}_0 \simeq k[[x_1,\dots,x_n]]/(f)$ and $(\partial f/\partial x_1,\dots,\partial f/\partial x_n) = (x_1,\dots,x_n)$ in $k[[x_1,\dots,x_n]]$. It follows readily that $[\underline{M}_{\hat{X}_0}(k[\epsilon])] \simeq k$ and, by 4.14, we get

$$R \simeq \Lambda[[t]] \quad\text{and}\quad \hat{A} \simeq R[[x_1,\dots,x_n]]/(F - t)$$

where F is a quadratic polynomial in $\Lambda[x_1,\dots,x_n]$ ($\subset R[[x_1,\dots,x_n]]$) obtained from f by lifting the coefficients to Λ . Now

$$0 \to \hat{A} \xrightarrow{df} \hat{A}dx_1 \oplus \dots \oplus \hat{A}dx_n \longrightarrow \hat{\Omega}_{\hat{A}|R} \to 0$$

is exact where $df : \hat{A} \to \hat{A}dx_1 \oplus \dots \oplus \hat{A}dx_n$ is given by $1 \to (\partial f/\partial x_1)dx_1 + \dots + (\partial f/\partial x_n)dx_n$, and therefore $I_{\hat{A}}(\hat{A}|R) = (\partial F/\partial x_1,\dots,\partial F/\partial x_n) = (x_1,\dots,x_n)$, and hence $\hat{A}/I_{\hat{A}}(\hat{A}|R) = \hat{A}/(x_1,\dots,x_n) = R[[x_1,\dots,x_n]]/(x_1,\dots,x_n,F - t) \simeq \Lambda$ i.e., the canonical map $\Lambda \to \hat{A}/I_{\hat{A}}(\hat{A}|R)$ is an isomorphism. It follows that $\hat{A} \to A/I_A(A|R)$ is an isomorphism, and consequently $\Lambda \to R/I_R(A|R)$ is also an isomorphism. This proves (1) and (2). On the other hand, (3) follows from $\hat{A} \simeq R[[x_1,\dots,x_n]]/(F - t) \simeq \Lambda[[t,x_1,\dots,x_n]]/(F - t) \simeq \Lambda[[x_1,\dots,x_n]]$ as Λ-algebras.

Corollary 6.3. Let X_0 be a k-scheme of finite type, and let x in X_0 be a non-degenerate quadratic singularity rational over k .

(1) Let $(R, \mathfrak{X})$, $(R, \mathfrak{Y})$ be any two deformations of X_0 where $R \in ob(\hat{\underline{C}}_\Lambda)$. Assume that $k = k^2$. Then $\mathfrak{X}_{(\mathfrak{x})} \simeq \mathfrak{Y}_{(\mathfrak{x})}$ as R-adic formal schemes if and only if $I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{Y}_{(x)}|R)$.

(2) Let $R \in ob(\hat{\underline{C}}_\Lambda)$ be a regular local ring, and let $(R, \mathfrak{X})$ be a deformation of X_0 . Then $O_{\mathfrak{X},x}$ is a regular local ring if and only if $I_R(\mathfrak{X}_{(x)}|R) \not\subset \underline{m}_R^2$ i.e., if and only if $I_R(\mathfrak{X}_{(x)}|R)$ is a regular divisor in $Spf(R)$.

Proof: Let (R_0, Z) be a formal versal deformation of $Spec(\hat{\underline{O}}_{X_0,x})$. Since x in X_0 is a non-degenerate quadratic singularity rational over k , we may assume that $R_0 \simeq \Lambda[[t]]$ and $Z \simeq Spf(R_0[[x_1,\dots,x_n]]/(F - t))$ as in 6.2.

(1) $(R, \mathfrak{X}_{(\mathfrak{x})})$, $(R, \mathfrak{Y}_{(\mathfrak{x})})$ are deformations of $Spec(\hat{\underline{O}}_{X_0,x})$, and therefore they are induced from (R_0, Z) by continuous local Λ-algebra maps $\varphi, \psi : R_0 \to R$ respectively, i.e., if we set $a = \varphi(t)$, $b = \psi(t)$, then

$$\mathfrak{X}_{(\mathfrak{x})} \simeq Spf(R[[X_1,\dots,X_n]]/(R - a)) \quad ,$$
$$\mathfrak{Y}_{(\mathfrak{x})} \simeq Spf(R[[X_1,\dots,X_n]]/(F - b))$$

as R-adic formal schemes. Assume that $I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{Y}_{(x)}|R)$ i.e.,

$(a) = (b)$ in R. Then $a = ab$ for some unit u in R. Since $k = k^2$ and R is complete, we can write $u = v^2$ for some unit v in R, i.e., $a = v^2 b$. Then $(F - b) = (v^2 F - a)$, so that $X_i \to vX_i$ establishes an R-algebra isomorphism $R[[X_1,\dots,X_n]]/(F - a) \xrightarrow{\sim} R[[X_1,\dots,X_n]]/(F - b)$ since F is a quadratic form. The other direction is trivial.

(2) There exists a continuous local Λ-algebra map $\varphi : R_0 = \Lambda[[t]] \to R$ such that $\mathfrak{X}_{(\mathfrak{x})} \simeq \mathrm{Spf}(R[[X_1,\dots,X_n]]/(F - \varphi(t)))$. We note that $I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{X}_{(\mathfrak{x})}|R) = (\varphi(t))$, and that $R[[X_1,\dots,X_n]]/(F - \varphi(t)) = \hat{\underline{O}}_{\mathfrak{X},x}$ is regular if and only if $\underline{O}_{\mathfrak{X},x}$ is regular. Now R is regular and hence $R[[X_1,\dots,X_n]]$ is also a regular local ring. Thus if we set M to be the maximal ideal of $R[[X_1,\dots,X_n]]$, then $R[[X_1,\dots,X_n]]/(F - \varphi(t))$ is regular if and only if $F - \varphi(t)$ is not in M^2, i.e. if and only if $\varphi(t) \notin \underline{m}_R{}^2$, i.e., if and only if $I_R(\mathfrak{X}_{(x)}|R) \notin \underline{m}_R{}^2$.

Theorem 6.4. *Let* X_0 *be a k-scheme of finite type with isolated non-smooth points only, and let* $(R, \mathfrak{X})$ *be a formal versal deformation of* X_0. *Denote by* $\{z_1,\dots,z_m\}$ *the set of non-smooth points in* X_0. *Assume that* $H^2(X_0, \underline{D}^0_{X_0}) = 0$, *and all* z_i*'s are non-degenerate quadratic singularity rational over* k. *Then*

(1) R *is formally smooth over* Λ i.e., R *is a formal power-series ring over* Λ;

(2) $J_{\mathfrak{X}}(\mathfrak{X}|R) = \coprod_{1 \le i \le m} J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R)$ *(disjoint union), and* $J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) \xrightarrow{\sim} J_R(\mathfrak{X}_{(z_i)}|R)$ *is an isomorphism*;

(3) $I_R(\mathfrak{X}_{(z_i)}|R)$ *is a principal ideal and in turn so is* $I_R(\mathfrak{X}|R) = \prod_{1 \le i \le m} I_R(\mathfrak{X}_{(z_i)}|R)$. *Furthermore if we set* $I_R(\mathfrak{X}_{(z_i)}|R) = (t_i)$, *then* $t_1,\dots,t_m$ *can be extended to a system of free formal generators of* R *over* Λ *(as formal power-series ring over* Λ *).*

(4) $\mathfrak{X} \to \mathrm{Spec}(\Lambda)$ *is smooth*, i.e. $J_{\mathfrak{X}}(\mathfrak{X}|\Lambda) = \varphi$.

(5) *If* X_0 *is a complete irreducible curve, then* $\dim \underline{M}_{X_0} = 3g - 3 + a$ *where* $g = \dim_k H^1(X_0, \underline{O}_{X_0})$ *and* $a = \dim_k H^0(X_0, \underline{D}^0_{X_0})$.

Proof: (1) follows from (2) and (3): The first statement of (2) follows from 5.7(2). Now let $(R_i, \mathfrak{Y}_i)$ be a formal versal deformation of $\mathrm{Spec}(\hat{\underline{O}}_{X_0,z_i})$. Since $(R, \mathfrak{X}_{(\hat{z}_i)})$ is a deformation of $\mathrm{Spec}(\hat{\underline{O}}_{X_0,z_i})$, there exists a continuous local Λ-algebra maps $\varphi_i = R_i \to R$ with a cartesian diagram

$$\begin{array}{ccc} \mathfrak{X}_{(\hat{z}_i)} & \longrightarrow & \mathfrak{Y}_i \\ \downarrow & & \downarrow \\ \mathrm{Spf}(R) & \xrightarrow[\mathrm{Spf}(\varphi_i)]{} & \mathrm{Spf}(R_i) \end{array}$$

Therefore we have

$$J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) = J_{\mathfrak{X}_{(\hat{z}_i)}}(\mathfrak{X}_{(\hat{z}_i)}|R) \simeq J_{\mathfrak{Y}_i}(\mathfrak{Y}_i|R_i) \underset{\mathrm{Spf}(R_i)}{\times} \mathrm{Spf}(R)$$

$$J_R(\mathfrak{X}_{(z_i)}|R) = J_R(\mathfrak{X}_{(\hat{z}_i)}|R) \simeq J_{R_i}(\mathfrak{Y}_i|R_i) \underset{\mathrm{Spr}(R_i)}{\times} \mathrm{Spf}(R) .$$

Since $J_{\mathfrak{Y}_i}(\mathfrak{Y}_i|R_i) \to J_{R_i}(\mathfrak{Y}_i|R_i)$ is an isomorphism by 6.2(1), we have that the canonical morphism $J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) \to J_R(\mathfrak{X}_{(z_i)}|R)$ is an isomorphism. Now $I_{R_i}(\mathfrak{Y}_i|R_i) = (u_i) = (u_i)$ with $R_i = \Lambda[[u_i]]$ by 6.2(2), and therefore

$$I_R(\mathfrak{X}_{(z_i)}|R) = I_R(\mathfrak{X}_{(\hat{z}_i)}|R) = (t_i)$$

where $t_i = \varphi_i(u_i)$, and therefore $I_R(\mathfrak{X}|R) = \prod_{1 \le i \le m} I_R(\mathfrak{X}_{(z_i)}|R) = (\prod_{i=1}^{n} t_i)$.

Now $\underline{M}_{X_0} \to \underline{M}_{X_0,(\hat{z}_1)} \times \dots \times \underline{M}_{X_0,(\hat{z}_m)}$ is smooth by and therefore $R_1 \hat{\otimes}_\Lambda \dots \hat{\otimes}_\Lambda R_m \to R$ is formally smooth by 4.13, which implies that $t_1,\dots,t_m$ is extended to a system of free formal generators of R over Λ .

(4) The projection morphism $\mathfrak{X}_{(\mathfrak{z}_i)} \cong \mathfrak{U}_i \times_{\mathrm{Spf}(R_i)} \mathrm{Spf}(R) \to \mathfrak{U}_i$ is formally smooth since $\mathrm{Spf}(R) \xrightarrow{\mathrm{Spf}(\varphi_i)} \mathrm{Spf}(R_i)$ is formally smooth. On the other hand, $\mathfrak{U}_i \to \mathrm{Spec}(\Lambda)$ is formally smooth by 6.2(3) and therefore $\mathfrak{X}_{(\mathfrak{z}_i)} \to \mathrm{Spec}(\Lambda)$ is formally smooth. Consequently $\mathfrak{X} \to \mathrm{Spec}(\Lambda)$ is formally smooth since $\{z_1, \dots, z_m\}$ is the set of non-smooth points of X_0 .

(5) Since R is formally smooth over Λ and

$$0 \to H^1(X_0, \underline{D}^0_{X_0}) \to [\underline{M}_{X_0}(k[\epsilon])] \to H^0(X_0, \underline{D}^1_{X_0}) \to 0$$

is exact by 4.4 and 4.11, we have

$$\begin{aligned}\dim \underline{M}_{X_0} &= \dim k \otimes R = \dim_k H^1(X_0, \underline{D}^0_{X_0}) + \dim H^0(X_0, \underline{D}^1_{X_0}) \\ &= \dim_k H^1(X_0, \underline{D}^0_{X_0}) + m \ .\end{aligned}$$

Then it suffices to show that

$$\dim_k H^1(X_0, \underline{D}^0_{X_0}) + m = 3g - 3 + \dim_k H^0(X_0, \underline{D}^0_{X_0})$$

i.e., $\chi(\underline{D}^0_{X_0}) = 3 - 3g + m$

where we set $\chi(\underline{E}) = \dim_k H^0(X_0, \underline{E}) - \dim_k H^1(X_0, \underline{E})$ for any coherent X_0-module $\underline{E}$. Since $\lambda(\underline{D}^0_{X_0})$ does not change when one replaces the base field k by its algebraic closure, we may assume that k is algebraically closed. Set $A = \underline{O}_{X_0, z}$, where z is a non-smooth point of X_0 , and let A' be its normalization. By hypothesis, we have $A \simeq k[[x,y]]/(f)$ where f is a non-degenerate quadratic form. Since k is algebraically closed, we may assume that $f = xy$, i.e., $A \simeq k[[x,y]]/(xy)$. From this we conclude immediately that $\ell(A'/A) = 1$ and $\mathrm{Der}_k(A, \underline{m}_A) = \mathrm{Der}_k(A, A)$. Let C be the conductor of A in A' . Since A is a complete-intersection we must have that $\ell(A/C) = \ell(A'/A) = 1$ so that $C = \underline{m}_A$. We thus obtain that $A' = \{a' \in K \mid a'\underline{m}_A \subset \underline{m}_A\}$ where K is the fraction field of A . We now claim that $\mathrm{Der}_k(A, A) \subset \mathrm{Der}_k(A', A')$ as A-modules contained in $\mathrm{Der}_k(K, K)$. Indeed, let $X \in \mathrm{Der}_k(A, A)$. Then for any $a' \in A'$ and

$x \in \underline{m}_A$ we have $xa' \in \underline{m}_A$ so that $X(xa') \in \underline{m}_A$ and $a'X(x) \in \underline{m}_A$ since $\mathrm{Der}_k(A, \underline{m}_A) = \mathrm{Der}_k(A, A)$. Consequently $X(xa') = a'X(x) + xX(a')$ entails that $xX(a') \in \underline{m}_A$ for all $x \in \underline{m}_A$, and therefore $X(a') \in A'$. Furthermore, the exact sequence

$$0 \to \mathrm{Der}_k(A, A) \to \mathrm{Der}_k(A', A') \to A'/\underline{m}_A A' \to 0$$

entails the exact sequence

$$0 \to \mathrm{Der}_k(A, A) \to \mathrm{Der}_k(A', A') \to A'/C \to 0$$

Therefore if we set X_0' to be the normalization of X_0, then

$$0 \to \underline{D}^0_{X_0} \to \underline{D}^0_{X_0'} \to \underline{O}_{X_0'}/C \to 0$$

is exact. Since X_0 is locally a complete-intersection, we have $\ell(\underline{O}_{X_0'}/C) = 2\ell(\underline{O}_{X_0'}/\underline{O}_{X_0}) = 2m$, whereas $\chi(\underline{D}^0_{X_0'}) = 3 - 3(g-m)$ by Riemann-Roch theorem. Consequently

$$\chi(\underline{D}^0_{X_0}) = \chi(\underline{D}^0_{X_0'}) - \ell(\underline{O}_{X_0'}/C) = 3 - 3(g-m) - 2m = 3 - 3g + m .$$

This completes our proof.

Remark 6.5. In the formula 6.4(5), the number $a = H^0(X_0, \underline{D}^0_{X_0})$ can be easily computed in terms of Riemann-Roch theorem. Thus

$$a = \begin{cases} 0 & \text{if } g \geq 2 \\ 1 & \text{if } g = 1 \\ 3 & \text{if } g = 0 \end{cases}$$

Finally we add a remark on characteristic 2. If the characteristic of the base field k is 2, there is no non-degenerate quadratic form on n variables if n is odd. In other words there does not exist a non-degenerate quadratic singularity of even dimension if $\mathrm{char}\, k = 2$. In this case one may consider a quadratic form of maximum non-degeneracy:

Definition 6.6. *Let* X *be a* k*-scheme of finite type. A closed point* $z \in X$ *is called an ordinary quadratic singularity rational over* k *if the following holds:*

(i) *If either* $\operatorname{char} k \neq 2$ *or* $\operatorname{char} k = 2$ *and* $\dim O_{X,z}$ *is odd, then* a *is a non-degenerate quadratic singularity rational over* k .

(ii) *If* $\operatorname{char} k = 2$ *and* $\dim O_{X,z}$ *is even, then* $\bar{k} \otimes_k \hat{O}_{X,z} \simeq \bar{k}[[X_0, X_1, \dots, X_{2n}]]/(f)$ *with*

$$f = X_0^2 + X_1X_2 + X_3X_4 + \dots + X_{2n-1}X_{2n}$$

where $\bar{k}$ *is the algebraic closure of* k .

Lemma 6.7. *Let* k *be a field of characteristic* 2 *and set* $A_0 = k[[X_0, X_1, \dots, X_{2n}]]/(f)$ *where*

$$f = X_0^2 + X_1X_2 + X_3X_4 + \dots + X_{2n-1}X_{2n} \quad ,$$

and let $(R, \mathfrak{X})$ *be a formal versal deformation of* $X_0 = \operatorname{Spec}(A_0)$. *Then* $R = \Lambda[[t, t_1]]$, *where* t, t_1 *are free formal generators of* R *over* Λ , *such that* $I_R(\mathfrak{X}|R) = (t^2)$, *and* $\mathfrak{X} \to \operatorname{Spec}(\Lambda)$ *is formally smooth.*

Proof: We set $\mathfrak{X} = \operatorname{Spf}(A)$. Since

$$(f, \frac{\partial f}{\partial X_0}, \dots, \frac{\partial f}{\partial X_{2n}}) = (X_0^2, X_1, \dots, X_{2n}) \quad ,$$

it follows that $D^1(A_0|k, A_0) \simeq k[X_0]/(X_0^2)$ and therefore $[\underline{M}_{X_0}(k[\epsilon])]$ is 2-dimensional vector space over k . Furthermore, it follows from 4.14 that

$$R \simeq \Lambda[[t, t_1]] \quad \text{and} \quad A \simeq R[[X_0, X_1, \dots, X_{2n}]]/(F)$$

where $F = f + t_1 + tX_0$. Now the exact sequence

$$0 \to A \xrightarrow{dF} AdX_0 \oplus \dots \oplus AdX_{2n} \to \hat{\Omega}_{A|R} \to 0$$

where

$$dF = 1 \to \sum_{0 \le i \le 2n} \frac{\partial f}{\partial X_i} dX_i + dX_0 \quad ,$$

entails that

$$I_A(A|R) = (t, x_1, x_2, \dots, x_{2n}) = (x_0^2 + t_1, t, x_1, \dots, x_{2n})$$

so that

$$A/I_A(A|R) \simeq A/(x_0^2 + t_1, t, x_1, \dots, x_{2n}) \simeq \Lambda[[t_1, x_0]]/(x_0^2 - t_1) \quad .$$

Since $\Lambda[[t_1, x_0]]/(x_0^2 - t_1) \simeq R/(t) \oplus R/(t)$ as R-modules, it follows that $I_R(A|R) = (t^2)$, which proves our first assertion. On the other hand,

$$A \simeq \Lambda[[t, t_1, x_o, x_1, \dots, x_{2n}]]/(x_0^2 + X_1X_2 + \dots + X_{2n-1}X_{2n} + tX_0 + t_1)$$
$$\simeq \Lambda[[t, X_0, X_1, \dots, X_{2n}]]$$

and therefore A is formally smooth over Λ .

The above statement 6.7 yields a similar result as 6.4, proof of which we leave to the readers.

Corollary 6.8. Let X_0 be a k-scheme of finite type with isolated non-smooth points only such that $H^2(X_0, \underline{D}^0_{X_0}) = 0$, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0 . Denote by $\{z_1, \dots, z_m\}$ the set of non-smooth points in X_0 , and assume that char $k = 2$, dim X_0 is even, and all z_i's are ordinary quadratic singularities. Then

(1) R is formally smooth over Λ

(2) $I_R(\mathfrak{X}_{(z_i)}|R)$ is a principal ideal and in turn so is $I_R(\mathfrak{X}|R) = \prod_{1 \leq i \leq m} I_R(\mathfrak{X}_{(z_i)}|R)$. Furthermore, $I_R(\mathfrak{X}_{(z_i)}|R) = (t_i^2)$ and $t_1, \dots, t_m$ can be extended to a system of free formal generators of R over Λ .

(3) $\mathfrak{X} \to \operatorname{Spec}(\Lambda)$ is formally smooth

(4) If k is algebraically closed and X_0 is a complete irreducible curve, then $\dim \underline{M}_{X_0} = 3g - 3 + a$ where $g = \dim_k H^1(X_0, \underline{O}_{X_0})$ and $a = \dim_k H^0(X_0, \underline{D}^0_{X_0})$.